REPUBLIC OF INDIA

BANGALORE WATER SUPPLY AND SEWERAGE BOARD

BENGALURU WATER SUPPLY AND SEWERAGE PROJECT (PHASE 3)

IN

THE STATE OF KARNATAKA, INDIA

FINAL REPORT

(Data Report)

NOVEMBER 2017

JAPAN INTERNATIONAL COOPERATION AGENCY (JICA)

NJS CONSULTANTS CO., LTD. (NJS)

4R
CR(5)
17-062

1 USD = 112 Yen
1 INR = 1.74 Yen
(As of June 2017)

Data Report

Natural Condition Survey

Summary

* Note: All the raw data are compiled in the soft copy.

Kindly refer to the data file.

Item	Point	Period/Frequency	Source	Data Type	Idea & Comment from Surveyor
(I) Natural Conditions					
Temperature	Inside of BBMP If there are, multi-point is desirable The List and Detail of Observation Stations in BBMP are also required.	Monthly maximum/average/minimum data in past 5 years	Meteorological Department	Excel	Data will be available from Meteorological Dept., Bangalore
Rainfall, Precipitation					
Wind Direction		Monthly typical direction			
Wind Speed		Monthly typical speed			
Other data (Humidity, etc.)		Monthly average data			
(II) Water Source/Environment					
Cauvery River Flow Volume and Water Quality	Before and after the existing intake point	a) Monthly/Annual flow in past 5 years	WRDO	Paper	Usually Paper Format is provided. We will check if Excel Soft Copy data is available
		b) Parameter-BOD, COD, SS, DO. Etc.	PCB	Paper	Usually Paper Format is provided. We will check if Excel Soft Copy data is available
Water Level Fluctuation at Intake Point	SBR = Shiva Balancing Reservoir NBR = Netkal Balancing Reservoir	Annual Fluctuation (HWL/LWL)	BWSSB / WRDO	Paper	Data should be available with BWSSB. If data not available with BWSSB, we will check with WRDO
Abolished Arkavathi River	Before Hesaraghatta/TG Halli intake (now abandoned)	c) Monthly/Annual flow in past 5 years	BWSSB / WRDO	Paper	Data should be available with BWSSB. If data not available with BWSSB, we will check with WRDO
		d) Parameter-BOD, COD, SS, DO. Etc.	PCB / BWSSB	Paper	Usually Paper Format is provided. We will check if Excel Soft Copy data is available
Another available/promising surface source (Lake/Dam)	If any, one or two point	ditto			Not Available
Ground Water Quality	Around 10 points of existing groundwater pumping stations in BBMP	Water Quality: BOD,COD, SS, NH^{4+}-N, NO^{3-}--N	PCB / CGWB	Paper	PCB usually collects data once in 6 months or 12 months. If not available, We will have to collect the sample and get it analyzed.
Lake Water Quality	Around 10 points of swamp/lake in BBMP	Water Quality: BOD,COD, SS, NH^{4+}-N, NO^{3-}--N	BWSSB / PCB / Lake Authority	Paper	If not available, We will have to collect the sample and get it analyzed.
(III) Others					
Survey on Rainwater Harvesting (groundwater recharge)	Record/Information of harvest project in BBMP	In cooperation with BWSSB	BBMP / BWSSB / CGWB / Mines & Geology	Paper	Ready reports will be available and can be procured from any of these departments
Survey on STP Effluent Reuse project	Record/information of effluent reuse project in BBMP	In cooperation with BWSSB	BWSSB	Paper	Data and information available locally
Water Right of Cauvery River	Water allocated to domestic use, irrigation, industry, commercial, tourism, etc. if available	In cooperation with State agency	WRDO	Paper	From WRDO - Cauvery unit, Cauvery river water allocation notification will be available and data on actual utilization may also be available with this department
Note : WRDO Hydrology Unit publishes water year book, wherein sufficient information is available.					
PCB = Pollution Control Board WRDO = Water Resources Development Organization CGWB = Central Ground Water Board BWSSB = Bangalore Water Supply & Sewerage Board BBMP = Bruhat Bengaluru Mahanagara Palike					

Bangalore Rural District Average Rainfall Data.

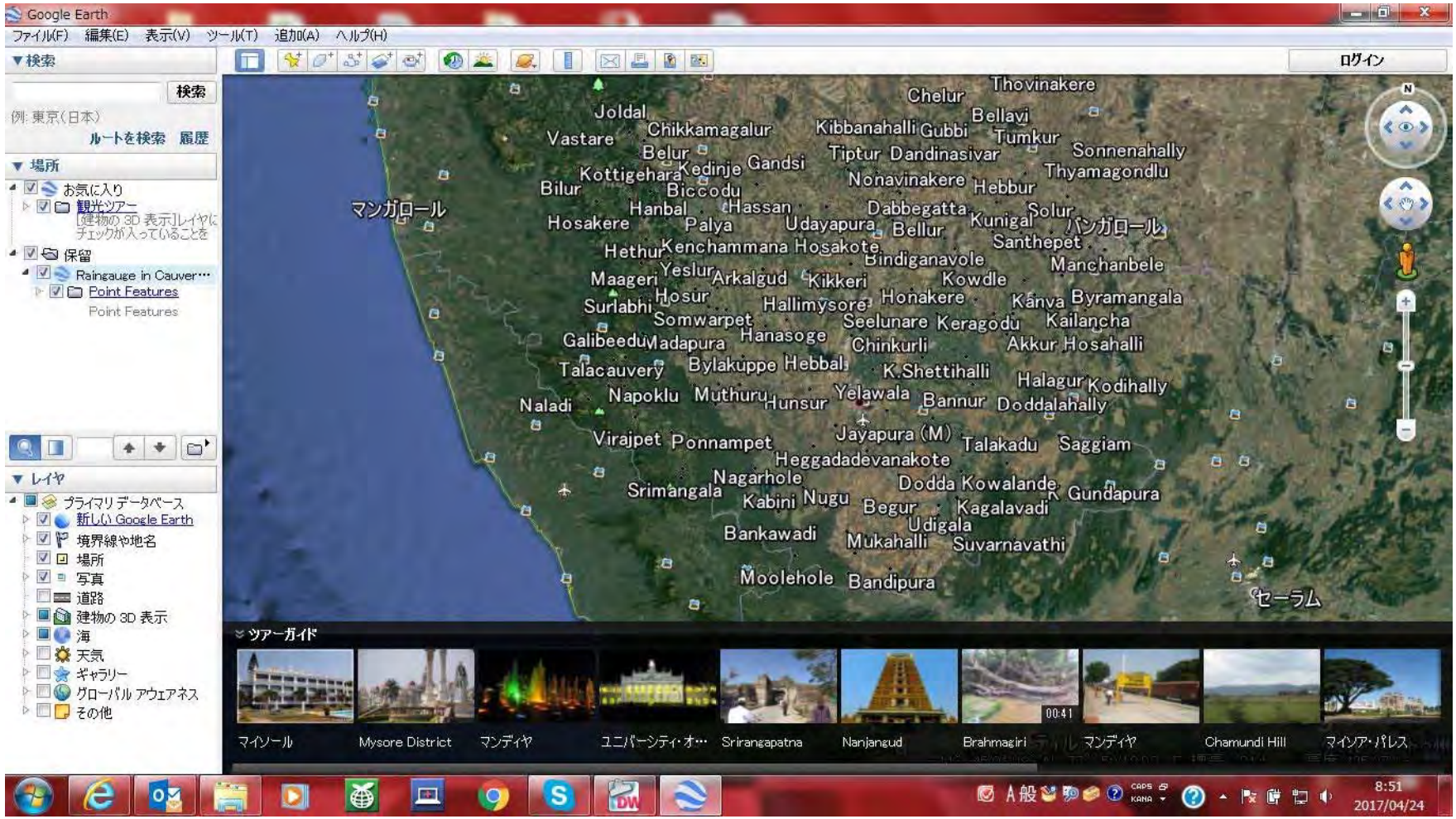
YEAR	JAN		FEB		MAR		APR		MAY		JUN		JUL		AUG		SEPT		OCT		NOV		DEC	
	R/F	%DEP	R/F	%DEP	R/F	%DEP	R/F	%DEP	R/F	%DEP	R/F	%DEP	R/F	%DEP	R/F	%DEP	R/F	%DEP	R/F	%DEP	R/F	%DEP	R/F	%DEP
2010	0	-100	0	-100	18.5	161	42.4	18	114	10	141.4	136	145.6	56	148.4	30	108.9	-33	86.1	-47	149.1	169	11.2	-31
2011	0	-100	2.5	-57	2.7	-64	88.7	145	97.5	5	47.4	-33	106.1	16	230.6	95	61.9	-64	130.2	-24	26.7	-52	5.2	-61
2012	0	-100	0	-100	4.3	-43	57.7	59	108.2	16	20.4	-71	73.2	-20	138.3	17	47.6	-72	79.6	-54	106.8	93	14.4	7
2013	0	-100	6.9	19	2.3	-70	38.9	7	119.6	28	81.5	14	83.7	-9	75.8	-36	268.5	57	130.7	-24	59.8	8	5.1	-62
2014	0	-100	0.1	-98	19.6	158	11	-70	92.5	-1	61.4	-14	90.1	-2	162.6	38	202.3	18	267.2	56	35.4	-36	3.9	-71

The District Rainfall in millimeters (R/F) shown above are the arithmatic averages of Rainfall of all Stations under the District.
% Dep. are the Departures of rainfall from the long period averages of rainfall for the District.
Blank Spaces show are non-availability of Data

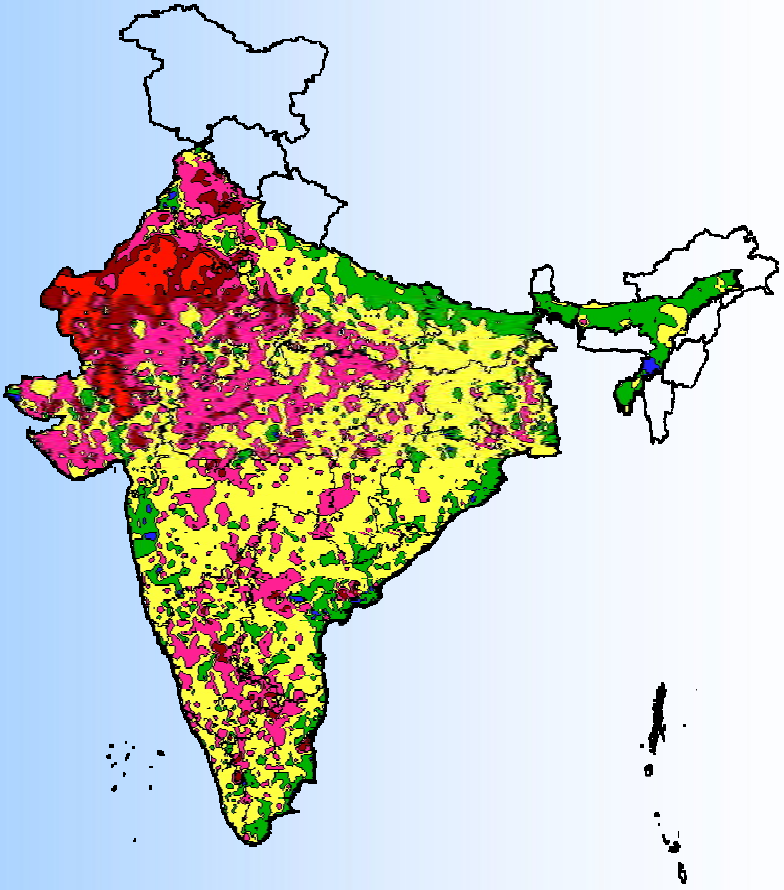
Bangalore Urban District Average Rainfall Data.

YEAR	JAN		FEB		MAR		APR		MAY		JUN		JUL		AUG		SEPT		OCT		NOV		DEC	
	R/F	%DEP	R/F	%DEP	R/F	%DEP	R/F	%DEP	R/F	%DEP	R/F	%DEP	R/F	%DEP	R/F	%DEP	R/F	%DEP	R/F	%DEP	R/F	%DEP	R/F	%DEP
2010	5.4	575	0	-100	17.4	100	97.5	170	123.9	19	67.3	-4	112.3	18	121.1	-4	149.9	-14	95.8	-34	166.8	207	2.7	-84
2011	0	-100	18.2	172	2	-76	157.8	303	139.1	31	54.5	-26	92.6	1	237.5	84	89.3	-51	170.8	7	64.8	20	5.9	-67
2012	0.1	-96	0	-100	0.6	-93	21.1	-46	108.1	2	9.2	-88	63.7	-30	130.5	1	41.3	-77	60	-62	100.6	87	17.2	-4
2013	0	-100	4	-40	2.6	-68	28.9	-26	110.1	4	93.1	26	76.2	-17	72.6	-44	249.3	38	95.5	-40	63.3	17	1.4	-92
2014	0	-100	0.1	-99	12.9	57	7.2	-82	83.2	-22	93.7	27	80.6	-12	118.1	-8	181.8	0	247.3	55	27.8	-48	2	-89

The District Rainfall in millimeters (R/F) shown above are the arithmatic averages of Rainfall of all Stations under the District.
% Dep. are the Departures of rainfall from the long period averages of rainfall for the District.
Blank Spaces show are non-availability of Data


Google Earth
ファイル(F) 編集(E) 表示(V) ツール(T) 追加(A) ヘルプ(H)
ログイン
検索
例: 東京(日本)
ルートを検索 履歴
場所
お気に入り
観光ツアー
保留
Point Features
レイヤ
プライマリデータベース
新しい Google Earth
境界線や地名
場所
写真
道路
建物の 3D 表示
海
天気
ギャラリー
グローバル アウェアネス
その他
マンガロール
Chelur
Thovinakere
Joldal
Bellavi
Vastare
Chikkamagalur
Kibbanahalli
Gubbi
Tumkur
Belur
Tiptur
Dandinasivar
Sonnenahally
Kottigehara
Kedinje
Gandsi
Thyamagondlu
Nonavinakere
Hebbur
Bilur
Biccodu
Hanbal
Hassan
Dabbegatta
Solur
Hosakere
Palya
Udayapura
Bellur
Kunigal
バンガロール
Hethur
Kenchammana Hosakote
Santhepet
Bindiganavole
Manchanbele
Maageri
Yeslur
Arkalgud
Kikkeri
Kowdle
Surlabhi
Hosur
Hallimysore
Honakere
Kanva
Byramangala
Somwarpet
Seelunare
Keragodu
Kailancha
Galibeedu
Madapura
Hanasoge
Chinkurli
Akkur
Hosahalli
Talacauvery
Bylakuppe
Hebbal
K.Shettihalli
Halagur
Kodihally
Napoklu
Muthuru
Hunsur
Yelawala
Bannur
Doddalahally
Naladi
Virajpet
Ponnampet
Jayapura (M)
Talakadu
Saggiam
Heggadadevanakote
Nagarhole
Srimangala
Kabini
Nugu
Dodda Kowalande
Gundapura
Begur
Kagalavadi
Udigala
Bankawadi
Mukahalli
Suvarnavathi
Moolehole
Bandipura
セーラム
ツアーガイド
マイソール
Mysore District
マンディヤ
ユニバーシティ・オ…
Srirangapatna
Nanjangud
Brahmagiri
マンディヤ
Chamundi Hill
マイソア・パレス
00:41
8:51
2017/04/24

Ground Water Scenario of India 2009-10



Central Ground Water Board
Ministry of Water Resources
Government of India
Faridabad
June 2010

Ground Water Scenario of India 2009-10

Central Ground Water Board
Ministry of Water Resources
Government of India
Faridabad
2010

CONTENTS

TOPIC	Page No.
1.0 Hydrogeological Setup of the Country	1-2
1.1 Ground Water Occurrence	
1.1.1 Porous Formations	
1.1.2 Fissured Formations (Consolidated Formations)	
1.2 Hydrogeological Units and their Ground Water Potential	
2.0 Ground Water Level Scenario	3-6
2.1 Introduction	
2.2 Ground Water Level Scenario – Pre-Monsoon, 2009	
2.3 Ground Water Level Scenario – August, 2009	
2.4 Ground Water Level Scenario – Post-Monsoon, 2009	
2.5 Ground Water Level Scenario – January, 2010	
2.6 Pre – Post Water Level Fluctuation Scenario – 2009	
3.0 Ground Water Resource Availability and Development Status	7-12
3.1 Dynamic Fresh Ground Water Resource	
3.2 Stage of Ground Water Development	
3.3 Categorisation of Assessment Units	

LIST OF PLATES

PLATES	No.
Hydrogeological Map of India	I
Location of Ground Water Monitoring Wells	II
Depth to Water Level, Pre-Monsoon 2009	III
Water Level Fluctuation, Pre-Monsoon 2009 Vs Decadal Mean (1999-2008)	IV
Depth to Water Level, August 2009	V
Water Level Fluctuation, August 2009 Vs Decadal Mean (1999-2008)	VI
Depth to Water Level, Post-Monsoon 2009	VII
Water Level Fluctuation, Post-Monsoon 2009 Vs Decadal Mean (1999-2008)	VIII
Depth to Water Level, January 2010	IX
Water Level Fluctuation, January 2010 Vs Decadal Mean (1999-2008)	X
Water Level Fluctuation, Pre-Monsoon 2009 Vs Post-Monsoon 2009)	XI
Ground Water Level of India At A Glance	XII
Annual Rainfall Map 2009	XIII
Ground Water Resource Availability and Utilisation In India	XIV
Categorisation of Ground Water Assessment Units	XV

LIST OF TABLES

TABLES	No.
Ground Water System In The Country	1
State Wise Distribution of Observation Wells	2
State Wise Ground water Resources Availability, Utilisation and Stage of Development	3
Categorisation of Ground Water Assessment Units	4

LIST OF ANNEXURES

ANNEXURES	No.
State - Wise Distribution Of Percentage Of Observation Wells - 2009 May (Pre-Monsoon)	I
State - Wise Distribution Of Percentage Of Observation Wells - 2009 August	II
State - Wise Distribution Of Percentage Of Observation Wells - 2009 November (Post-Monsoon)	III
State - Wise Distribution Of Percentage Of Observation Wells - 2010 January	IV
State - Wise Fluctuation And Frequency Distribution From Different Ranges From 2008 May To 2009 May	V
State - Wise Fluctuation And Frequency Distribution From Different Ranges From 2008 August To 2009 August	VI
State - Wise Fluctuation And Frequency Distribution From Different Ranges From 2008 November To 2009 November	VII
State - Wise Fluctuation And Frequency Distribution From Different Ranges From 2009 January To 2010 January	VIII
State - Wise Fluctuation And Frequency Distribution From Different Ranges From 2009 May To 2009 August	IX
State - Wise Fluctuation And Frequency Distribution From Different Ranges From 2009 May To 2009 November	X
State - Wise Fluctuation And Frequency Distribution From Different Ranges From 2009 May To 2010 January	XI
State - Wise Fluctuation And Frequency Distribution With Decadal Mean (1999-2008) And 2009 May	XII
State - Wise Fluctuation And Frequency Distribution With Decadal Mean (1999-2008) And 2009 August	XIII
State - Wise Fluctuation And Frequency Distribution With Decadal Mean (1999-2008) And 2009 November	XIV
State - Wise Fluctuation And Frequency Distribution With Decadal Mean (2000-2009) And 2010 January	XV

Data Report

Topographical Survey

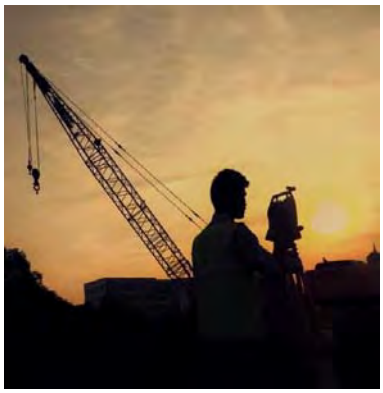
Report, Data and Drawings

* Note: All the raw data are compiled in the soft copy.

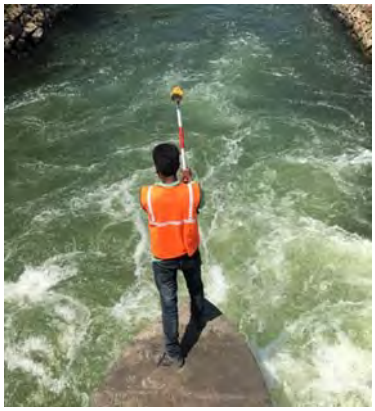
Kindly refer to the data file.

NJS CONSULTANTS CO., LTD.

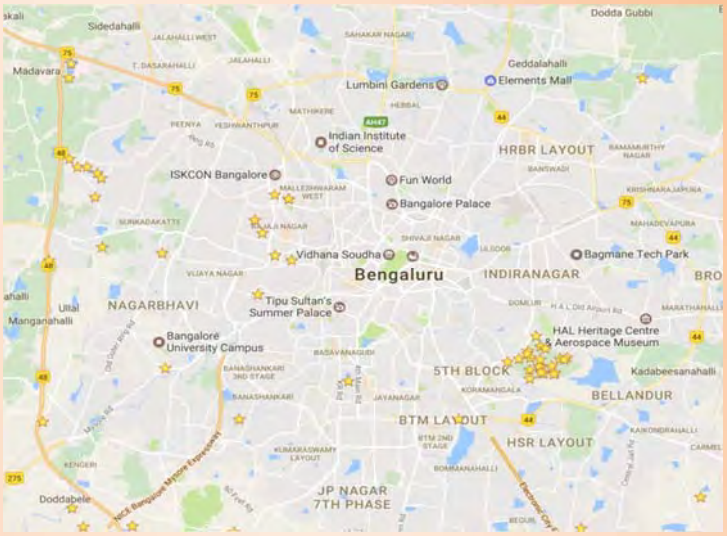
BENGALURU WATER SUPPLY & SEWERAGE PROJECT (PHASE III)

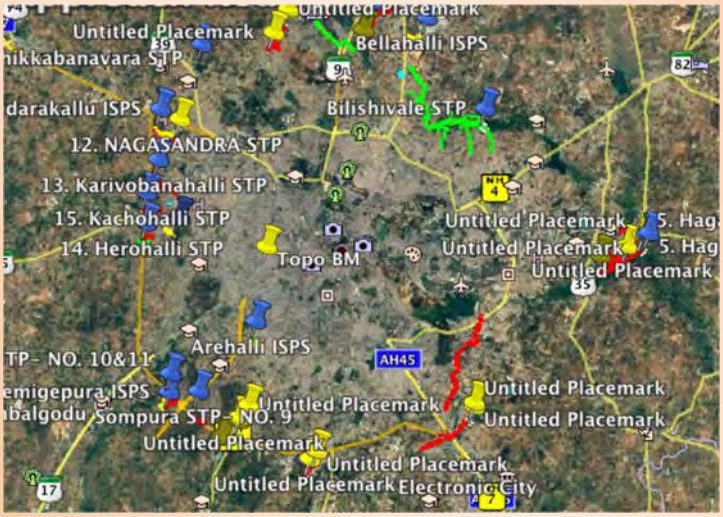




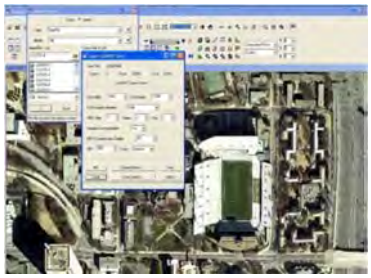


Topographical Survey Report





Feb 2017



Submitted By

Abel Engineering Consultants, Bangalore

TECHNICAL SPECIFICATION ADOPTED FOR CONDUCTING TOPOGRAPHICL SURVEY FOR BANGALORE WATER SUPPLY AND SEWERAGE PROJECT (PHASE III)

Scope of work

1. Centerline longitudinal profile survey for pipeline route.
2. Plane table survey for facilities construction site

Specifications

The work was executed according to the specifications and good standard of practice necessary to fulfil the objective of the survey work.

Carrying GTS Benchmark

Carrying out the benchmark by deploying GNSS GPS System levelling from the nearest SOI Benchmark. The BM location with value was Obtained from Survey Of India. The Following SOI BM was Used to establish BM,s Throughout the Project area.



G.T.S. Bench Mark of RL +903.2285m Carried from end of 3rd step leading to Trinity Church Bangalore, under porch on West face of the Building. It is Situated on Mahatma Gandhi Road.

Fly levelling was also carried out using two good quality Electronic Digital Levels and Bar Coded Leveling Staffs and Reading levels simultaneously. The levelling instruments was always kept free of collimation error, which was checked and adjusted, before start of work every day.

Closing error in levelling was limited to 10 $\sqrt{}$ L mm. where L is the length of the route in km.

List of Benchmarks

Benchmark Description						
SL No.	**Facility Name**	**Point ID**	**RL**	**Easting**	**Northing**	**Source BM**
1	Tataguni GLR	T1	825.472	773155.229	1421686.594	BWSSB BM RL = 822.500 m Marked on a 600mm Dia Pipe inside Tataguni Pumping Station
2	Pillegenahalli	TBM1	884.165	778992.934	1420202.349	
3	Thalaghattapura	TBM2	843.834	774218.867	1423082.293	
4	Sompura	TBM3	807.510	771515.978	1424731.706	
5	Hemmigepura	TBM4	780.393	769535.841	1424746.927	
6	Arehalli	TBM5	845.647	775471.996	1429303.080	
7	GKVK GLR	TBM6	932.488	777509.915	1448533.164	BWSSB BM RL = 933.405m Marked on steps heading to the top of GLR inside GKVK GLR Compound.
8	Kattigenahalli	TBM7	904.442	777532.598	1450628.393	
9	Vasudevpura	TBM8	920.185	778989.260	1452562.984	
10	Jakkur	TBM9	921.061	780365.164	1449888.826	
11	Bellahalli	TBM10	897.960	786653.200	1449966.023	
12	Chokkanahalli	TBM11	920.973	785411.253	1447204.799	
13	Bilishivile	TBM12	912.431	786119.643	1447835.360	
14	Chickbanavara	TBM13	860.759	771568.395	1448017.367	
15	TS 426	TBM14	908.326	788579.096	1424768.152	Survey of India BM = 903.229m On 3rd Step Leading to Trinity Church on M G Road, Bangalore.
16	Naganathapura	TBM15	900.874	790173.860	1423861.129	
17	Doddakkanahalli	TBM16	907.310	793188.649	1425940.764	
18	Kadugodi	TBM17	896.555	798100.748	1437136.181	Survey of India BM = 868.0418m Inside Whitefield Railway Station. Next to Toilet block present near old Railway Station Building.
19	Hagadur	TBM18	853.502	802000.469	1435538.874	
20	Doddabidirakallu	T0	861.849	768736.098	1443704.939	BWSSB BM RL = 869.569m On top of plinth of Contractor's site office at BWSSB Nagasandra STP
21	Kachohalli	BM1	857.141	768045.177	1438167.176	
22	Lingadeeranahalli	BM2	906.917	769900.225	1438538.550	
23	Karivobanahalli	BM3	872.668	768852.685	1440213.295	
24	Herohalli ISPS	BM4	890.572	770068.186	1436503.479	
25	Herohallli STP	BM5	851.659	768150.372	1435938.458	

Traversing



Triangulation or traversing or a combination of the two methods, was adopted for the purpose of establishing horizontal control and in order to determine the exact relationship between various existing points on ground, so that surveys required under the present scope of work and in future may be co-related and tied together.

GPS MSL values were used for the entire survey work.

Total station instruments were deployed to achieve the specified accuracy of the work, Proper precautions for avoiding graduation errors and other instrumental and personal errors was scrupulously observed.

Main traverse/triangulation station, subsidiary station was established at suitable intervals to cover the entire project area. Traverse Points were established throughout the project area. Traversing was Conducted by Total Station and Levels for these stations were provided by Digital Levels.

The closing error in traverse was checked to be within permissible limits and the permissible error shall not exceed one in twenty-five thousand (1:25000) in terms of length, or L√N seconds total in angular measurement, whichever is less (where L is the least count of the instrument and N is the number of stations).

Contouring

Spot level surveying at an interval of ***5 meters*** along all roads, and Natural Nallas was carried out. The contours were interpolated at ***0.50 meters'*** intervals based on Leveling Conducted

Topographical Survey and Mapping



Position, both in plan and elevation, of all natural and artificial features of the area like waterways, railway tracks, trees, cultivation, houses, fences, pucca and kutcha roads including culverts, crossings, foot tracks, other permanent objects like telephone posts, and transmission towers, etc are to be established and subsequently shown on survey maps by means of conventional symbols, all hills and valleys within in the area/areas are to be surveyed and plotted on maps by contours. Necessary levelling work of the entire area/areas was surveyed and plotted on maps by establishing horizontal location so that location and sketching of contours for the area/areas can be done at specified intervals and in specified scales on maps. Method of survey, contour intervals etc. shall be decided by engineer-in-charge on site in case of steep slopes and dense jungle etc. where griding is not possible. Any unusual condition or formations on the ground, locations of rock outcrops (if visible on the surface), and springs/falls, possible aggregate deposits etc. shall be noted and plotted on the maps.
The fieldwork was done with Total Station Equipment in the following steps:

i) Establishing horizontal and vertical controls and locating reference grids and benchmark in the area.

ii) Surveying for establishing spot levels and plotting contours.

iii) Surveying for locating the natural and manmade details as described earlier.

Deployment of Resources

For the scope of work under the contract, have mobilized the following instruments as minimum requirement: -

1.	Total Station	:	3 No.
2.	GNSS	:	2 No.
3.	Electronic Digital Levels	:	2 No.
4.	Computer with AutoCAD 2000 software	:	2 No.
5.	Plotter	:	2 No.
6.	Vehicle	:	2 (Jeep)

Make of Instruments

1. Total Staion – Sokkia SET 510K
2. Digital Level – Sokkia SDL 50 Power Level
3. GNSS GPS – Leica DGPS 1200

Brochures of Instruments has been Provided Separately.

Calibration Certificate of Instruments

CERTIFICATE OF CONFORMITY

CERTIFICATE NO: 360203556-1174

TOSHNI-TEK INTERNATIONAL

267 Kilpauk Garden Road, Chennai 600 010

Sold To : ABEL ENGINEERING CONSULTANTS
BANGALORE

Date : February 3, 2017

Instrument Type

SOKKIA ELECTRONIC TOTAL STATION

Model	Serial Number	Test Report
SET510K	**208578**	**TKV-1174**

condition of instrument as despatched

CALIBRATED TO SPECIFICATION

Test Results

Meets all specifications detailed in the catalogue

Certified that the instrument detailed hereon have been inspected, tested and calibrated in accordance with the documented procedures using standard test equipment from self-calibration technicques and collimator system from SOKKIA, JAPAN.

Signed:.....
QUALITY ASSURANCE AUTHORISED SIGNATORY
For and behalf of **Toshni-Tek International**

Calibration Date: **February 3, 2017** Calibration Due Date: **February 3, 2018**

CERTIFICATE OF CONFORMITY

CERTIFICATE NO: 360207556-1182

TOSHNI-TEK INTERNATIONAL

267 Kilpauk Garden Road, Chennai 600 010

Sold To : ABEL ENGINEERING CONSULTANTS
BANGALORE

Date : February 7, 2017

Instrument Type

SOKKIA ELECTRONIC TOTAL STATION

Model	Serial Number	Test Report
SET510K	**209956**	**TKV-1182**

condition of instrument as despatched

CALIBRATED TO SPECIFICATION

Test Results

Meets all specifications detailed in the catalogue

Certified that the instrument detailed hereon have been inspected, tested and calibrated in accordance with the documented procedures using standard test equipment from self-calibration techniques and collimator system from SOKKIA, JAPAN.

TOSHNI-TEK INTERNATIONAL CHENNAI 600 010

Signed:.....
QUALITY ASSURANCE AUTHORISED SIGNATORY
For and behalf of **Toshni-Tek International**

Calibration Date: **February 7, 2017** Calibration Due Date: **February 7, 2018**

TIN : 29470204860

Phone : Off : 23527932 / 22740497
Telefax : 23527932
Mobile : 09343747335
E-mail: aptengg.blr@gmail.com

aptec

Apt Engineering Corporation

Dealers in : 'SOKKIA, TOPCON, aptec & KOLIDA' Survey Instruments
(Total Station, Digital Theodolite, Auto-Level), Construction Testing Equipments
(Soil, Cement, Concrete, Bitumen) and Garmin's (USA) GPS & Construction Testing Equipments
SPECIALIST IN : CALIBRATION OF COMPRESSION TESTING MACHINE GAUGES &
WE UNDERTAKE REPAIRS OF SURVEYING & MATERIAL TESTING EQUIPMENTS

No.1, 1st Main Road, Maruthi Extension, Devaiah Park, Bangalore - 560 021.
Visit us at : www.aptec-blr.com

TO,
M/S:ABLE ENGINEERING,
BANGALORE.

CALIBRATION CERTIFICATE

APT ENGINEERING CORPORATION'S CALIBRATION LABORATORY CERTIFIES THAT THE INSTRUMENT HAS BEEN INSPECTED,TESTED AND CALIBRATED IN ACCORDANCE WITH THE DOCUMENTED PROCEDURES USING MEASURING AND TEST EQUIPMENT, WHICH ARE TRACEABLE TO NATIONAL STANDARDS AND OF THE INTERNATIONAL ACCEPTED STANDARD.

WE APT ENGINEERING CORPORATION HERE BY CERTIFY THAT THE EQUIPMENT MENTIONED BELOW MEETS THE SPECIFICATION AND THE RESULT OF THE TRACEABILITY IS CARRIED OUT IN ACCORDANCE TO OUR COMPANY'S STANDARD.

INSTRUMENT TYPE	:	TOTAL STATION
MAE & MODEL	:	SOKKIA SET510K
SERIAL NUMBER	:	147738
CALIBRATION DATE	:	12th SEPTEMBER 2016
NEXT CALIBRATION DATE	:	11th MARCH 2017

THIS CERTIFICATE IS VALID FOR SIX MONTHS ONLY.

FOR APT ENGINEERING CORPORATION

CALIBRATED BY

(SERVICE ENGINEER)

SOKKIA TOPCON APTEC KOLIDA SUN GARMIN

... come on let us build our path to future

CERTIFICATE OF CONFORMITY

CERTIFICATE NO: 360207556-1182

TOSHNI-TEK INTERNATIONAL

267 Kilpauk Garden Road, Chennai 600 010

Sold To : ABEL ENGINEERING CONSULTANTS
BANGALORE

Date : February 7, 2017

Instrument Type

SOKKIA ELECTRONIC TOTAL STATION

Model	Serial Number	Test Report
SET510K	**209956**	**TKV-1182**

condition of instrument as despatched

CALIBRATED TO SPECIFICATION

Test Results

Meets all specifications detailed in the catalogue

Certified that the instrument detailed hereon have been inspected, tested and calibrated in accordance with the documented procedures using standard test equipment from self-calibration techniques and collimator system from SOKKIA, JAPAN.

TOSHNI-TEK INTERNATIONAL CHENNAI 600 010

Signed:.....
QUALITY ASSURANCE AUTHORISED SIGNATORY
For and behalf of **Toshni-Tek International**

Calibration Date: **February 7, 2017** Calibration Due Date: **February 7, 2018**

CERTIFICATE OF CONFORMITY

CERTIFICATE NO: 351208462-1565

TOSHNI-TEK INTERNATIONAL

267 Kilpauk Garden Road, Chennai 600 010

Sold To : ABEL ENGINEERING CONSULTANTS
BANGALORE

Date : December 8, 2015

Instrument Type

SOKKIA ELECTRONIC TOTAL STATION

Model	Serial Number	Test Report
SET510K	**208578**	**TKV-1565**

condition of instrument as despatched

CALIBRATED TO SPECIFICATION

Test Results

Meets all specifications detailed in the catalogue

Certified that the instrument detailed hereon have been inspected, tested and calibrated in accordance with the documented procedures using standard test equipment from self-calibration technicques and collimator system from SOKKIA, JAPAN.

TOSHNI-TEK INTERNATIONAL Chennai 600 010

Signed:.....

QUALITY ASSURANCE AUTHORISED SIGNATORY

For and behalf of **Toshni-Tek International**

Calibration Date: **December 8, 2015** Calibration Due Date: **December 7, 2016**

CERTIFICATE OF CONFORMITY

CERTIFICATE NO: 351208462-1564

TOSHNI-TEK INTERNATIONAL

267 Kilpauk Garden Road, Chennai 600 010

Sold To : ABEL ENGINEERING CONSULTANTS
BANGALORE

Date : December 8, 2015

Instrument Type

SOKKIA ELECTRONIC TOTAL STATION

Model	Serial Number	Test Report
SET510K	**147738**	**TKV-1564**

condition of instrument as despatched

CALIBRATED TO SPECIFICATION

Test Results

Meets all specifications detailed in the catalogue

Certified that the instrument detailed hereon have been inspected, tested and calibrated in accordance with the documented procedures using standard test equipment from self-calibration technicques and collimator system from SOKKIA, JAPAN.

TOSHNI-TEK INTERNATIONAL Chennai 600 010

Signed:.....
QUALITY ASSURANCE AUTHORISED SIGNATORY
For and behalf of **Toshni-Tek International**

Calibration Date: **December 8, 2015** Calibration Due Date: **December 7, 2016**

CERTIFICATE OF CONFORMITY

CERTIFICATE NO: 351209462-1571

TOSHNI-TEK INTERNATIONAL

267 Kilpauk Garden Road, Chennai 600 010

Sold To : ABEL ENGINEERING CONSULTANTS
BANGALORE

Date : December 9, 2015

Instrument Type

SOKKIA ELECTRONIC TOTAL STATION

Model	Serial Number	Test Report
SET510K	**209956**	**TKV-1571**

condition of instrument as despatched

CALIBRATED TO SPECIFICATION

Test Results

Meets all specifications detailed in the catalogue

Certified that the instrument detailed hereon have been inspected, tested and calibrated in accordance with the documented procedures using standard test equipment from self-calibration technicques and collimator system from SOKKIA, JAPAN.

Signed:.....
QUALITY ASSURANCE AUTHORISED SIGNATORY
For and behalf of **Toshni-Tek International**

Calibration Date: **December 9, 2015** Calibration Due Date: **December 8, 2016**

Drawings

The drawings were prepared directly on computers using standard software like Autodesk AutoCAD 2017 and Autodesk Civil 3D.

Area and Length of Survey.

Length of Natural Nella's and Road Net Work ----------88.85Km.
Block level Survey----------123.32 ha.

Preparation & Submission of Topographical Survey Maps

Survey maps of the site in ***suitable scale*** indicating grid lines and contour lines, demarcating all permanent features like roads, railways, waterways, natural etc. Plan of each area will contours with grid lines and permanent features.

All the maps are prepared in digital format using AutoCAD Software. Lots of layers have been used in the Soft Copies for clear understanding.

List of Drawings Submitted.

1. Base Map
2. LS of Alignment of Pipeline
3. Block Level and Contours at STP/ISPS Locations

The Following are the Layers used.

Layer Name	Color	Linetype
0	white	Continuous
0	white	Continuous
Asphalt	cyan	DASHED
ATM	white	Continuous
BANK	white	Continuous
benchmark	white	Continuous
BLOCK LEVEL 5x5m	white	Continuous
Bore Well	green	Continuous
BOUNDARY	white	Continuous
Building	red	Continuous
Building Hatch	white	Continuous
BUILDING LINE	red	Continuous

Layer Name	Color	Linetype
FLYOVER	magenta	Continuous
FOOT OVER BRIDGE	green	Continuous
Footpath	141	Continuous
GATE	white	Continuous
Generator	white	Continuous
Govt_txt	white	Continuous
GPS	white	Continuous
GRID	white	Continuous
GRID TEXT	white	Continuous
Hatch_Build	11	Continuous
Hospital	white	Continuous
Hostel	white	Continuous

Layer Name	Color	Linetype
Bunk	white	Continuous
Bus Stop	white	Continuous
canal	white	Continuous
canal hatch	white	Continuous
canal text	white	Continuous
Chamber	cyan	Continuous
Church	white	Continuous
Circle_	green	Continuous
CMRL Office	white	Continuous
CONCRETE ROAD CENTER	250	Continuous
Construction Building	white	Continuous
CONT-MJR	30	Continuous
CONT-MJR-TEXT	white	Continuous
CONT-MNR	9	HIDDEN
CONT-MNR-TEXT	white	Continuous
controlpoint	white	Continuous
Counter	white	Continuous
COURT	white	Continuous
CULVERT	magenta	Continuous
Cutting	8	Continuous
CUTTING SYMBLE	white	Continuous
Cwall	blue	Continuous
Defpoints	white	Continuous
Destination	white	Continuous
DIMENSION	white	Continuous
DRAIN HATCH	white	Continuous
Drain Width	white	Continuous
Drain_txt	white	Continuous
EL_PO	yellow	Continuous
Electrical	white	Continuous
ESCALATOR	magenta	Continuous
Fence	yellow	Continuous
flow	white	Continuous
flyovear	red	Continuous
Railway Line	blue	Continuous
Railway Station	white	Continuous

Layer Name	Color	Linetype
Hotel	white	Continuous
Katti	blue	Continuous
key	white	Continuous
KEY TEXT	white	Continuous
Lamp Pole	yellow	Continuous
Layout of Sheet	white	Continuous
legend	white	Continuous
lLEVELS	white	Continuous
Madeon	green	Continuous
Man Hole	magenta	Continuous
Market	white	Continuous
Mdical	white	Continuous
minitank	blue	Continuous
Mosque	white	Continuous
Nala-txt	blue	Continuous
new - culvert	10	Continuous
new - Lamp Pole	yellow	Continuous
new- Circle_txt	green	Continuous
new- Lamp Pole	yellow	Continuous
new- Nala-txt	blue	Continuous
new newCodes	yellow	Continuous
new- RCC Edge	177	Continuous
new- Road-S	30	Continuous
Office	white	Continuous
OPEN	yellow	Continuous
PALM TREE	green	Continuous
Park and Garden	white	Continuous
Parking Area	white	Continuous
PIPE	green	Continuous
PLATFARM	green	Continuous
Police Station	white	Continuous
Power Box	white	Continuous
Property-Bry	210	Continuous
Pylon	yellow	Continuous
RAILWAY CENTER	white	Continuous
STEPS	green	Continuous

Layer Name	Color	Linetype
RCC Edge	177	Continuous
RCC-Drain	249	Continuous
Retained Wall	8	Continuous
River	white	Continuous
road	red	Continuous
road center	white	Continuous
Road Hatch	white	Continuous
Road Name	white	Continuous
Road-S	30	Continuous
Shed	red	DASHED
Shelter	white	Continuous
Shop	white	Continuous
Slab	white	Continuous
Stadium	white	Continuous

Layer Name	Color	Linetype
SUBWAY	green	Continuous
Telephone Box	white	Continuous
Temple	red	Continuous
Text	white	Continuous
TITLE	white	Continuous
Toilet	red	Continuous
TRANSFORMER	yellow	Continuous
TREE	green	Continuous
trench	white	Continuous
water edge	cyan	Continuous
Water Tank	blue	Continuous
Well	blue	Continuous
Width in Meter	white	Continuous
Work Area	white	Continuous

Digital Survey Photos

During conducting Survey lot of digital photos were captured which have been provided in Separate Folders

Leica GPS1200+ Series
High performance
GNSS System



- when it has to be right

Leica Geosystems

Leica GPS1200+
The only future proof GNSS

When it says future proof GNSS, it means maximum productivity and reliability. More satellites, more GNSS signals. Today and tomorrow! With Leica GPS1200+ you can be certain that you're ready for the future. Invest today in future proof GNSS technology and be sure that your equipment can track all satellites today and tomorrow. GPS1200+ is the only future proof GNSS System.

Best GNSS and RTK technology
Fast satellite acquisition, high accuracy measurements, tracking to low elevations, the world's first phase multipath mitigation technology, jamming resistant, high up-date rate, low latency, and fast, reliable, long-range RTK.

GNSS/TPS: standardized interface
Keyboard and touch screen, intuitive interface, powerful data management, on-board routines and programs: all easy to use and identical for GNSS and TPS.

SmartRover – extremely light weight
SmartRover weighs just 2.7 kg for a complete cable free all on the pole RTK GNSS rover. Work the complete day in comfort and enjoy full compatibility with SmartStation and SmartPole.

Fully waterproof, incredibly robust
GPS1200+ receivers are designed to work anywhere under the roughest conditions imaginable. They float, withstand falls, jolts and vibrations, operate in rain, dust, sand and snow, at temperatures from –40°C to + 65°C.

Totally versatile
GPS1200+ can be used as a reference or rover in any mode from static to RTK. Small, light, and supporting all formats and communication devices, it can be used on a pole, in a minipack, on a tripod, or even on a construction machine, survey boat or aircraft.

For all applications
You can use GPS1200+ for everything: control, topo, engineering, cadastre, stake out, monitoring, seismic – whatever you want.



FUNCTION integrated

Combine GNSS and TPS. Use them in the same way.
Change easily from one to the other.
Work faster, more accurately and more efficiently.
Enjoy all the freedom, flexibility and power of System 1200.



Leica SmartStation

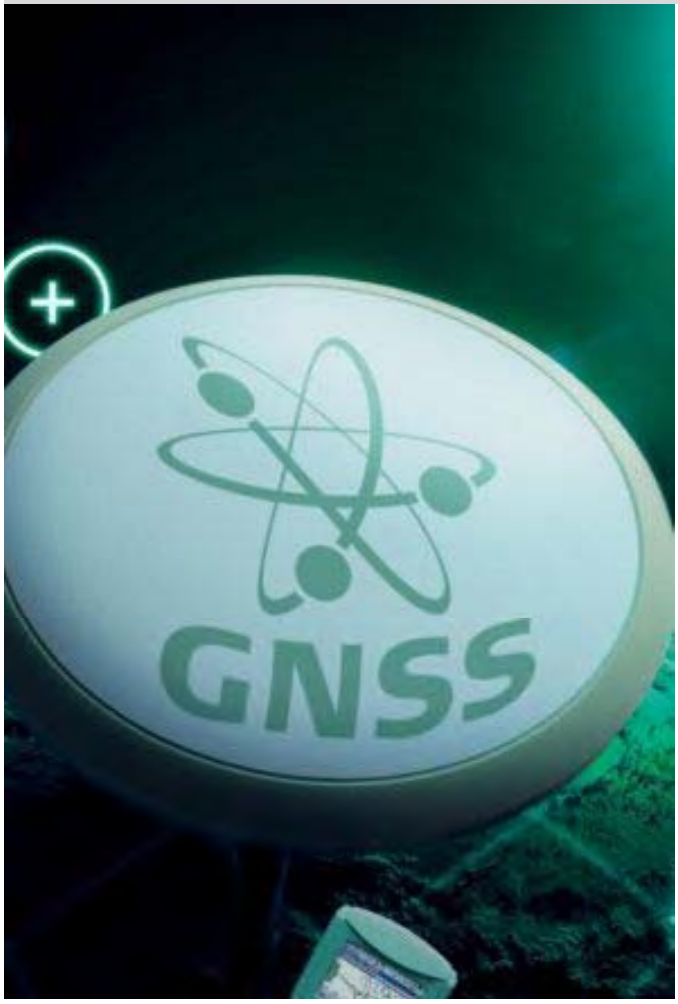
TPS1200+ with integrated GNSS. All TPS1200+ can be ugraded to Smart Station.



Leica GPS1200+

Unites top GNSS technology with powerful data management. Perfect for all GNSS applications.





Leica System 1200

GNSS and TPS
Working together
For all applications
Today and in the future

Designed and built to the most stringent standards with the latest measurement technologies, Leica System 1200 instruments are extremely efficient and reliable, and stand up to the severest environments.

A highly intuitive user interface, a multitude of functions and features, powerful data management, and user-programming capabilities are common to both System 1200 GNSS and TPS instruments.

Operators can switch instantly between GNSS and TPS and use whichever is the most convenient and suitable; extra training is not required.

The new high-tech GNSS and TPS instrument of the System 1200 series with identical operation enable you to do every type of job, faster, more accurately and more efficiently than ever before.

And most important, you reduce your costs and increase your profits.

Leica TPS1200+

Top performance, high accuracy total stations do everything you want and much more.



Leica SmartPole

Save time with SmartPoles' setup On-the-fly and easily swap between GNSS and TPS when needed.



Leica SmartWorx

SmartWorx TPS/GNSS application software is both easy-to-use and extremely powerful.

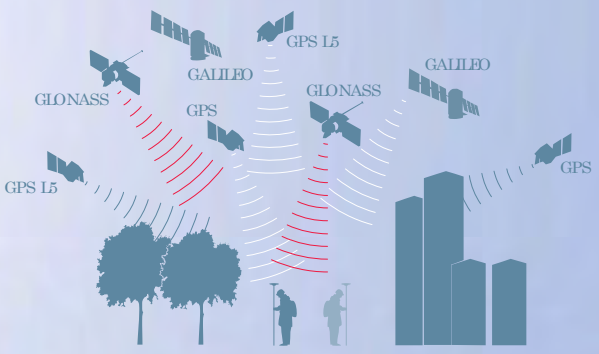


Leica Geo Office

Everything you need in a single package for TPS and GNSS: import, visualization, conversions, quality control, processing, adjustment, reporting, export etc.



Leica GPS1200+
Fast, accurate, rugged and reliable



SmartTrack+

GNSS technology
GPS1200+ means newest GNSS technology. The third generation of SmartTrack+ measurement engine tracks all existing satellite signals and those planned for the future. This includes GPS L5, Galileo, GAGAN, WAAS, EGNOS, MSAS and Compass signals. More satellites means higher productivity, accuracy and reliability. SmartTrack+ acquires satellites within seconds and is ideal in urban canyons and obstructed areas where other receivers often fail. Uniquely, older GPS1200 receivers can be upgraded with the new SmartTrack+ measurement engine.

SmartCheck+
Continuously checking provides the highest possible reliability. A unique, built-in integrity monitoring system checks all results immediately. SmartCheck+ processes all available GNSS measurements simultaneously for centimeter-accuracy, 20 Hz RTK at 40 km and more. Initialize within seconds and survey in obstructed areas with a GX1230+ (GPS only) sensor or increase productivity with a GX1230+ GNSS/AX1230+ GNSS (support all available GNSS systems).

SmartRTK
With Leica Geosystems SmartRTK and RTCM 3.1 correction data, performance and peace-of-mind is guaranteed. Never again you will need to worry about loosing consistency and traceability because of a moving virtually computed reference station. SmartRTK uses fixed reference station monuments that surveyors can trust. SmartRTK does not only give benefits with RTCM 3.1, the new atmospheric decorrelator technology provides precise positioning in all Networks regardless of the correction data.

SmartRTK





Exceptionally rugged
Don't worry about how your crews handle GPS1200+. It's built to MIL specs to withstand the roughest use. With its strong, precision-machined magnesium housing, GPS1200+ stands up to drops and falls and the jolts and vibrations of machines.



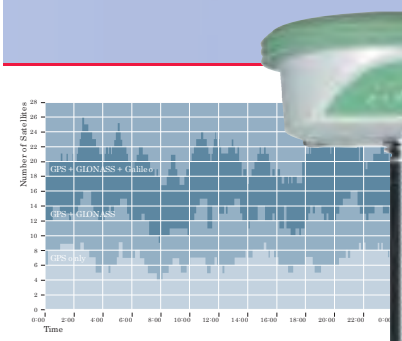
Immune to bad weather
Designed for temperatures from –40° C to +65° C (storage +80° C), GPS1200+ shrugs off arctic cold and blistering heat.
Fully waterproof – withstands immersion to 1 m – sand and dustproof, it operates perfectly in any conditions from tropical rainfall to desert sandstorms. GPS1200+ just keeps on working.

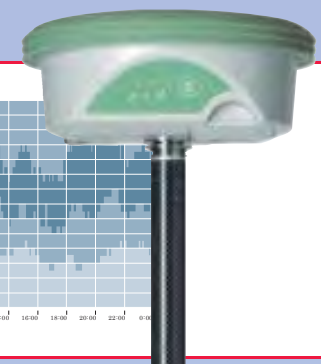
High contrast touch screen
The high quality 1/4 VGA (11 lines by 32 characters) with optional colour option (RX1250) touch screen guarantees perfect clarity and contrast. Whether in fading light or bright sunshine, you can always read the display perfectly. Operate using the touch screen or the QWERTY keyboard, which-ever you prefer.

With or without controller
Connect the controller to the receiver when you need to input information and make full use of the on-board functions and programs.

RTK/DGPS communication
Radio modems, GSM, GPRS and CDMA modules fit in waterproof housings attached to the receiver. Attach either one or two devices for RTK/DGPS reference and rover applications.

With Bluetooth® Wireless-Technology built in to the RX1250 controller complete cable free operation and connectivity to compatible wireless products is available.











GNSS Modernization
When is the right time to invest in a new hybrid GNSS receiver? The answer is when the investment brings significant productivity gains. GLONASS has already proven such gains. GPS L5 and Galileo will bring even more advantages, such as allowing instantaneous ambiguity resolution and longer baseline ranges. An investment in GPS1200+ effectively increases the value of your equipment - a receiver guaranteed to track all satellite signals of today and tomorrow will remain competitive well into the future.

GPS1200+ receivers:
GX1230+ GNSS/ ATX1230+ GNSS

- Triple frequency
- GPS/ GLONASS/ Galileo/ Compass[1]
- 120 Channels
- L1/L2/L5 GPS
- L1/L2 GLONASS
- E1/ E5a/ E5b /Alt-BOC Galileo
- 4 SBAS
- Full Real Time RTK
- Use as rover or reference

GX1230+/ GX1220+

- Dual frequency, GPS only geodetic receiver
- Easily upgradeable to GNSS receiver
- 16 L1 + 16 L2 GPS
- 4 SBAS
- Real-Time RTK (or DGPS option)
- GPS L5 and Galileo ready

SmartStation with SmartAntenna
SmartStation is a TPS1200+ with a ATX1230+ GNSS SmartAntenna. All GNSS and TPS operations are controlled from the TPS keyboard, all data are in the same database, all information is shown on the TPS screen. Touch the GPS key, let RTK determine the position to centimeter accuracy, then survey and stake out with the total station. You can do anything with SmartStation. You can also use SmartAntenna independently on a pole with a RX1250 controller.

- Light, modular equipment
  Use it the way that suits you best.
- All on the pole
  Light weight with excellent balance. Ideal for stakeout on construction sites and other demanding conditions.
- Pole and minipack
  Minimum weight in your hand when surveying for hours on end.
- On a tripod or pillar
  For geodetic control and reference stations.
- All in the minipack
  For 30 cm DGPS, GIS and seismic surveys.



Seamless dataflow



WORKING TOGETHER

FUNCTION integrated

LEICA SYSTEM 1200

Keyboard illumination
Switch on the display and keyboard illumination when working at night. All the keys light up.

Use GPS1200+ for everything

- For RTK, DGPS, and static data logging
- As a rover or reference
- On a pole, tripod, pillar, or in a minipack
- On construction machines, survey boats, or planes
- For every type of application

Choice of RTK pole
Carbon fiber or aluminum pole with adjustable, ergonomic handgrip.

Leica Geo Office
Software support package for GNSS and TPS with tools and components for import, visualization, conversions, quality control, processing, adjustment, reporting, export etc.

CompactFlash cards
Same CompactFlash cards for GNSS and TPS.

Plug-in Li-Ion batteries
For reliable, long-lasting power, GPS1200+ uses the best, high-capacity batteries available. Work for up to 17 hours with just two plug-in, Lithium-ion batteries.



TPS1200+ Total Stations
GNSS and TPS use the same CompactFlash cards, formats and data management. Transfer cards from one to the other and continue working in the same way.



[1] The Compass signal is not finalized, although, test signals have been tracked with GPS1200+ receivers in a test environment. As changes in the signal structure may still occur, Leica Geosystems cannot guarantee full Compass compatibility.

Leica GPS1200+

Extremely powerful
Yet very easy to use



GPS1200+ is loaded with a multitude of features and functions to meet the many different needs of users all over the world, yet it is remarkably easy to use.

GPS1200+'s graphical operating concept is self-explanatory and guides you straight to what you need.

You can use the default settings or, if you prefer, you can set GPS1200+ to operate, display and output data in exactly the way you require.

When you use GPS1200+, you'll find that everything is very easy to understand.

Even better, you'll notice that GPS1200+ and TPS1200+ are fully compatible with the same CompactFlash cards, data management, displays and keyboards.

Depending on the jobs you do, you can switch easily from GNSS to TPS and continue working in exactly the same way.

Operate GPS1200+ using the QWERTY keyboard or the large graphic touch screen, whichever you prefer.

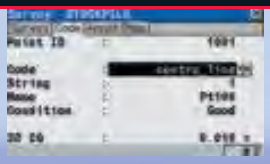
Graphic view mode



Graphic views show your work. Zoom in for details and out for the entire survey. Use the touch screen or keyboard to access data related to points and objects.

With graphical views you can check quickly in the field for completeness and correctness.

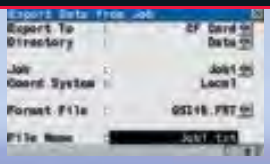
Coding and plan of your work



Define points, lines and areas to build up a plan in the display as you survey. You see immediately what you've done. Attach the codes, attributes and information needed for input into your office or mapping software.

System 1200 has all types of tools and is incredibly versatile.

Data export in any format



Data can be exported directly from GPS1200+ or via Leica Geo Office in various standard formats or in your own userdefined formats for direct input into any type of processing, office, CAD or mapping software.

System 1200 interfaces easily to third-party software packages.



Status icons
Indicate the current measurement and operation modes, recording and battery status, instrument settings etc.

Definable function keys
Allocate commands, functions, displays etc. to these keys for immediate access.

Configurable user menu
Set up your own user menu for the way you and your crews operate. Show what you need and hide the rest.

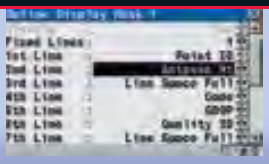
QWERTY keyboard
The standard QWERTY layout of the controller keyboard facilitates fast, easy input of alphanumeric data and information.

Program menu
Direct access to all loaded application programs such as survey, stakeout, COGO etc. and optional application programs.

Large graphic display
1/4 VGA high-resolution LCD with optional colour display (RX1250), easy to read in any light. Display and keyboard light up for work in the dark.

Touch screen
The controller's touch screen provides immediate access without using the keyboard. You can view data and information related to points and objects and call up all types of functions directly via the screen. Use the touch screen and/or the keyboard whichever you prefer.

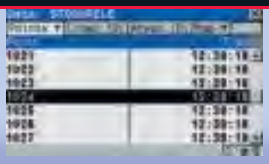
User definable displays



With GPS1200+ you can define different display masks so that the system shows exactly what you and your crews want to see when surveying in the field. Set the displays according to the jobs you do and the information required.

GPS1200+ adapts perfectly to your needs.

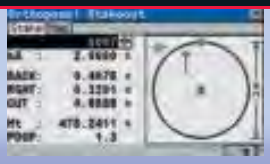
Data management



The powerful database manages data, files, jobs, quality checks etc. You can view, edit, delete, and search with or without filters. Coordinates of points measured more than once are averaged provided that they lie within specified tolerances.

Surveying is much easier and more reliable with System 1200.

Application programs



GPS1200+ is supplied with many useful programs such as Survey, Stakeout, COGO. Other programs such as RoadRunner, Reference Line and DTM Stakeout are optional. You can also write your own programs for special applications in Geo C++.

Most programs run on both GNSS and TPS.

WORKING TOGETHER
FUNCTION integrated
LEICA SYSTEM 1200

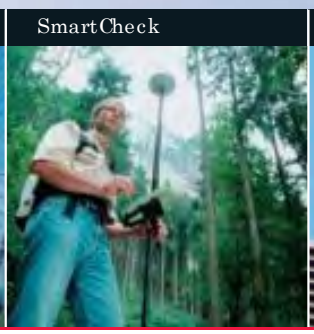
Leica GPS1200+
Superb measurement and RTK performance



SmartTrack+

World leading GNSS technology
Low noise, reliable, high accuracy code and phase measurements are the basis of all satellite surveying work. The better the raw data and the more satellites being tracked, the better the performance and the results. GPS1200+'s completely new SmartTrack+ measurement engine and triple frequency antenna are matched with 120 channels perfectly to each other for the best possible receiver performance:

- Acquisition within seconds
- Excellent signal strength
- Tracking to low elevations
- Suppresses phase and code multipath
- Jamming resistant
- Top quality GNSS measurements
- Perfect tracking in dynamic environments
- Totally reliable



SmartCheck

Fast, self-checking +40km RTK
The SmartCheck+ algorithms process all available signals and deliver fast, accurate RTK.
Centimeter accuracy positions are available continuously at rates of up to 20 Hz. Integrity monitoring runs in the background resolving the ambiguities and verifying the coordinates. Reliability is phenomenal – 99.99% for baselines up to 40 km – and the range is outstanding.

Whatever the work, whether the receiver is on a pole or vehicle, you'll find GPS1200+ RTK to be the perfect tool:

- Initializes within seconds
- Measures amongst trees and obstructions
- Position updates every 0.05 second (20 Hz)
- Latency less than 0.03 second
- Consistent cm-accuracy
- Total reliability



Reference stations

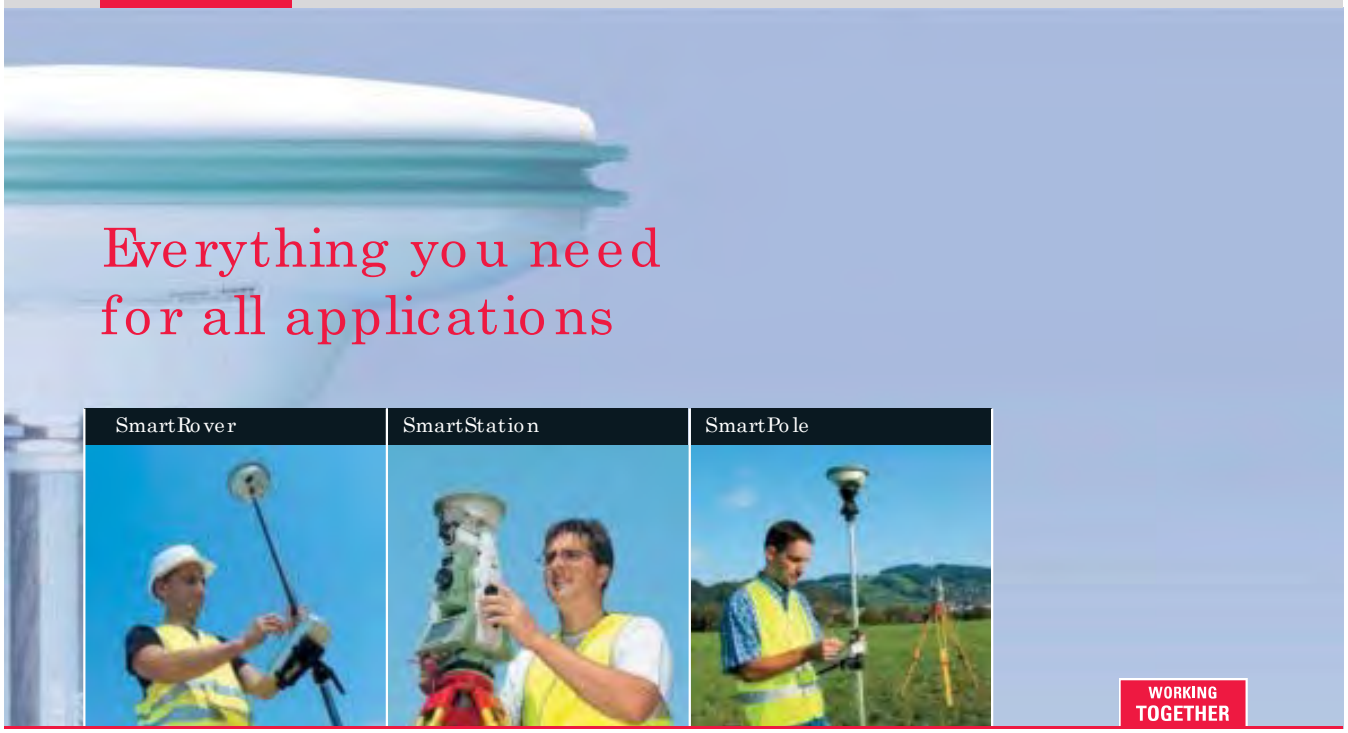
GPS1200+ at CORS sites

Organizations in many countries are establishing GNSS reference stations. GPS1200+ with a SmartTrack+ antenna or IGS/Dorne & Margolin choke-ring antenna is ideal for a Continuously Operating Reference Station (CORS). It logs data, streams data, outputs RTK and DGPS for transmission to RTK and GIS rovers, and is perfect for use with GNSS SPIDER, Leica's reference station software.

As GPS1200+ accepts all formats (Leica, CMR, RTCM) and outputs all standard messages (NMEA), GPS1200+ RTK rovers work perfectly with all reference station services all over the world.

- With single reference stations
- With networks of stations
- With MAX and i-MAX
- With area corrections (FKP) and virtual reference stations (VRS)

Everything you need for all applications



SmartRover

SmartRover – extremely light weight

SmartRover weighs just 2.7 kg for a complete cable free all on the pole RTK GNSS rover. Work the complete day in comfort and enjoy full compatibility with SmartStation and SmartPole.

SmartRover is fully compatible with SmartStation and SmartPole through the interchangeable SmartAntenna. Using Bluetooth® Wireless-Technology, the new light weight RX1250 colour display controller communicates with the SmartAntenna to provide RTK positioning to centimeter accuracy. SmartRover delivers many benefits:

- Weighs just 2.7 kg
- Interchange SmartAntenna between SmartStation, SmartPole and SmartRover
- Cable free all on the pole set-up is ideal for construction applications

SmartStation

GNSS & TPS perfectly combined

TPS1200+ total station with GNSS SmartAntenna combined in one easy-to-use instrument. Ideal for measuring to points that cannot be occupied by an RTK rover. Eliminates need for control points, traverses and resections when using a total station. Set up SmartStation and let RTK fix the position to centimeter accuracy, then survey and stake out with the TPS. Once SmartStation is positioned, use the SmartAntenna on a pole with controller and sensor as an RTK rover.

- Use TPS and GNSS together
- Fix the position with RTK, then survey with TPS
- Survey easier and faster
- Do any type of job
- Increase productivity and profits

SmartPole

Instantly switch between GNSS & TPS

Every survey site is different. Some sites are best suited to TPS and others to GNSS. With SmartPole both TPS and GNSS are available simultaneously. When GNSS is restricted by overhead obstructions use TPS: when no TPS line-of-sight is available use GNSS. No longer is it necessary to identify control points in the office and search for control in the field.

SmartPole is fully compatible with System 1200. The same light-weight GNSS SmartAntenna can be used together with a TPS1200+ as a SmartStation, together with a RX1250 controller as a SmartRover or together with the unique light-weight 360° reflector and RX1250 controller as a SmartPole.

- Higher accuracy & consistency of GNSS control
- Save time in planning and executing the survey
- Maximum flexibility and hence productivity

WORKING TOGETHER

FUNCTION integrated

LEICA SYSTEM 1200

Leica GPS1200+
Technical specifications and system features



GPS1200+ receivers	GX1230+ GNSS/ ATX1230+ GNSS	GX1220+ GNSS	GX1230+	GX1220+	GX1210+
GNSS technology	SmartTrack+	SmartTrack+	SmartTrack	SmartTrack	SmartTrack
Type	Triple frequency	Triple frequency	Dual frequency	Dual frequency	Single frequency
Channels	120 channels L1/L2/L5 GPS L1/L2 GLONASS E1/E5a/ E5b/ Alt-BOC Galileo Compass[1] 4 SBAS	120 channels L1/L2/L5 GPS L1/L2 GLONASS E1/E5a/ E5b/ Alt-BOC Galileo Compass[1] 4 SBAS (with DGPS option)	16 L1 + 16 L2 GPS 4 SBAS	16 L1 + 16 L2 GPS 4 SBAS (with DGPS option)	16 L1 GPS 4 SBAS (with DGPS option)
Upgrade to GX1230+ GNSS	-	Yes	Yes	Yes	Yes
RTK	SmartCheck+	No	SmartCheck	No	No
Status indicators	3 LED indicators for GX1200+: power, tracking, memory				

GPS1200+ receivers	GX1230+ (GNSS)/ GX1220+ (GNSS)	GX1210+	ATX1230+ GNSS
Ports	1 power port, 3 serial ports, 1 controller port, 1 antenna port		1 power/controller port, Bluetooth® Wireless-Technology port
Supply voltage, Consumption	Nominal 12 VDC 4.6 W receiver + controller + antenna		Nominal 12 VDC 1.8 W
Event input and PPS	Optional: 1 PPS output port 2 event input ports	Optional: 1 PPS output port 2 event input ports	
Standard antenna Built-in groundplane	SmartTrack+ AX1203+ GNSS Built-in groundplane	SmartTrack AX1201 Built-in groundplane	SmartTrack+ ATX1230+ GNSS Built-in groundplane

The following apply to all receivers except where stated.

Power supply	Two Li-Ion 4.4 Ah/7.4 V plug into receiver. One Li-Ion 2.2 Ah/7.4 V plugs into ATX1230+ GNSS and RX1250.
Plug-in Li-Ion batteries Same for GNSS and TPS	Power receiver + controller + SmartTrack antenna for about 17 hours (for data logging). Power receiver + controller + SmartTrack antenna + low power radio modem or phone for about 11 hours (for RTK/DGPS). Power SmartAntenna + RX1250 controller for about 6 hours (for RTK/DGPS)
External power	External power input 10.5 V to 28 V.
Weights	Receiver 1.20 kg. Controller 0.48 kg (RX1210) and 0.75 kg (RX1250). SmartTrack antenna 0.44 kg. SmartAntenna 1.12 kg. Plug-in Li-Ion battery 0.11 kg (2.2 Ah) and 0.2 kg (4.4 Ah) Carbon fiber pole with SmartTrack antenna and RX1210 controller: 1.80 kg. All on pole: carbon fiber pole with SmartAntenna, RX1250 controller and plug-in batteries: 2.74 kg.

Temperature ISO9022 MIL-STD-810F	Operation:	Receiver	−40° C to +65° C
		Antennas	−40° C to +70° C
		Controllers	−30° C to +65° C
		Controller RX1250c	−30° C to +50° C
	Storage:	Receiver	−40° C to +80° C
		Antennas	−55° C to +85° C
		Controllers	−40° C to +80° C
		Controller RX1250c	−40° C to +80° C
Humidity ISO9022, MIL-STD-810F	Receiver, antennas and controllers Up to 100% humidity.		
Protection against water, dust and sand IP67, MIL-STD-810F	Receiver, antennas and controllers: Waterpoof to 1 m temporary submersion. Dust tight		
Shock/drop onto hard surface	Receiver: withstands 1 m drop onto hard surface. Antennas: withstand 1.5 m drop onto hard surface.		
Topple over on pole	Receiver, antennas and controllers: withstand fall if pole topples over.		
Vibrations ISO9022 MIL-STD-810F	Receiver, antennas and controllers: withstand vibrations on large construction machines. No loss of lock.		

[1] The Compass signal is not finalized, although, test signals have been tracked with GPS1200+ receivers in a test environment. As changes in the signal structure may still occur, Leica Geosystems cannot guarantee full Compass compatibility.

SmartTrack+ Advanced GNSS measurement technology	Time needed to acquire all satellites after switching on: typically about 50 seconds. Re-acquisition of satellites after loss of lock (e.g. passing through tunnel): typically within 1 second. Very high sensitivity: acquires more than 99 % of all possible observations above 10 degrees elevation. Very low noise. Robust tracking. Tracks weak signals to low elevations and in adverse conditions. Multipath mitigation. Jamming resistant. Measurement precision: Carrier phase on L1: 0.2 mm rms. On L2: 0.2 mm rms. Code (pseudorange) on L1 and L2: 20 mm rms.
SmartCheck+ Advanced, long range RTK technology	Initialization typically 8 seconds. Position update rate selectable up to 20 Hz. Latency < 0.03 secs. Range 40 km or more in favorable conditions. Self checking.
Accuracies	Kinematic Horizontal: 10 mm + 1 ppm Vertical: 20 mm + 1 ppm Static (ISO 17123-8) Horizontal: 5 mm + 0.5 ppm Vertical: 10 mm + 0.5 ppm Reliability: 99.99 % for baselines up to 40 km.
Reference station networks	Formats supported for transmission and reception: Leica proprietary (Leica, Leica 4G), CMR, CMR+, RTCM **V2.1/2.2/2.**3/3.0/3.1. RTK rover fully compatible with Leica's Spider i-MAX & MAX formats, VRS and Area Correction (FKP) reference station networks.
DGPS GX1230+ (GNSS), ATX1230+ GNSS, GX1220+ (GNSS) – standard GX1210+ – optional	DGPS, includes support of MSAS, WAAS, EGNOS and GAGAN. RTCM **V2.1/2.2/2.**3/3.0/3.1. formats supported for transmission and reception. Baseline rms: typically 25 cm rms with suitable reference station.
Position update rate and latency	Applies to RTK, DGPS and navigation positions. Update rate selectable from 0.05 sec (20 Hz) to 1 sec. Latency less than 0.03 secs.
NMEA output	NMEA 0183 V3.00 and Leica proprietary.
Post-processing with Leica Geo Office software All GPS1200+ receivers	Horizontal: 10 mm + 1 ppm, kinematic Vertical: 20 mm + 1 ppm, kinematic Horizontal: 5 mm + 0.5 ppm, static Vertical: 10 mm + 0.5 ppm, static For long lines with long observations Horizontal: 3 mm + 0.5 ppm, static Vertical: 6 mm + 0.5 ppm, static
Notes on performance and on accuracies	Figures quoted are for normal to favorable conditions. Performance and accuracies can vary depending on number of satellites, satellite geometry, observation time, ephemeris, ionosphere, multipath etc.

Controllers RX1210/RX1250	High contrast, 1/4 VGA display with colour option (RX1250) Touch screen, 11 lines x 32 characters. Windows CE 5.0 on RX1250. Full alphanumeric QWERTY keypad. Function keys and user definable keys. Illumination for screen and keys. Can also be used with TPS1200+ for alphanumeric input and extensive coding.
Operation with controller Same for GNSS and TPS	Via keypad and/or via touch screen. Graphical operating concept. Function keys and user definable keys. All information displayed.
Displayed information Graphical display of survey Same for GNSS and TPS	All information displayed: status, tracking, data logging, database, RTK, DGPS, navigation, survey, stakeout, quality, timer, power, geographical, cartesian, grid coordinates etc. Graphical display (plan) of survey. Zooming. Can access surveyed points directly via touch screen.
Stakeout display Same for GNSS and TPS	Graphical with zoom. Digital, polar and orthometric. Accuracy: 10 mm + 1 ppm at 20 Hz (0.05 sec) update rate. No degradation with high update rates.
Operation without controller GX1200+ only	Automatic on switching on. LED status indicators. For reference stations and static measurements.
Data logging Same cards used for GNSS and TPS Capacity	On CompactFlash cards: 256 MB and 1 GB Optional internal receiver memory: 256 MB. 64 MB sufficient for (30 % less for GPS/GLONASS): About 500 hours L1 + L2 data logging at 15 sec rate. About 2 000 hours L1 + L2 data logging at 60 sec rate. About 90 000 RTK points with codes.
Data management Same for GNSS and TPS	User definable job management. Point identifiers, coordinates, codes, attributes etc. Search, filter and display routines. Multi point averaging. Five types of coding systems cover all requirements.
Coordinate systems Same for GNSS and TPS	Ellipsoids, projections, geoidal models, coordinate, transformations, transformation parameters, country specific coordinate systems. Fully support of RTCM 3.1 coordinate system transfer.
Application programs Same for GNSS and TPS	Standard: Full range of COGO functions. Hidden point. Optional: RoadRunner, Reference Line, DTM Stakeout, Reference Plane, Area Division and X-Section Survey, DXF Export, LandXML Export and Volume Calculations
Programmable Same for GNSS and TPS	User programmable in GeoC++. Users can write and upload programs for their own special requirements and applications.
Communication Data links	One or two of the following devices can be connected: Radio modem, GSM, GPRS, CDMA. Different frequencies and/or formats can be received and transmitted. Time slicing is supported.

Whether you want to survey a parcel of land or a construction site, a facade or indoors to create as-built plans or carry out high-precision measurements of bridge and tunnel constructions – Leica Geosystems' surveying instruments provide the right solution for all measuring tasks.

The System 1200 Series instruments as well as the software are designed to meet the daily challenges of modern surveying. They all have outstanding, easy to read and user-friendly interfaces. Their straightforward menu structures, their clearly outlined scope of functions and high technology perfectly mate GNSS and TPS applications in the field. Whether you use the advantages of both technologies combined or each separately – due to the exceptional flexibility of Leica Geosystems instruments, reliable and productive surveying is assured.

When it has to be right.

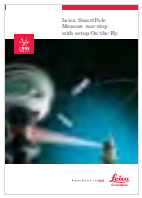
Illustrations, descriptions and technical specifications are not binding and may change.
Printed in Switzerland – Copyright Leica Geosystems AG, Heerbrugg, Switzerland, 2008.
738812en – XII.08 – RDV

TQM
ISO 9001 / ISO 14001

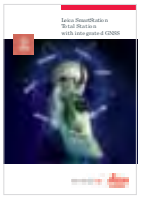
Total Quality Management – our commitment to total customer satisfaction.

Ask your local Leica Geosystems dealer for more information about our TQM program.

The Bluetooth® word mark and logos are owned by Bluetooth SIG, Inc. and any use of such marks by Leica Geosystems AG is under license. Other trademarks and trade names are those of their respective owners.



Leica SmartPole
Product brochure



Leica
SmartStation
Product brochure



Leica TPS1200+
Product brochure



Leica System 1200
Software
Product brochure



Leica GRX1200+
Product brochure

Leica Geosystems AG
Heerbrugg, Switzerland
www.leica-geosystems.com

- when it has to be **right**

Leica Geosystems

SURVEYING INSTRUMENTS

SOKKIA

SDL30/SDL50

DIGITAL LEVELS

Save Time with Innovative, Industry-Leading Technologies

2.5-second High-speed Measurement, 20 lux Minimum Brightness, Inverted Staff Recognition, Wave-and-Read, and the Highest Accuracy in its Class

■ 2.5 seconds – High-speed Measurement

Aim, focus, and press a key. Height and distance are simultaneously measured in 2.5 seconds, 20 percent faster than ordinary digital levels.





■ Wave-and-Read Technology

The innovative "Wave-and-Read" technology provides an additional survey style option. The SDL30/50 tracks the RAB-Code staff waved back and forth to read the correct height. The staff reading becomes the minimum when it is in vertical position. The SDL30/50 automatically detects the least value of staff readings.



■ Choice of Accuracy

SDL30: 0.4mm (New Super-Invar Staff) / 0.6mm (Invar) / 1.0mm (Fiberglass)
SDL50: 0.6mm (New Super-Invar Staff) / 0.8mm (Invar) / 1.5mm (Fiberglass)

Choose the digital level and staffs according to the accuracy* you need. Sokkia offers the top-of-the-line SDL1X model for higher accuracy of up to 0.2mm.

* 1km double-run leveling

■ Consistent Performance in Diverse Environments

The SDL30/50 provides the superior measurement capability under a variety of environmental conditions. Even when the staff surface is partially shaded, or in dim lighting conditions where the brightness at staff surface is as low as 20 lux*, the SDL30/50 consistently provides reliable measurement results without downtime. A small flashlight is enough to illuminate the staff in the dark.

*20 lux is defined as the minimum brightness with which human face can be recognized.



■ Internal Memory

Up to 2,000 measurement data of elevation or height difference can be recorded in the internal memory. Auto mode records data as soon as the measurement is taken, while manual mode allows you to check the measurement results before recording. Stored data can be exported using the "Spectrum Link" software.

■ Automatic Recognition of Inverted Staff

The SDL30/50 automatically recognizes directions of RAB-Code staffs and displays the results with a minus sign (-) when the staff is inverted. Height of ceiling, overpass, bridge, road sign, tree branch, tunnel crown, and other objects can be easily measured without a calculator.



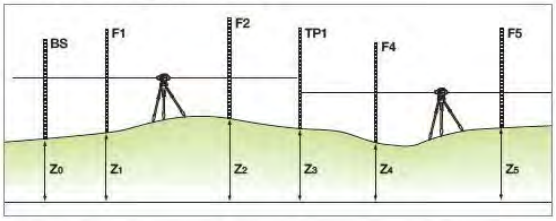


■ Convenient Onboard Programs

The onboard measurement programs of SDL30/50 facilitate leveling and setting-out tasks. Programs include:

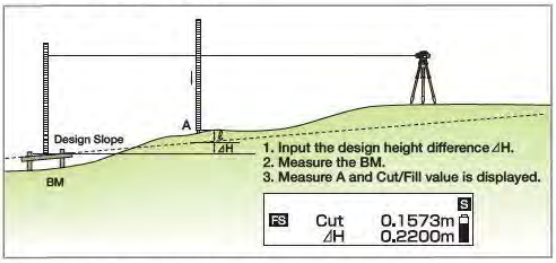
● Elevation ● Height Difference
● Cut and Fill Setting-out ● Setting-out in Distance

■ Elevation / Height Difference



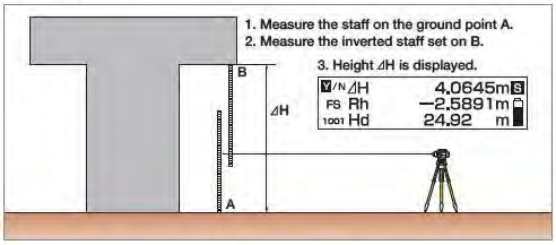
- The SDL30/50 calculates height difference between backsight (BS) and foresight (FS).
- Elevation of foresight can be calculated by inputting BS elevation.

■ Cut and Fill Setting-out



- Slope can be set using Cut and Fill Setting-out program.
- Leveling is possible by inputting the height difference zero (0).

■ Height Measurement



- Two measurements provide the height of point B.
- Elevation can be calculated by inputting BS elevation.

■ SDL30 / SDL50 Specifications

Model			SDL30	SDL50
Height accuracy (ISO 17123-2)*	Electronic	BIS30A staff	0.4mm (0.016in.)	0.6mm (0.024in.)
		BIS20/30 staffs	0.6mm (0.024in.)	0.8mm (0.03in.)
		BGS staffs	1.0mm (0.04in.)	1.5mm (0.06in.)
	Visual	BGS staffs	1.0mm (0.04in.)	2.0mm (0.08in.)
Distance accuracy (D: measuring distance)	Electronic		<±10mm (±0.4in.) [D≦10m (D≦33ft.)] <±0.1% x D [10<D≦50m (33<D≦164ft.) <±0.2% x D [50<D≦100m (164<D≦328ft.)]	
Measuring range	Electronic		1.6 to 100m (5.3 to 328ft.)	
	Visual		from 1.5m (5.0ft.)	
Measuring mode			Single / Repeat / Average / Tracking / Wave-and-Read	
Display resolution	Height		0.0001/0.001/0.01m (0.001/0.01/1ft., 1/8in.)	
	Distance		0.01/0.1m (0.1/1ft., 1in.)	
Measuring time	Single/Repeat		<2.5s	
	Average		<2.5s x [number of measurements]	
	Tracking		<1s	
Minimum brightness condition			20 lux at the surface of staff (with natural light)	
Telescope	Objective aperture		45mm (1.8in.)	36mm (1.4in.)
	Magnification / Resolving power		32x / 3"	28x / 3.5"
	Minimum focus / Field of view		1.5m (5ft.) / 1°20'	
Compensator	Type		Pendulum compensator with magnetic damping system	
	Working range		±15'	
Sensitivity of circular level			10'/2mm	
Horizontal circle			Diameter: 103mm (4in.), Graduation: 1° (1gon)	
Display			Dot matrix LCD (128 x 32 dot) with illuminator	
Keyboard			8 keys (7 keys on front panel, 1 key on side panel)	
Data storage			2,000 points internal memory	
Interface			RS-232C, baud rate 1,200 to 38,400bps	
Onboard programs			Elevation / Height difference / Cut & Fill setting-out / Setting-out distance / Height measurement	
Water resistance			IPX4 (IEC60529:2001)	
Operating temperature			-20 to +50°C (-4 to 122°F)	
Size			W158 x D257 x H182mm (W6.2 x D10.1 x H7.2in.)	
Weight with battery			2.4kg (5.3 lb.)	
Standard battery			BDC46B (Rechargeable Li-ion, 7.2V, 2.45Ah)	
Operating time			Approx. 16 hours at 25°C (77°F)	

* Standard deviation for 1km double-run leveling

■ Standard Configuration

SDL30/SDL50 digital level, BDC46B battery, CDC68 charger, EDC113A/B/C power cable, tool kit, dust cover, cleaning cloth, operator's manual, carrying case

■ Optional Accessories

DE23 Diagonal eyepiece
GS60L Circular level for staff

■ RAB-Code Staff

Material	Model	Length	Linear expansion	Sections	Weight
New Super-Invar	BIS30A	3m (9.9ft.)	±0.1ppm/°C	1	5.5kg (12.2 lb.)
Invar	BIS20	2m (6.6ft.)	1ppm/°C	1	4.3kg (9.5 lb.)
	BIS30	3m (9.9ft.)	1ppm/°C	1	5.5kg (12.2 lb.)

Material	Model	Length	Rear graduation	Sections	Weight
Fiberglass	BGS40	4m (13.2ft.)	Metric	3	2.4kg (5.3 lb.)
	BGS50	5m (16.4ft.)	Metric	4	3kg (6.6 lb.)
	BGS50G3	5m (16.4ft.)	feet/10th/100th	4	3kg (6.6 lb.)
Aluminum	BAS55	5m (16.4ft.)	Metric	5	1.9kg (4.2 lb.)

Product names mentioned in this brochure are trademarks of their respective holders.
Product colors in this brochure may vary slightly from those of actual products owing to limitations of the printing process.
Designs and specifications are subject to change without notice.

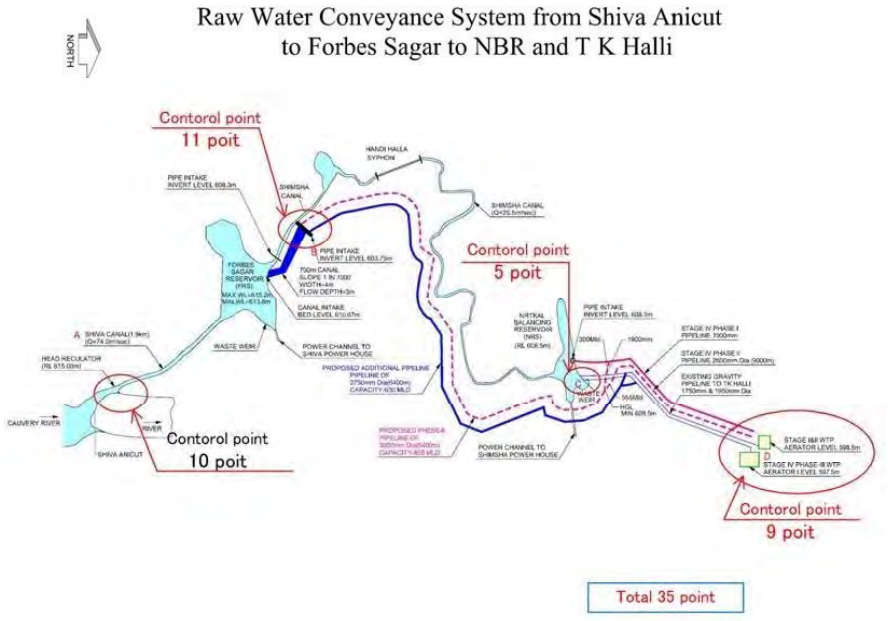
SOKKIA SINGAPORE POSITIONING SALES PTE LTD
60 ALEXANDRA TERRACE, #08-27 THE COMTECH SINGAPORE 118502
PHONE : +65 6479 3966 FAX : +65 6479 4966 WEBSITE : www.sokkia.com.sg
COMPANY REG. NO. : 201007531Z

© 2011 TOPCON CORPORATION. All rights reserved. SOKKIA is a trademark of SOKKIA TOPCON CO., LTD.

2011 06-1 P-74-2 GE Printed in Singapore

１． 水準点測量位置 Watermark Location

1）取水場・浄水場一図 Intake plants / Water purification plants 1 Map



A. Head Reculateor



天端高

Heaven
Tip Point

MSL = 620.961m

ｹﾞｰﾄ堰高

Gate weir
height

MSL = 621.046m

コンクリート高

Concrete height

MSL = 615.167m

水深高　水面高

Water depth MSL =611.770m
high water level MSL = 615.037m



Water depth high water level

High-end concrete MSL = 620.970m

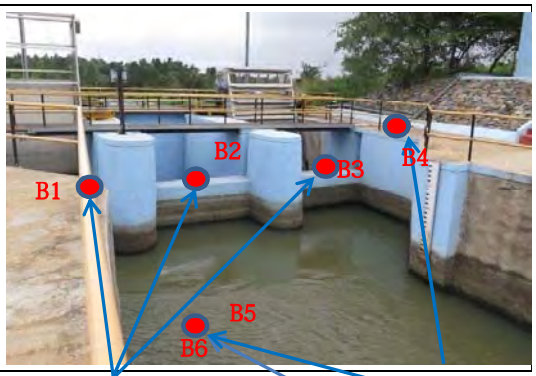
high concrete MSL = 616.010m



Cauvery river weir MSL = 615.180m

B. Pipe Intake



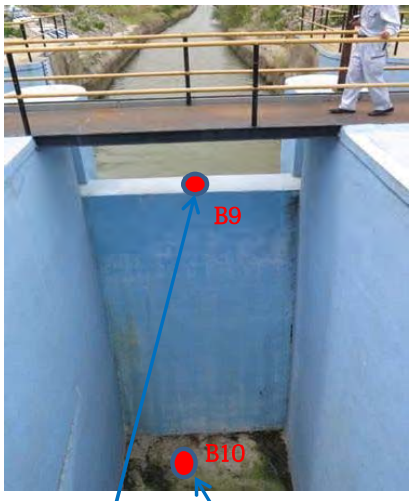


Weir height 3 locations

Bottom of the water surface

Top end





W e i r r i s e

Weir rise

Bottom plate



B1 = B7 =
B2 = B8 = B9 = 615.402m
B3 = 615.398m
B4 = 616.831m
B5 = 615.910m (Water Surface)
B6 = 610.012m (Bottom Surface)
B10 = 609.857m
B11 = B12 = 608.115m

Φ3000 pipe lower end　　Bottom plate

C. Waste Weir

NO.1 Intake

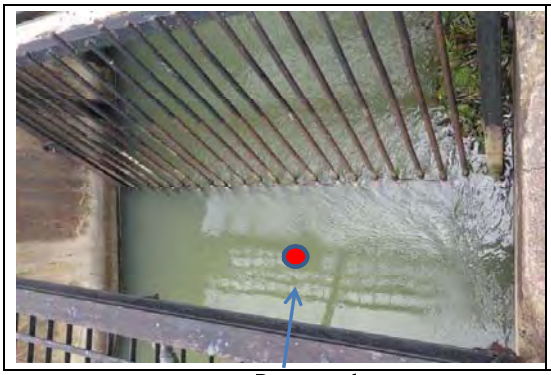




ground　　Bottom plate　　Top end　MSL = 610.831m

MSL = 610.151m　MSL = 603.399m

NO.2 Intake

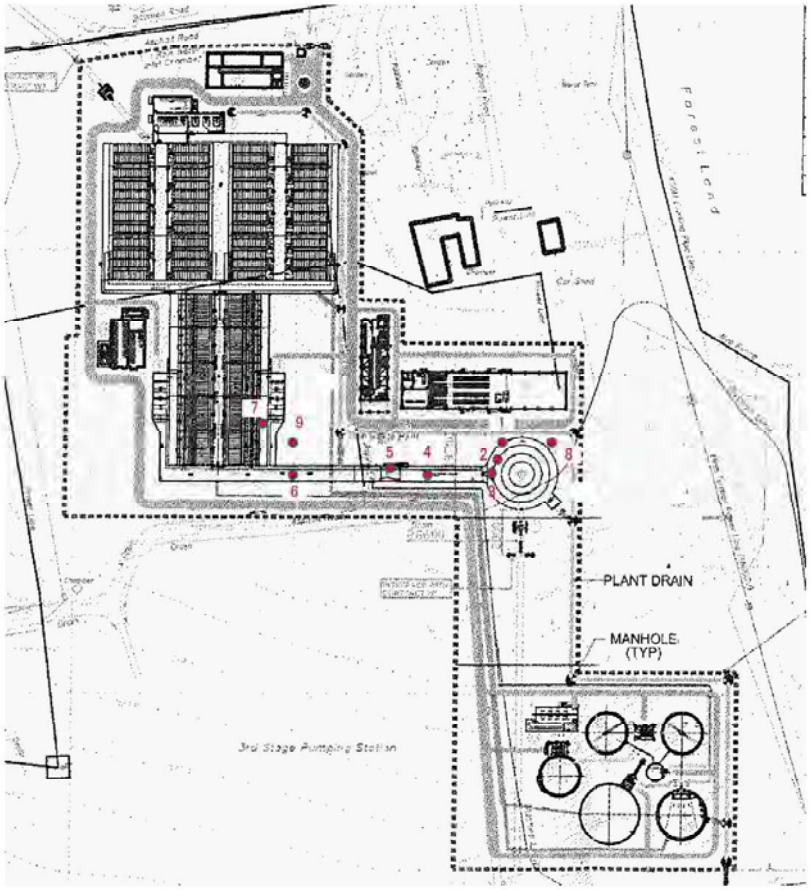




Bottom plate
MSL = 604.060m

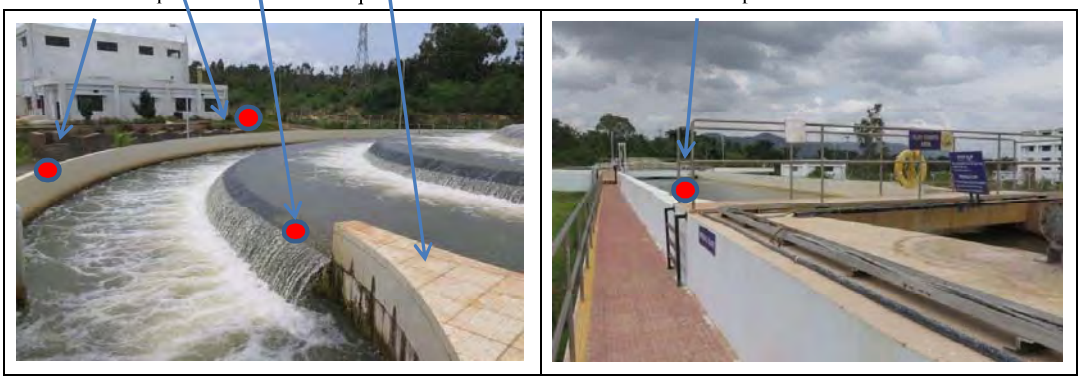
Top end
MSL = 610.041m

D WTS



8 ground

1 Sidewall top 2 Weir 3 Top end 4 Bottom plate



1. MSL = 599.316m 3. MSL = 599.289m

2. MSL = 598.985m 8. MSL = 596.429m

X. Areator Bottom MSL = 597.204m

4. MSL = 597.610m

5. MSL = 599.052m 5 Top end

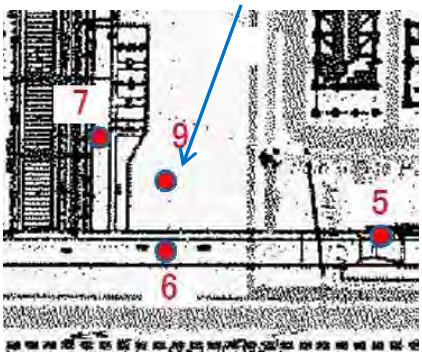


6 Bottom plate 6. MSL = 597.168m



9 Ground height 9. MSL = 596.610m





7 Top end

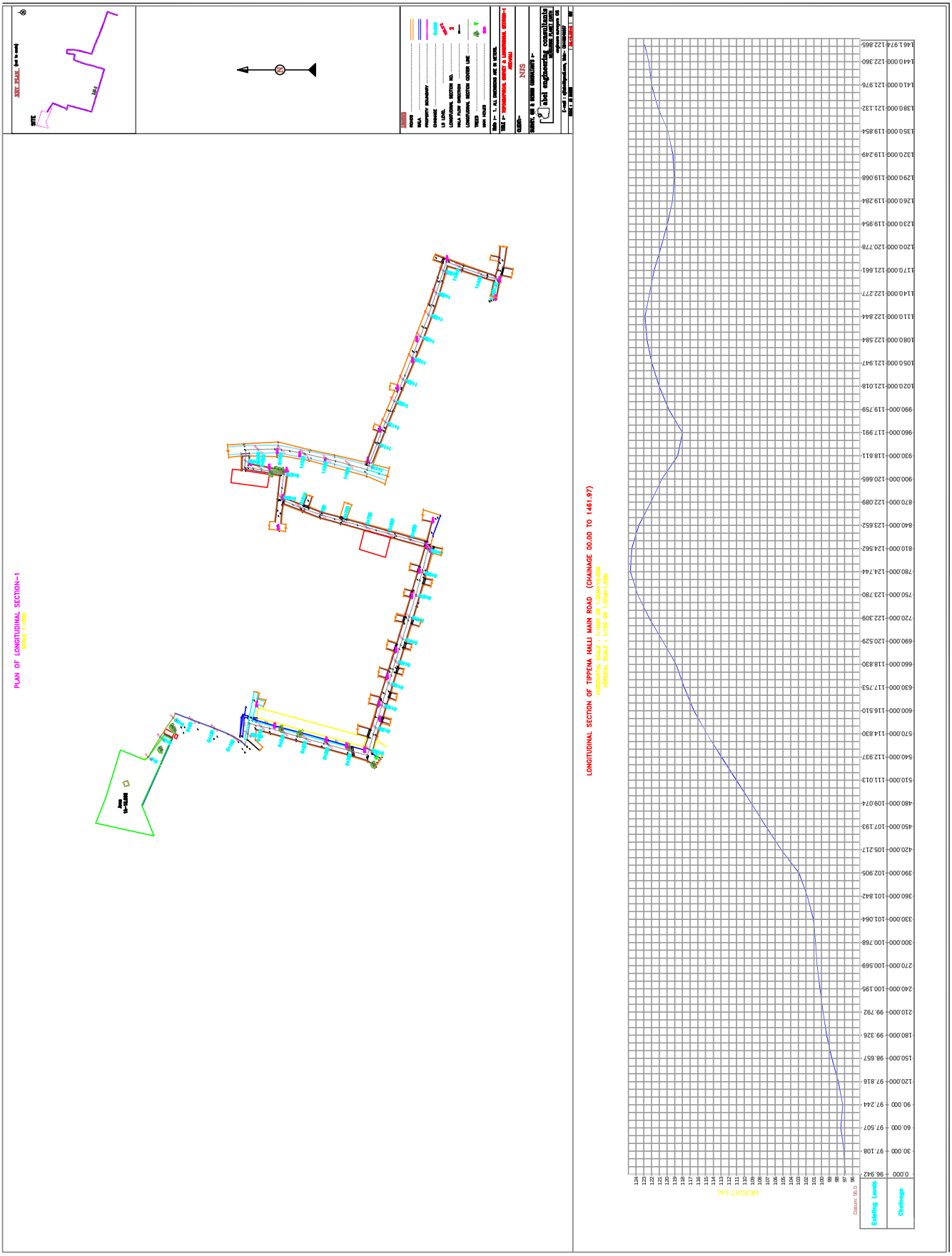
7. MSL = 598.111m

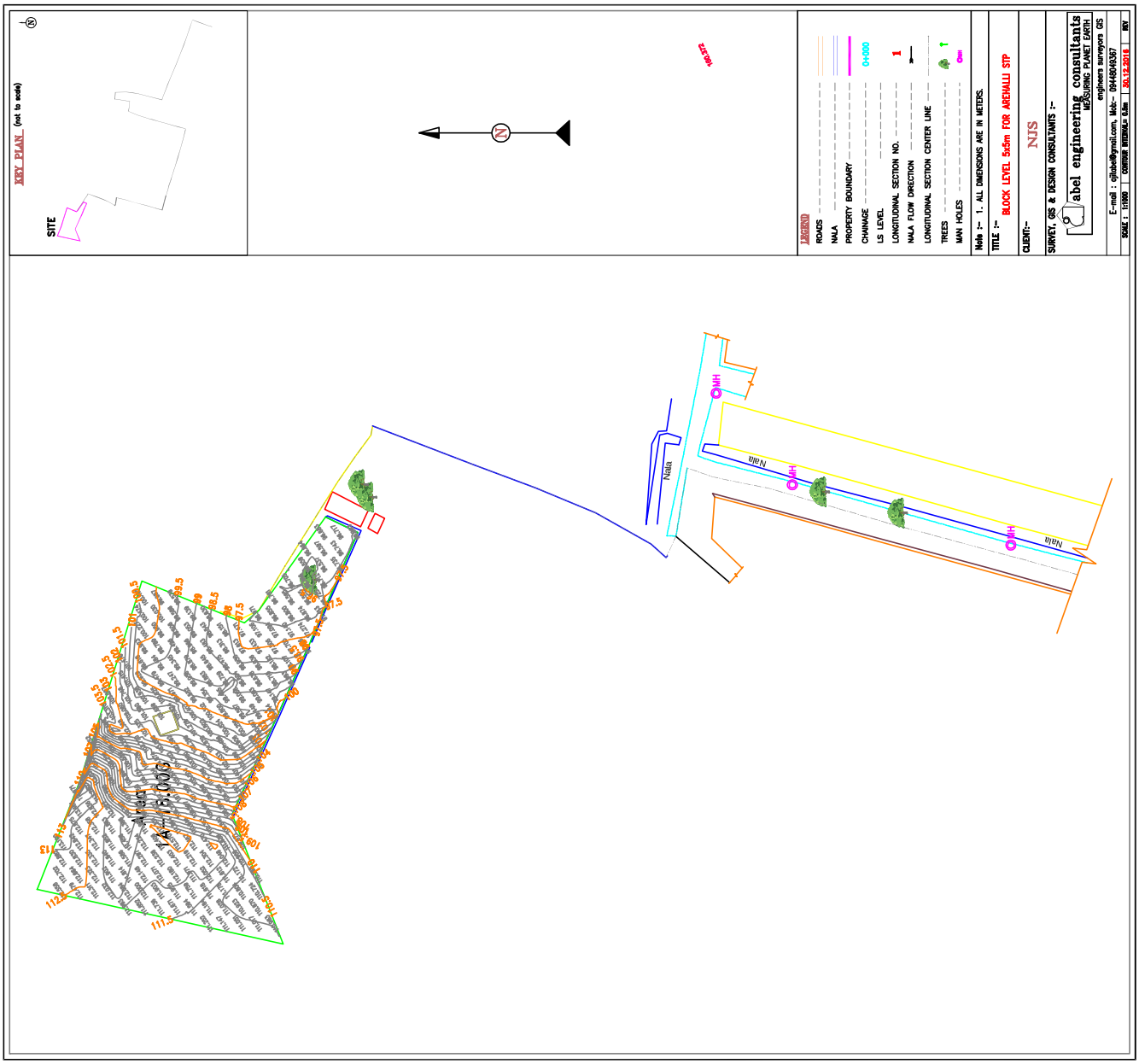
Valve Chamber

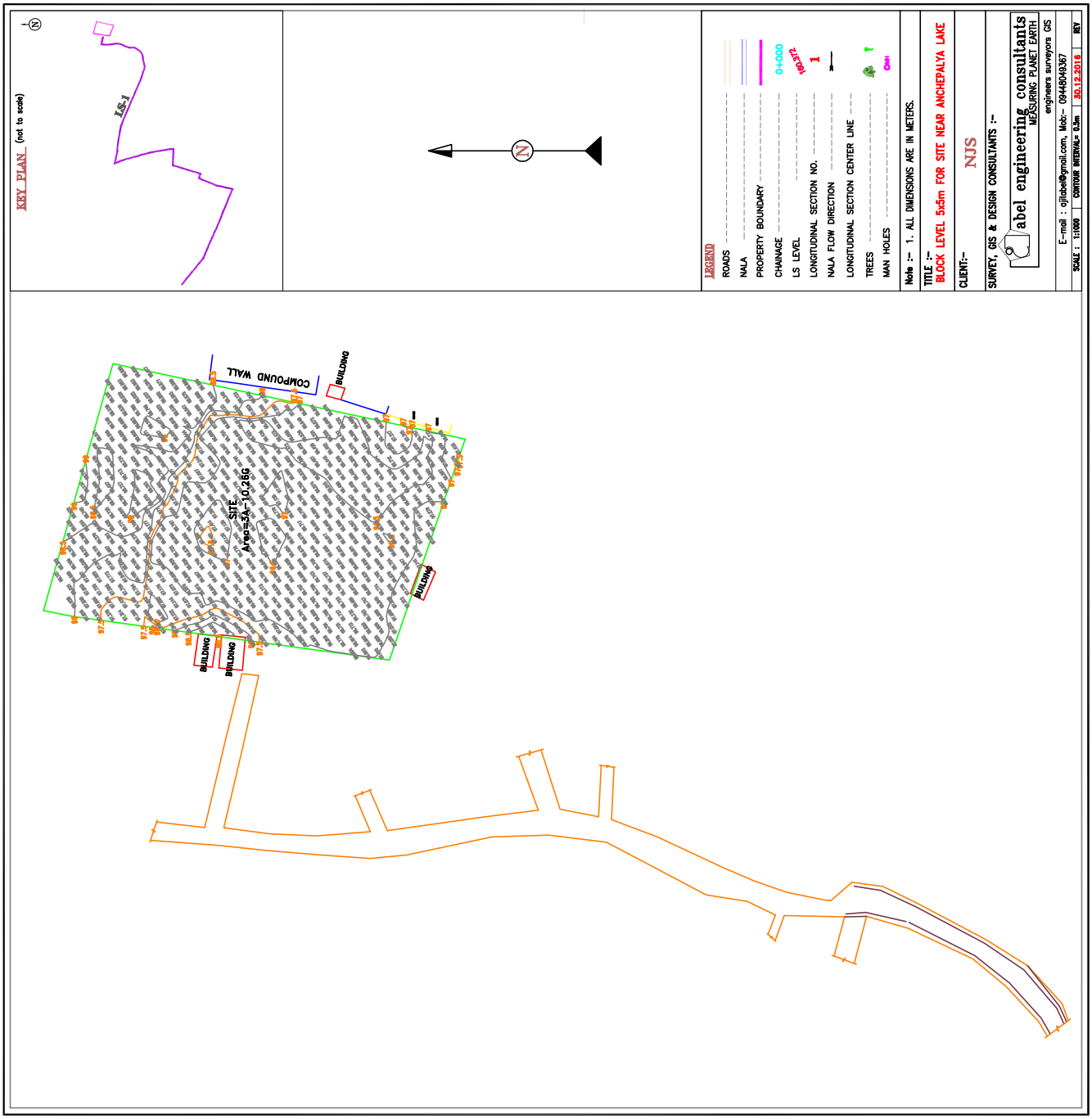
1. Chamber Top MSL = 608.724m

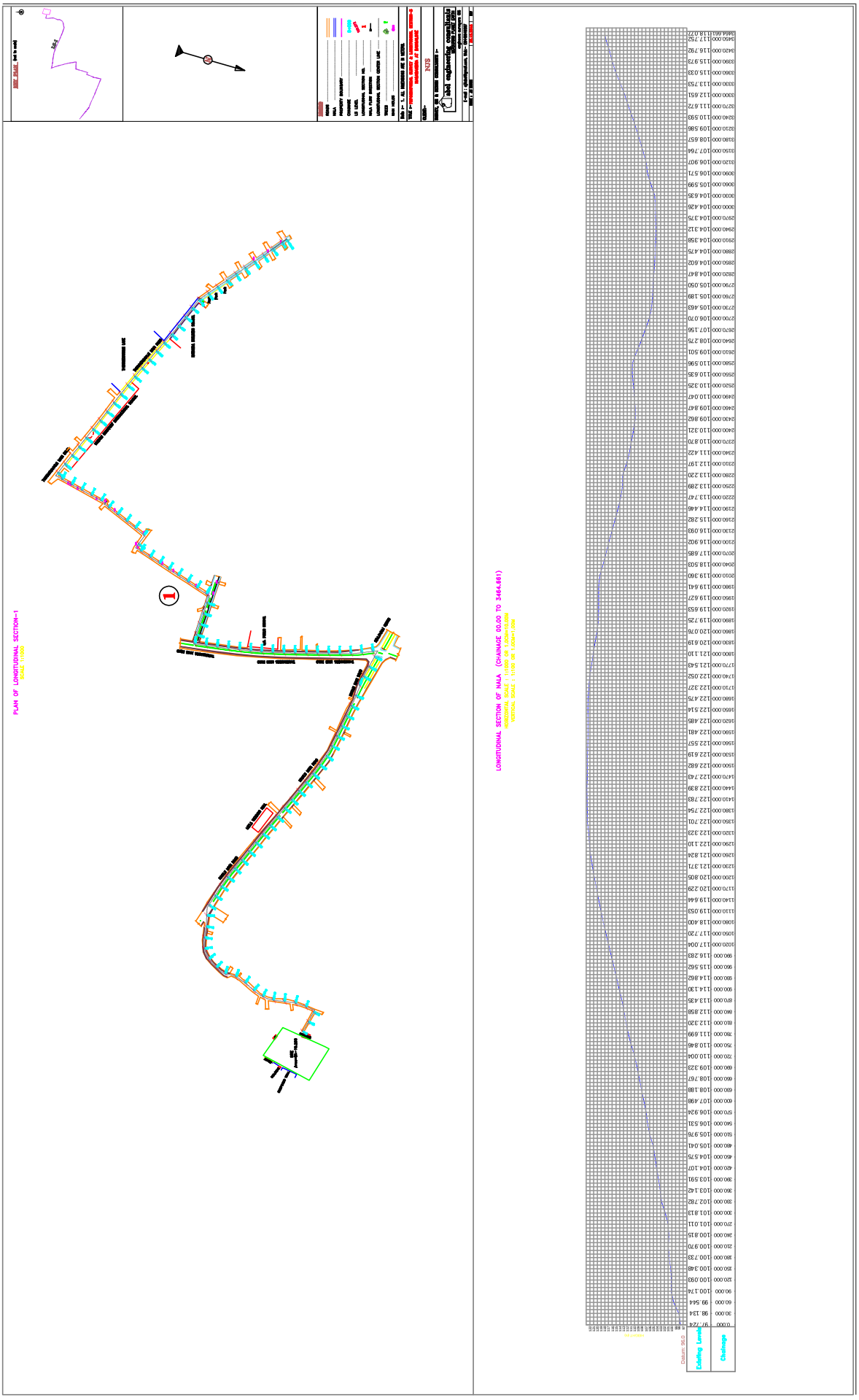
2. Chamber Bottom MSL = 602.413m

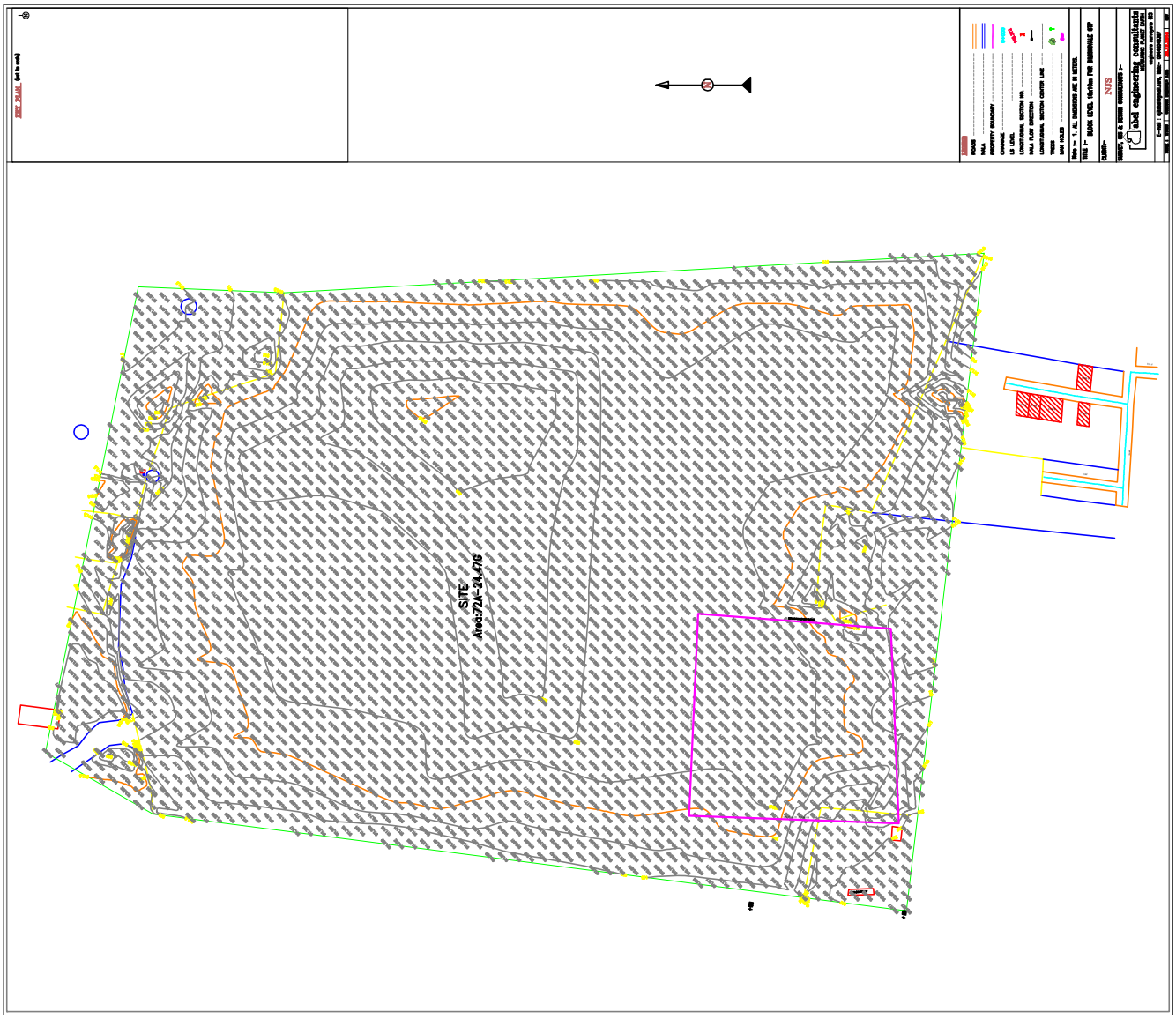
3. Pipe Top MSL = 604.000m

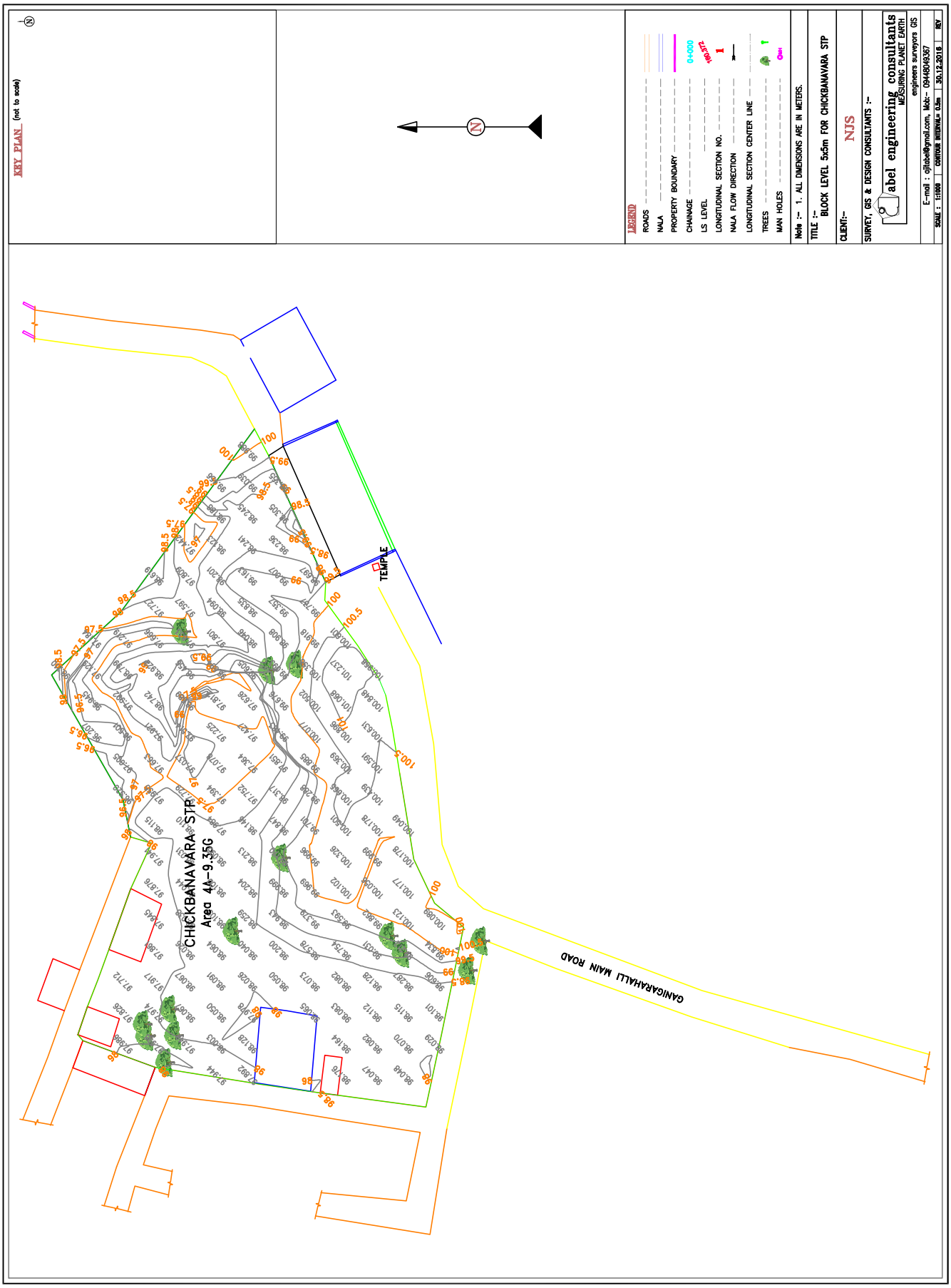

PLAN OF LONGITUDINAL SECTION-1
SCALE 1:1000
LONGITUDINAL SECTION OF TIPPENA HALLI MAIN ROAD (CHAINAGE 00.00 TO 1461.97)
HORIZONTAL SCALE : 1:1000 OR 1:2000
VERTICAL SCALE : 1:100 OR 1:200
HEIGHT (m)
Existing Levels
Chainage


KEY PLAN (not to scale)
SITE
LEGEND
ROADS
NALA
PROPERTY BOUNDARY
CHAINAGE
LS LEVEL
LONGITUDINAL SECTION NO.
NALA FLOW DIRECTION
LONGITUDINAL SECTION CENTER LINE
TREES
MAN HOLES
Note :- 1. ALL DIMENSIONS ARE IN METERS.
TITLE :- BLOCK LEVEL 5x5m FOR AREHALLI STP
CLIENT:-
SURVEY, GIS & DESIGN CONSULTANTS :-
abel engineering consultants
MEASURING PLANET EARTH
engineers surveyors GIS
E-mail : abelsurvey@gmail.com, Mob:- 09448049367
SCALE : 1:1000 | CONTOUR INTERVAL : 0.5m | DATE:24.2.2014
REV
Nala
MH
Area
1A 15.003

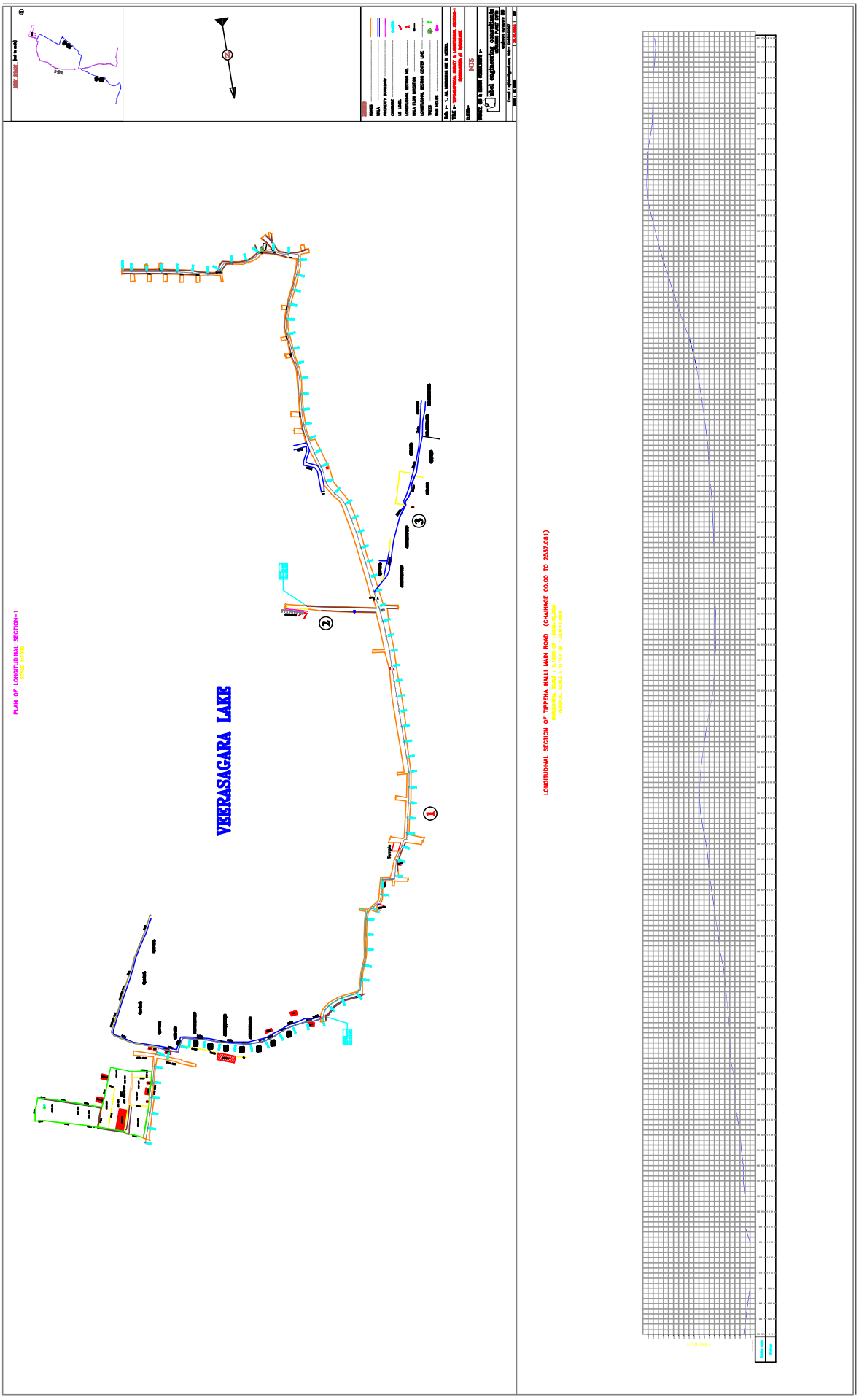

KEY PLAN (not to scale)
LEGEND
ROADS
NALA
PROPERTY BOUNDARY
CHAINAGE
LS LEVEL
LONGITUDINAL SECTION NO.
NALA FLOW DIRECTION
LONGITUDINAL SECTION CENTER LINE
TREES
MAN HOLES
Note :- 1. ALL DIMENSIONS ARE IN METERS.
TITLE :-
BLOCK LEVEL 5x5m FOR SITE NEAR ANCHEPALYA LAKE
CLIENT:-
NJS
SURVEY, GIS & DESIGN CONSULTANTS :-
abel engineering consultants
MEASURING PLANET EARTH
engineers surveyors GIS
E-mail : ajitabel@gmail.com, Mob:- 09448049367
SCALE : 1:1000
CONTOUR INTERVAL= 0.5m
30.12.2016
REV
COMPOUND WALL
BUILDING
SITE
Area=3A-10.28G
BUILDING
BUILDING
BUILDING

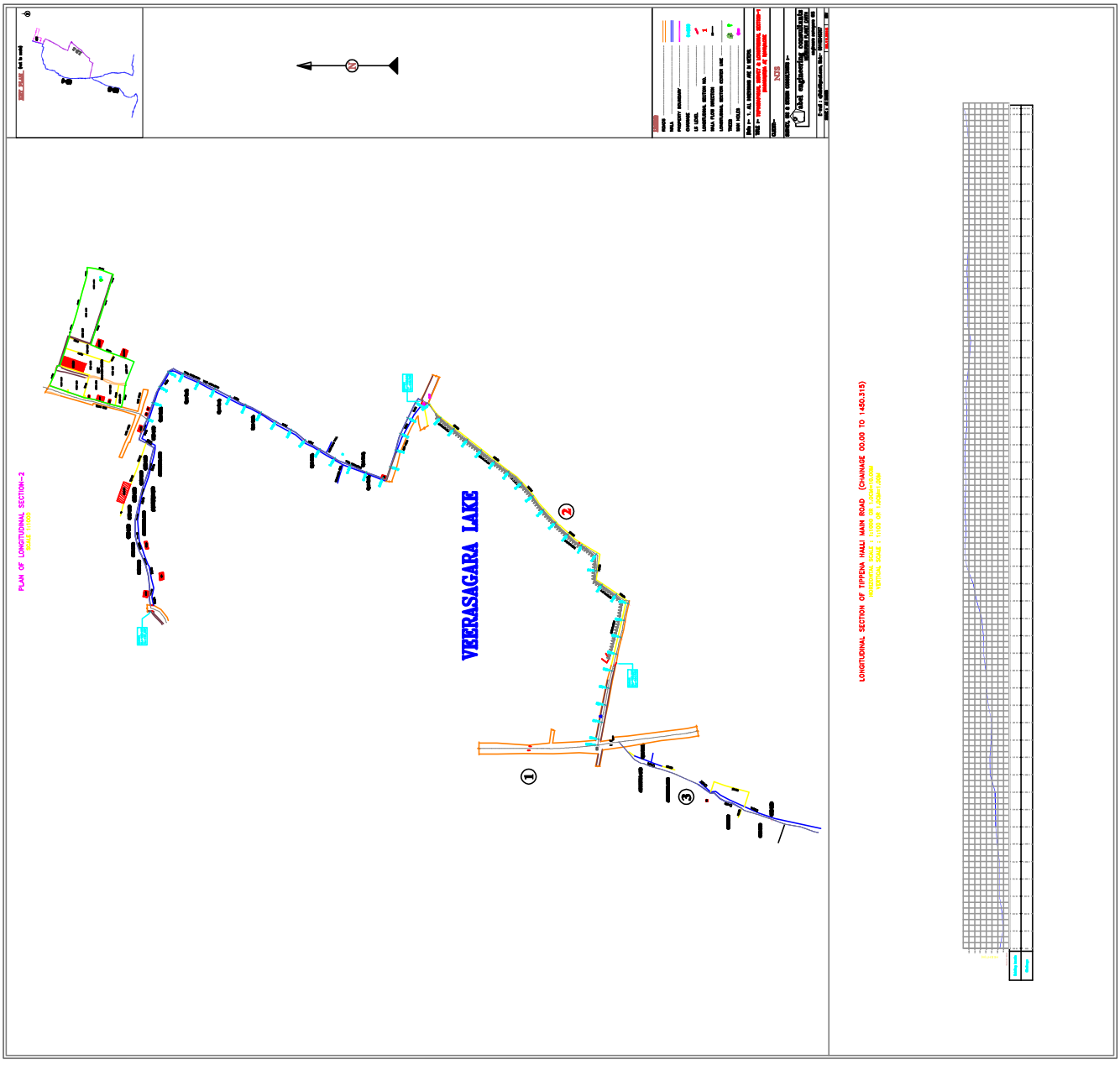

PLAN OF LONGITUDINAL SECTION-1
LONGITUDINAL SECTION OF NALA (CHAINAGE 00.00 TO 3464.681)
Existing Level
Chainage

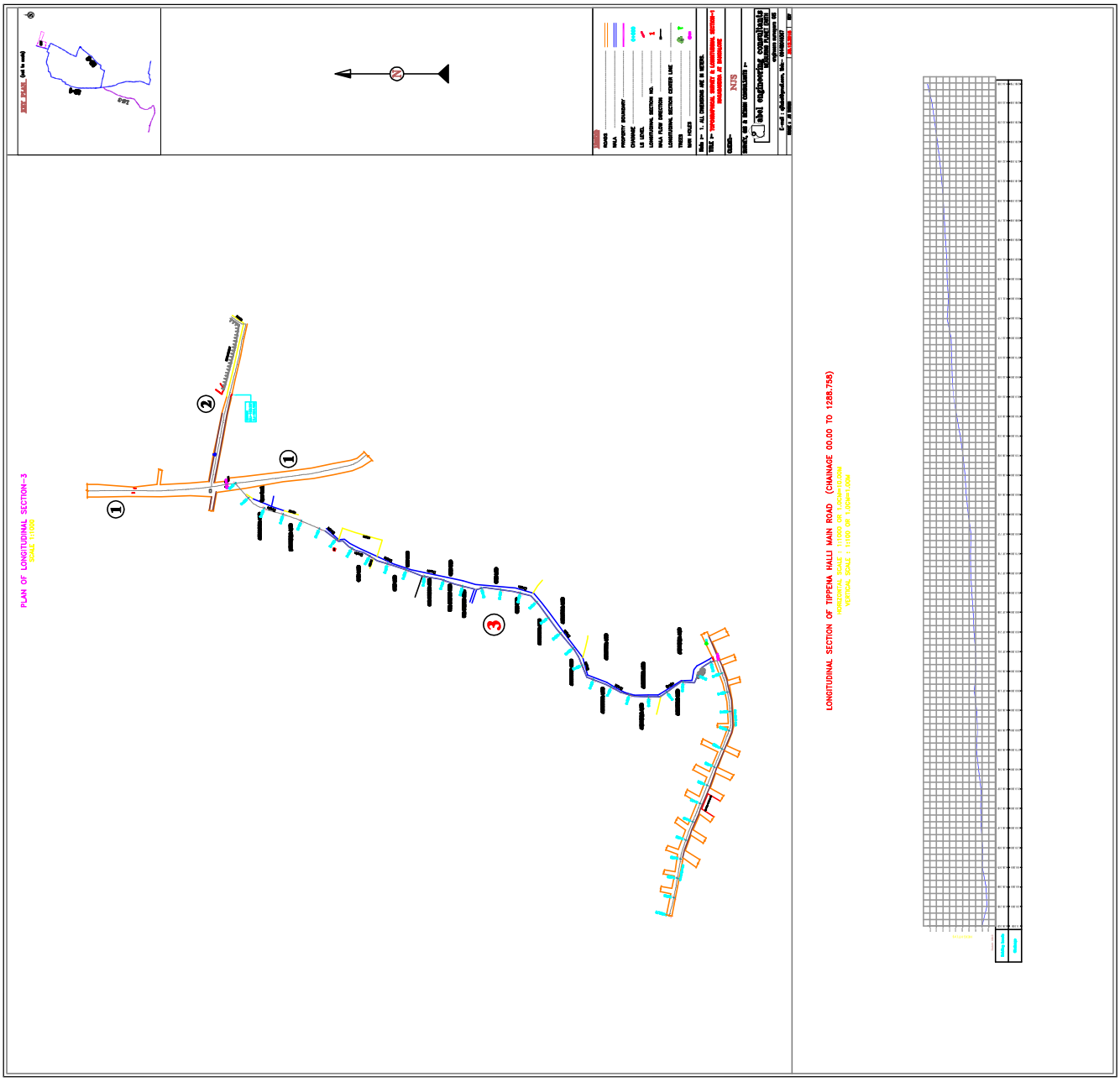

SITE
Area:72A-24.47G

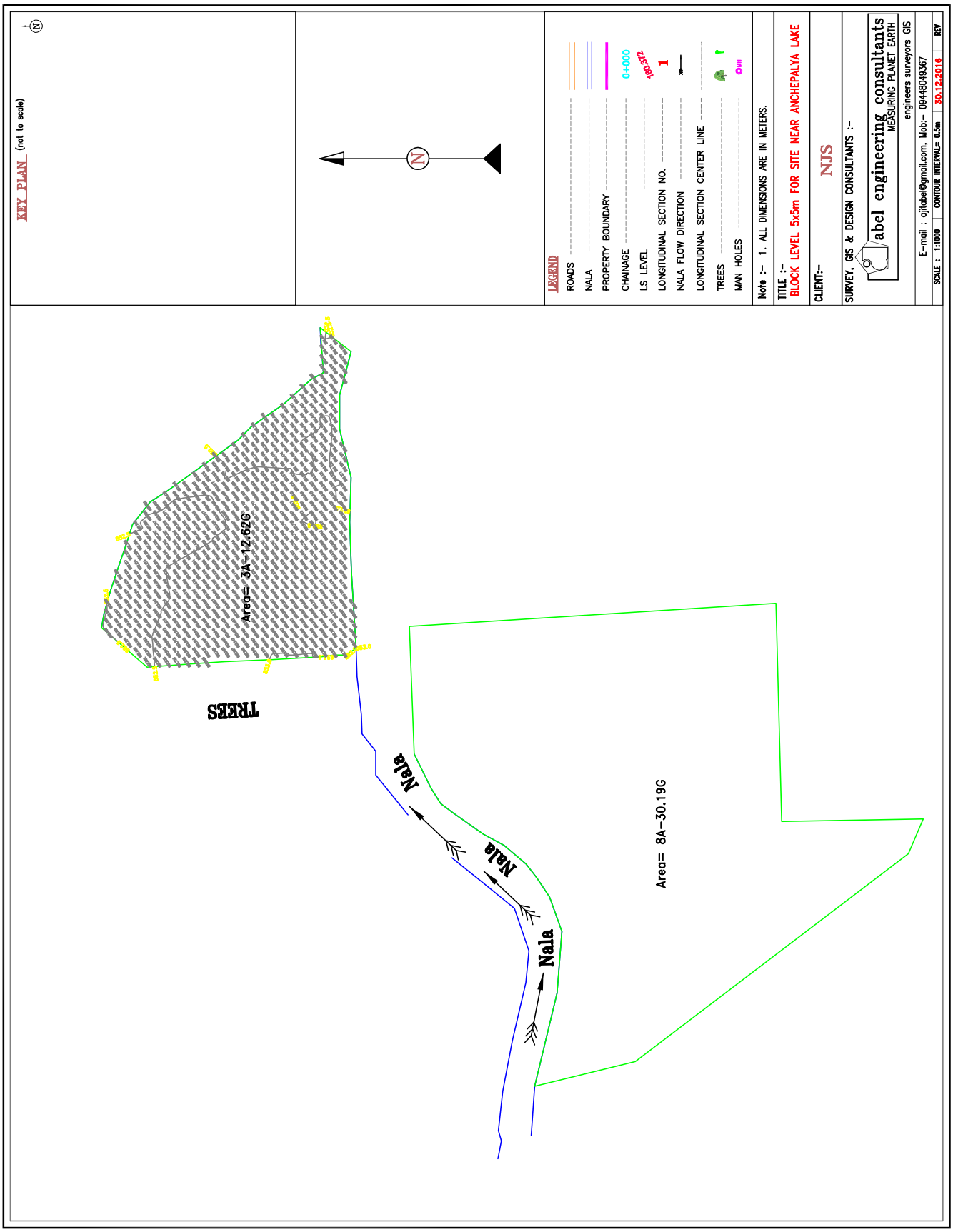

KEY PLAN (not to scale)
LEGEND
ROADS
NALA
PROPERTY BOUNDARY
CHAINAGE
LS LEVEL
LONGITUDINAL SECTION NO.
NALA FLOW DIRECTION
LONGITUDINAL SECTION CENTER LINE
TREES
MAN HOLES
Note :- 1. ALL DIMENSIONS ARE IN METERS.
TITLE :- BLOCK LEVEL 5x5m FOR CHICKBANAVARA STP
CLIENT:- NJS
SURVEY, GIS & DESIGN CONSULTANTS :- abel engineering consultants
MEASURING PLANET EARTH
engineers surveyors GIS
E-mail : ajilabel@gmail.com, Mob:- 09448049367
SCALE : 1:1000 | CONTOUR INTERVAL= 0.5m | 30.12.2016 | REV
TEMPLE
CHICKBANAVARA STP
Area 44-9.35G
GANIGARAHALLI MAIN ROAD

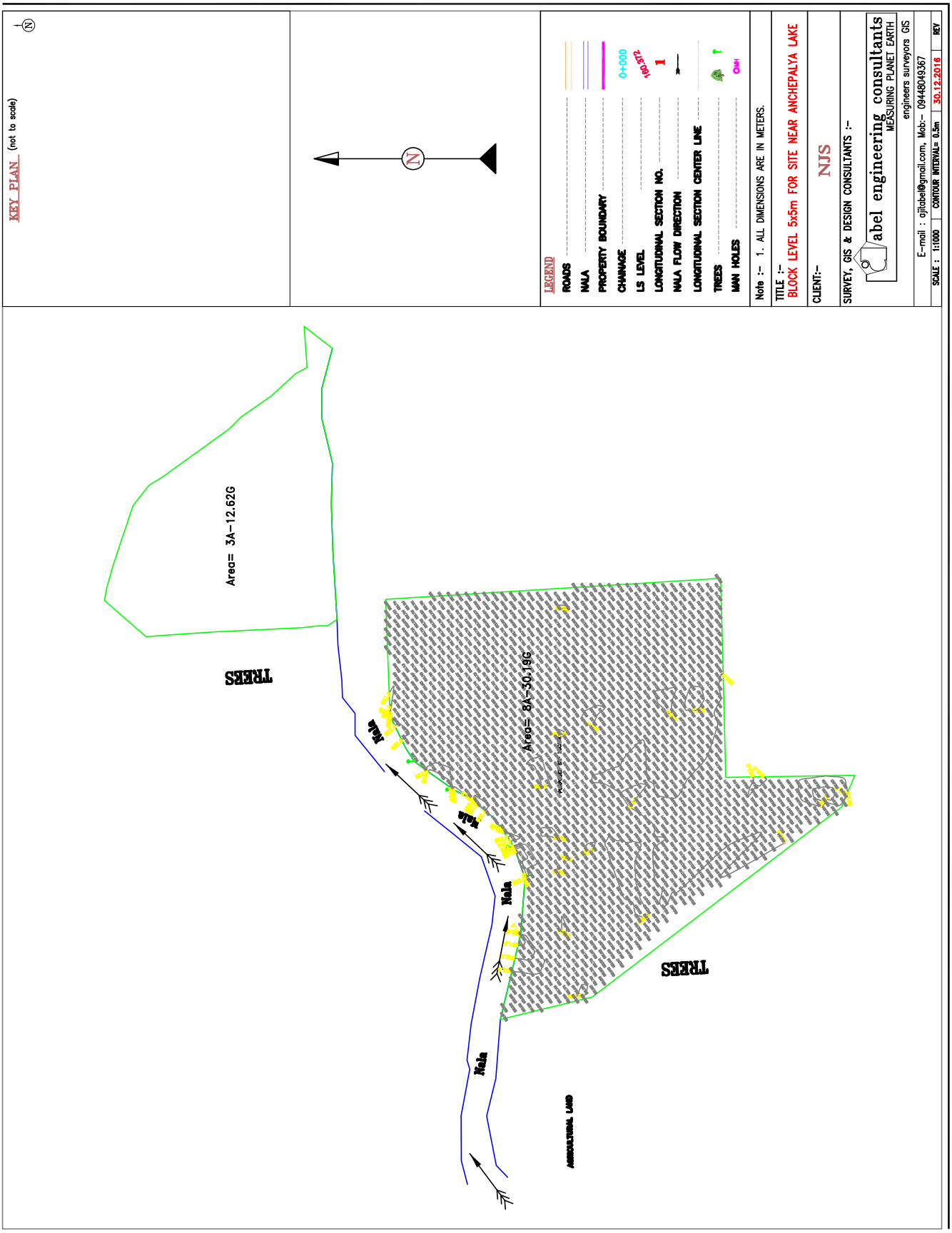

KEY PLAN (not to scale)
SITE
N
SITE
Area 3.0A-37.19G
BUILDING
FENCE
OPEN LAND
MUD ROAD
Nala
AGRICULTURAL LAND
Open Land,
COMPOUND WALL
LEGEND
ROADS
NALA
PROPERTY BOUNDARY
CHAINAGE
LS LEVEL
LONGITUDINAL SECTION NO.
NALA FLOW DIRECTION
LONGITUDINAL SECTION CENTER LINE
TREES
MAN HOLES
Note :- 1. ALL DIMENSIONS ARE IN METERS.
TITLE :-
BLOCK LEVEL 5x5m FOR SITE NEAR ANCHEPALYA LAKE
CLIENT:-
NJS
SURVEY, GIS & DESIGN CONSULTANTS :-
abel engineering consultants
MEASURING PLANET EARTH
engineers surveyors GIS
E-mail : ajitabel@gmail.com, Mob:- 09448049367
SCALE : 1:1000
CONTOUR INTERVAL= 0.5m
30.12.2016
REV

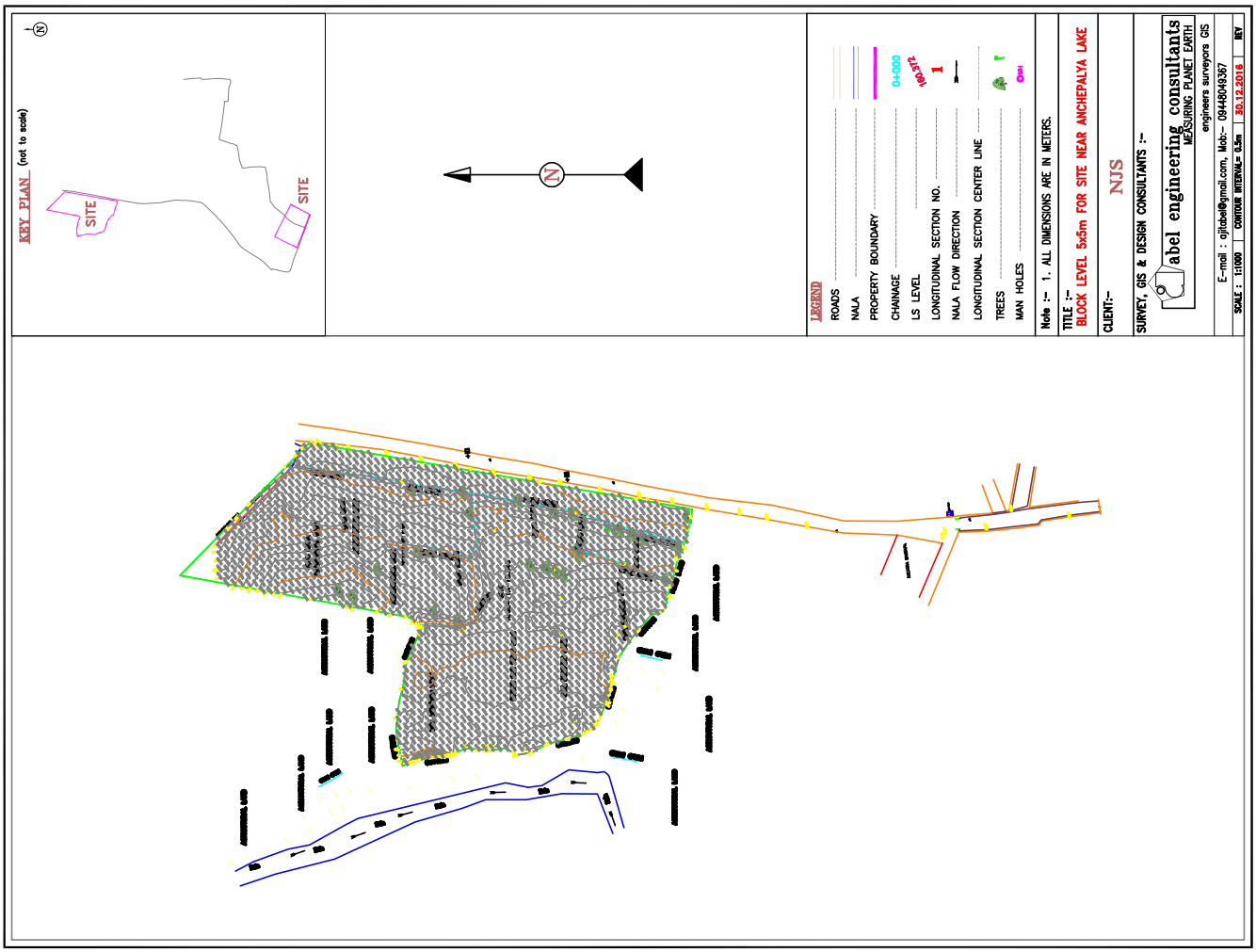

VEERASAGARA LAKE
PLAN OF LONGITUDINAL SECTION-1
LONGITUDINAL SECTION OF TIPPENA HALLI MAIN ROAD (CHAINAGE 00.00 TO 2837.081)

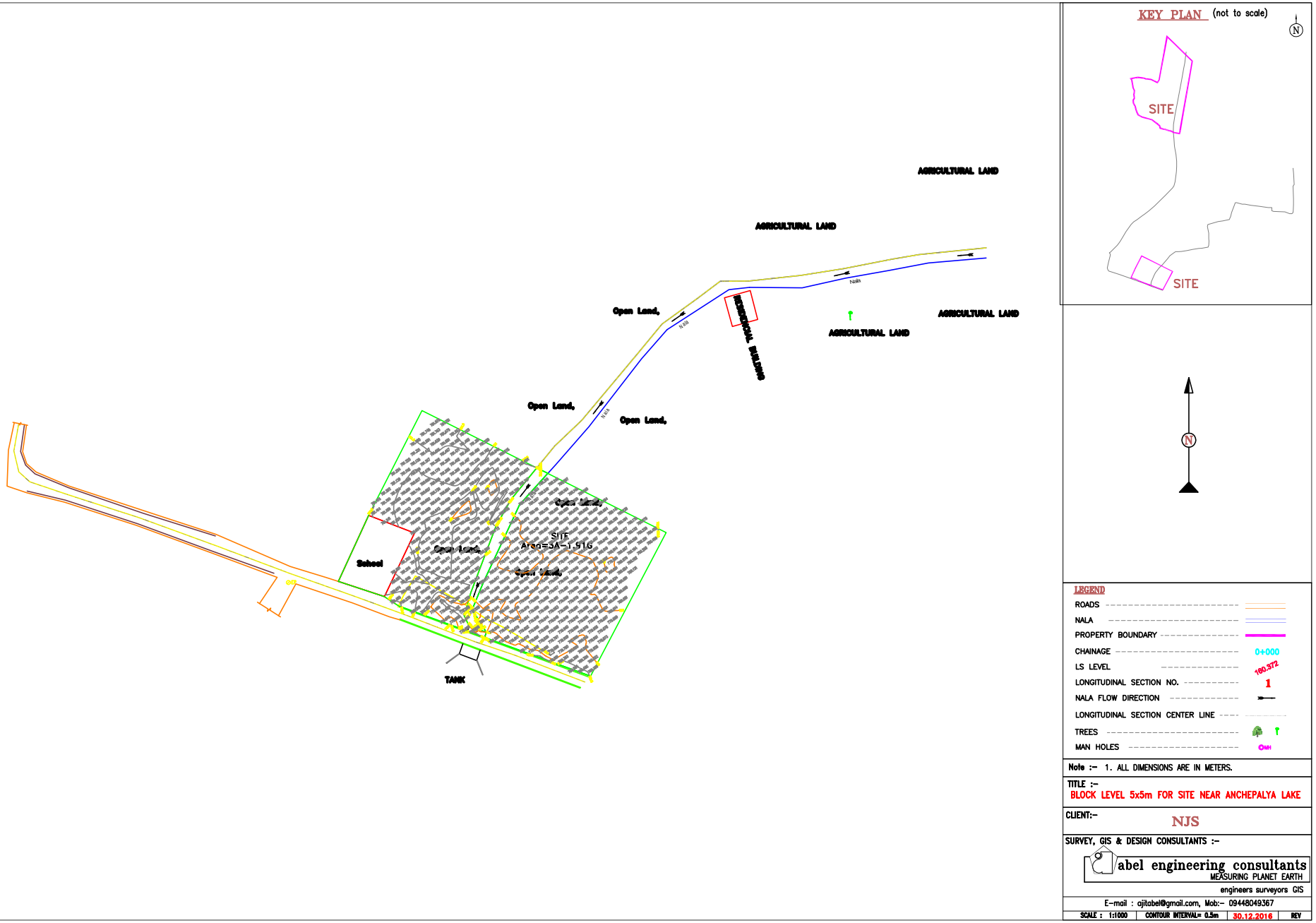

VEERASAGARA LAKE
PLAN OF LONGITUDINAL SECTION-2
LONGITUDINAL SECTION OF TIPPENA HALLI MAIN ROAD (CHAINAGE 00.00 TO 1450.315)

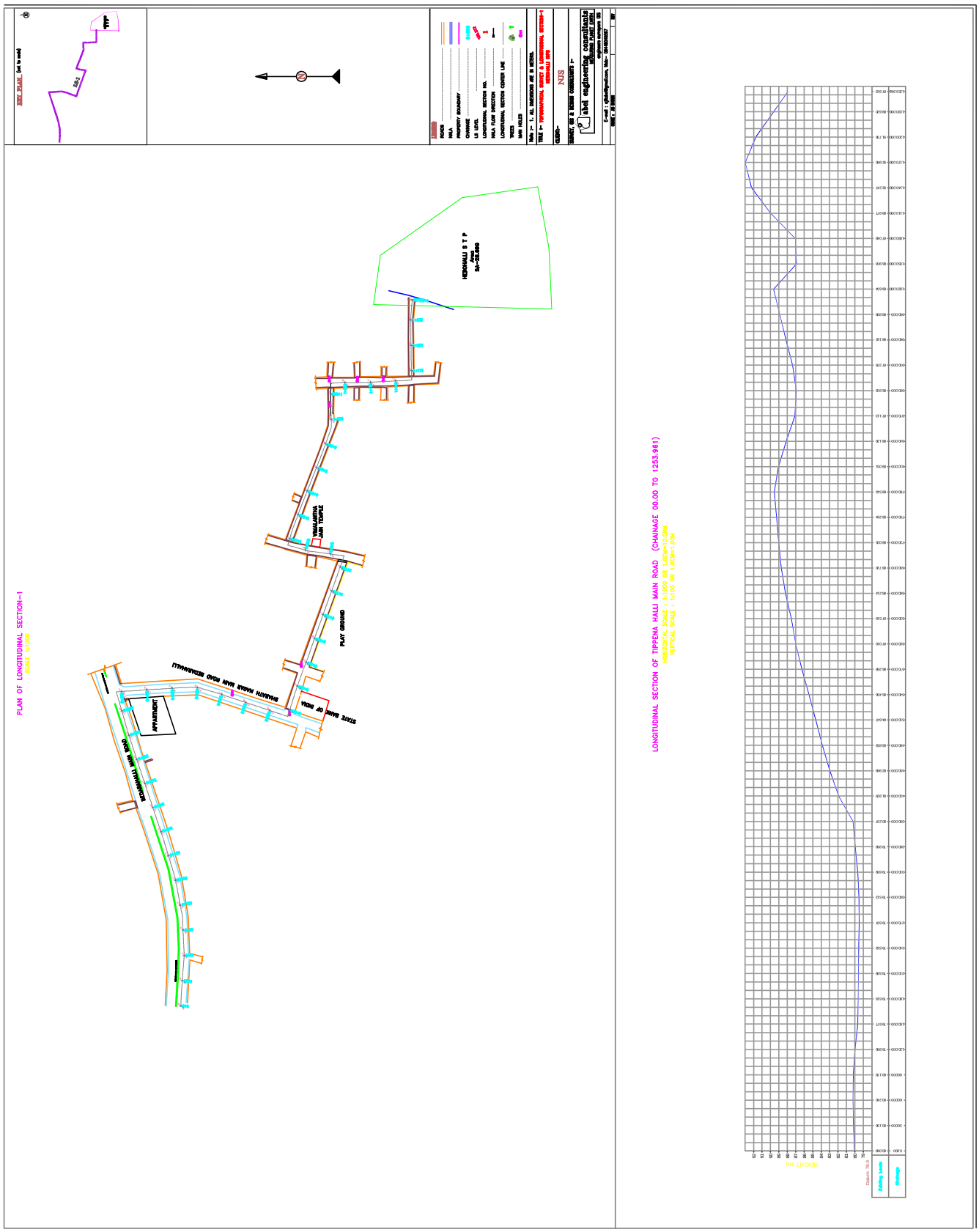

PLAN OF LONGITUDINAL SECTION-3
SCALE 1:1000
LONGITUDINAL SECTION OF TIPPENA HALLI MAIN ROAD (CHAINAGE 00.00 TO 1285.756)
HORIZONTAL SCALE : 1:1000 OR 1:1000mm
VERTICAL SCALE : 1:100 OR 1:100mm
NJS

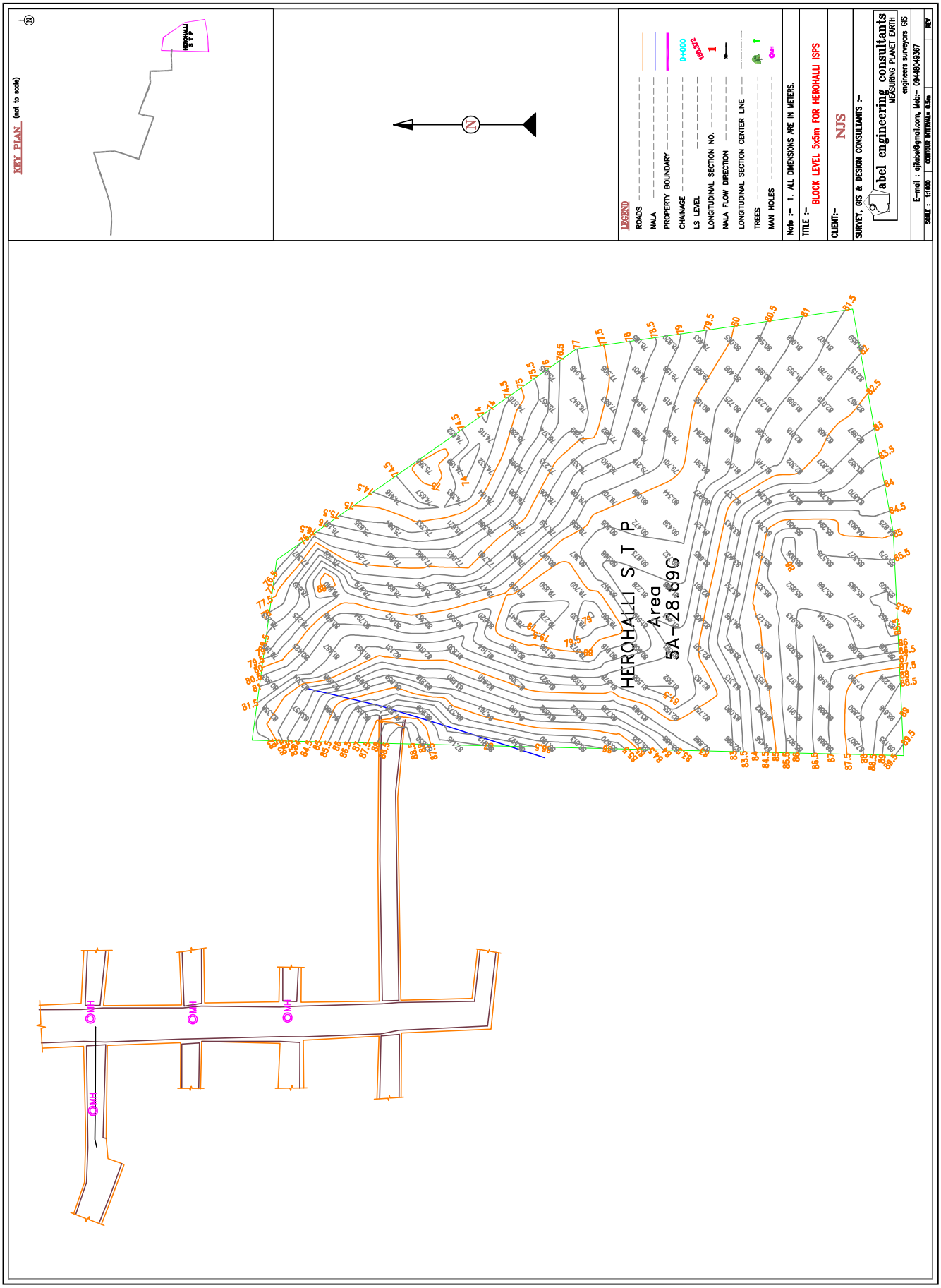

KEY PLAN (not to scale)
N
LEGEND
ROADS
NALA
PROPERTY BOUNDARY
CHAINAGE 0+000
LS LEVEL 225.681
LONGITUDINAL SECTION NO. 1
NALA FLOW DIRECTION
LONGITUDINAL SECTION CENTER LINE
TREES
MAN HOLES MH
Note :- 1. ALL DIMENSIONS ARE IN METERS.
TITLE :-
BLOCK LEVEL 5x5m FOR SITE NEAR ANCHEPALYA LAKE
CLIENT:-
NJS
SURVEY, GIS & DESIGN CONSULTANTS :-
abel engineering consultants
MEASURING PLANET EARTH
engineers surveyors GIS
E-mail : ajitabel@gmail.com, Mob:- 09448049367
SCALE : 1:1000
CONTOUR INTERVAL= 0.5m
30.12.2016
REV
Area= 3A-12.62G
TREES
Nala
Nala
Nala
Area= 8A-30.19G

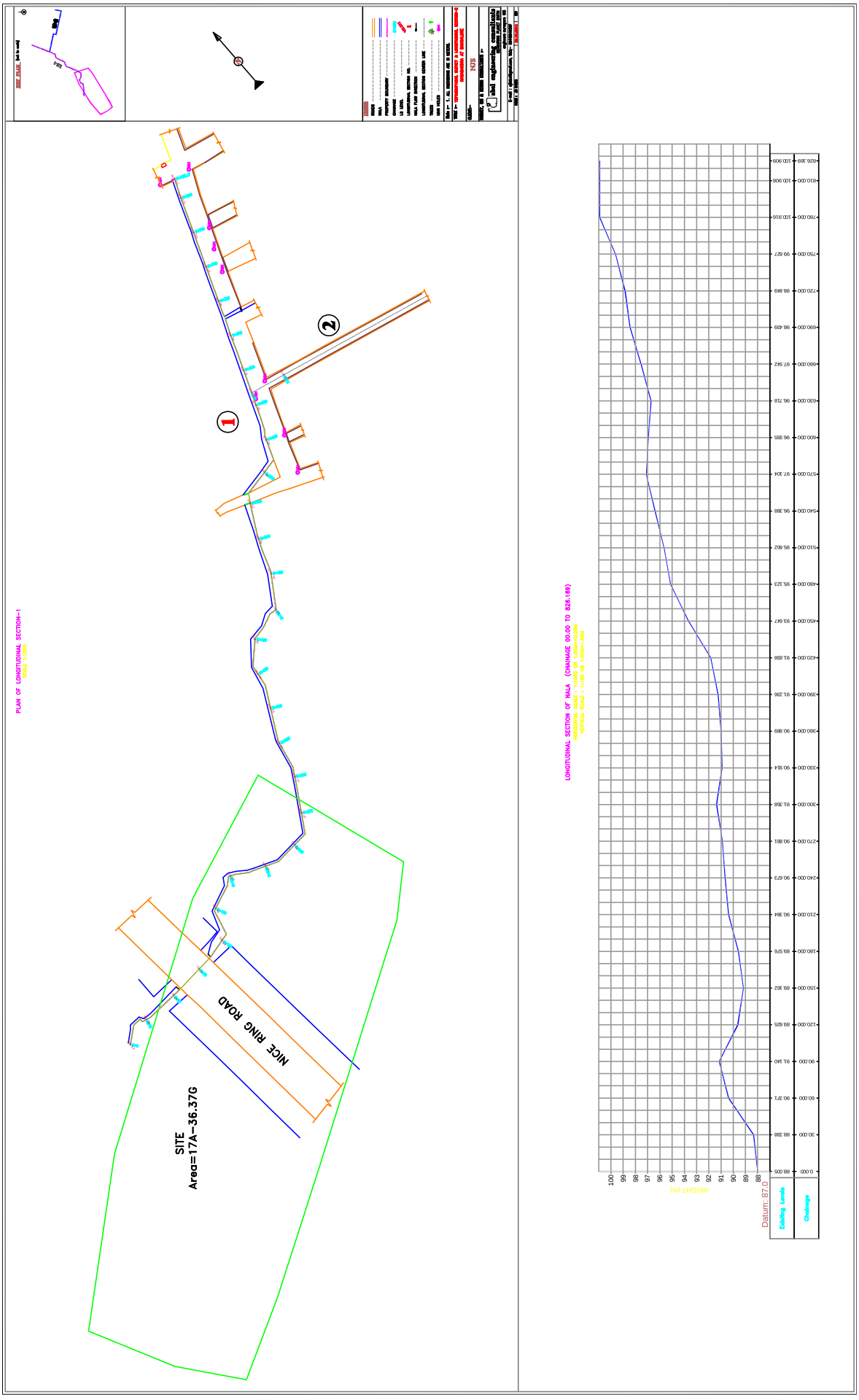

KEY PLAN (not to scale)
N
LEGEND
ROADS
NALA
PROPERTY BOUNDARY
CHAINAGE
0+000
LS LEVEL
100.372
LONGITUDINAL SECTION NO.
1
NALA FLOW DIRECTION
LONGITUDINAL SECTION CENTER LINE
TREES
MAN HOLES
OMH
Note :- 1. ALL DIMENSIONS ARE IN METERS.
TITLE :-
BLOCK LEVEL 5x5m FOR SITE NEAR ANCHEPALYA LAKE
CLIENT:-
NJS
SURVEY, GIS & DESIGN CONSULTANTS :-
abel engineering consultants
MEASURING PLANET EARTH
engineers surveyors GIS
E-mail : ajitabel@gmail.com, Mob:- 09448049367
SCALE : 1:1000
CONTOUR INTERVAL= 0.5m
30.12.2016
REV
Area= 3A-12.62G
Area= 8A-30.19G
TREES
TREES
Nala
Nala
Nala
AGRICULTURAL LAND

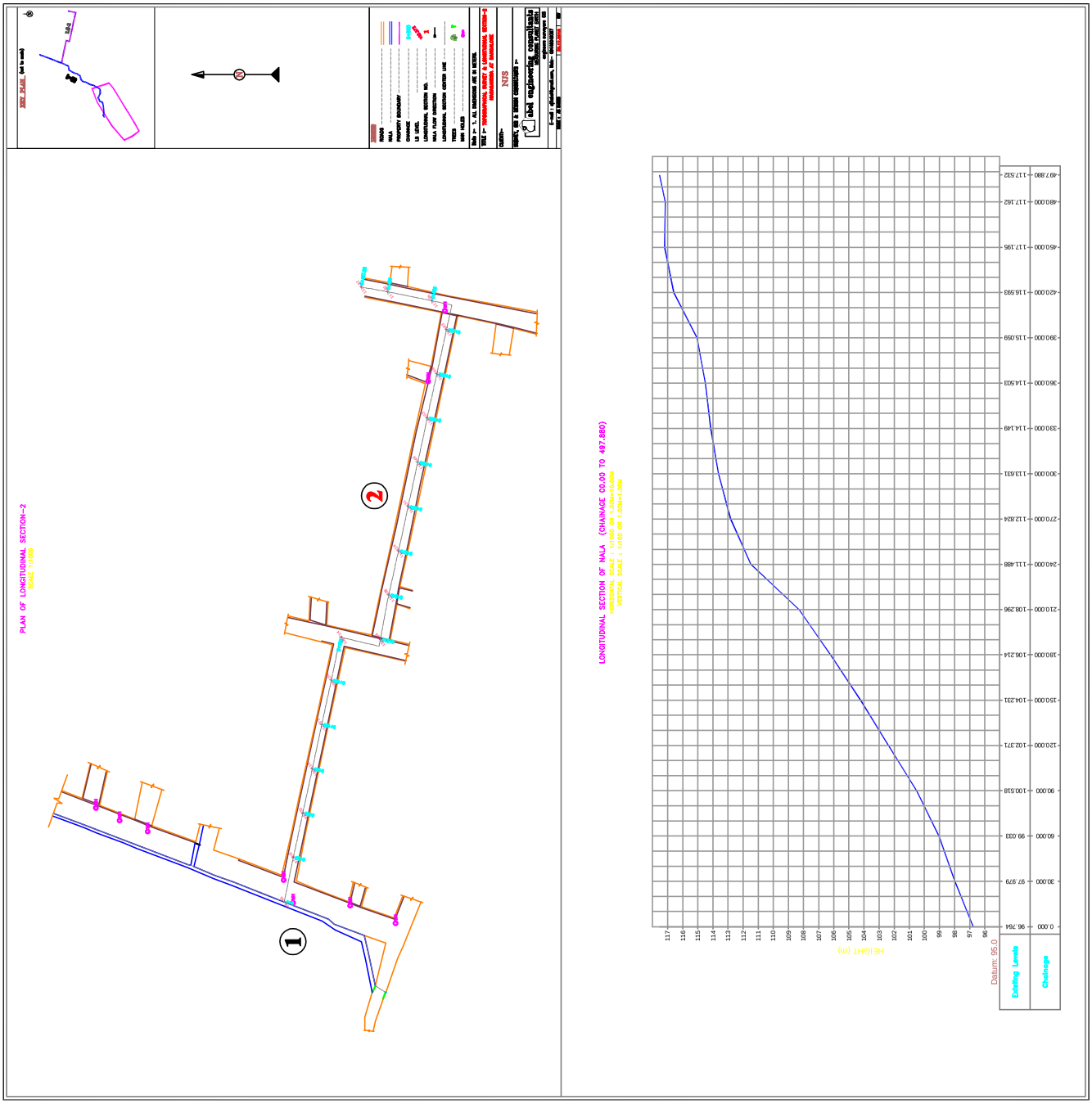

KEY PLAN (not to scale)
SITE
SITE
N
LEGEND
ROADS
NALA
PROPERTY BOUNDARY
CHAINAGE
0+000
LS LEVEL
+101.272
LONGITUDINAL SECTION NO.
1
NALA FLOW DIRECTION
LONGITUDINAL SECTION CENTER LINE
TREES
MAN HOLES
Note :- 1. ALL DIMENSIONS ARE IN METERS.
TITLE :-
BLOCK LEVEL 5x5m FOR SITE NEAR ANCHEPALYA LAKE
CLIENT:-
NJS
SURVEY, GIS & DESIGN CONSULTANTS :-
abel engineering consultants
MEASURING PLANET EARTH
engineers surveyors GIS
E-mail : ojtabel@gmail.com, Mob:- 09448049367
SCALE : 1:1000
CONTOUR INTERVAL= 0.5m
30.12.2016
KEY

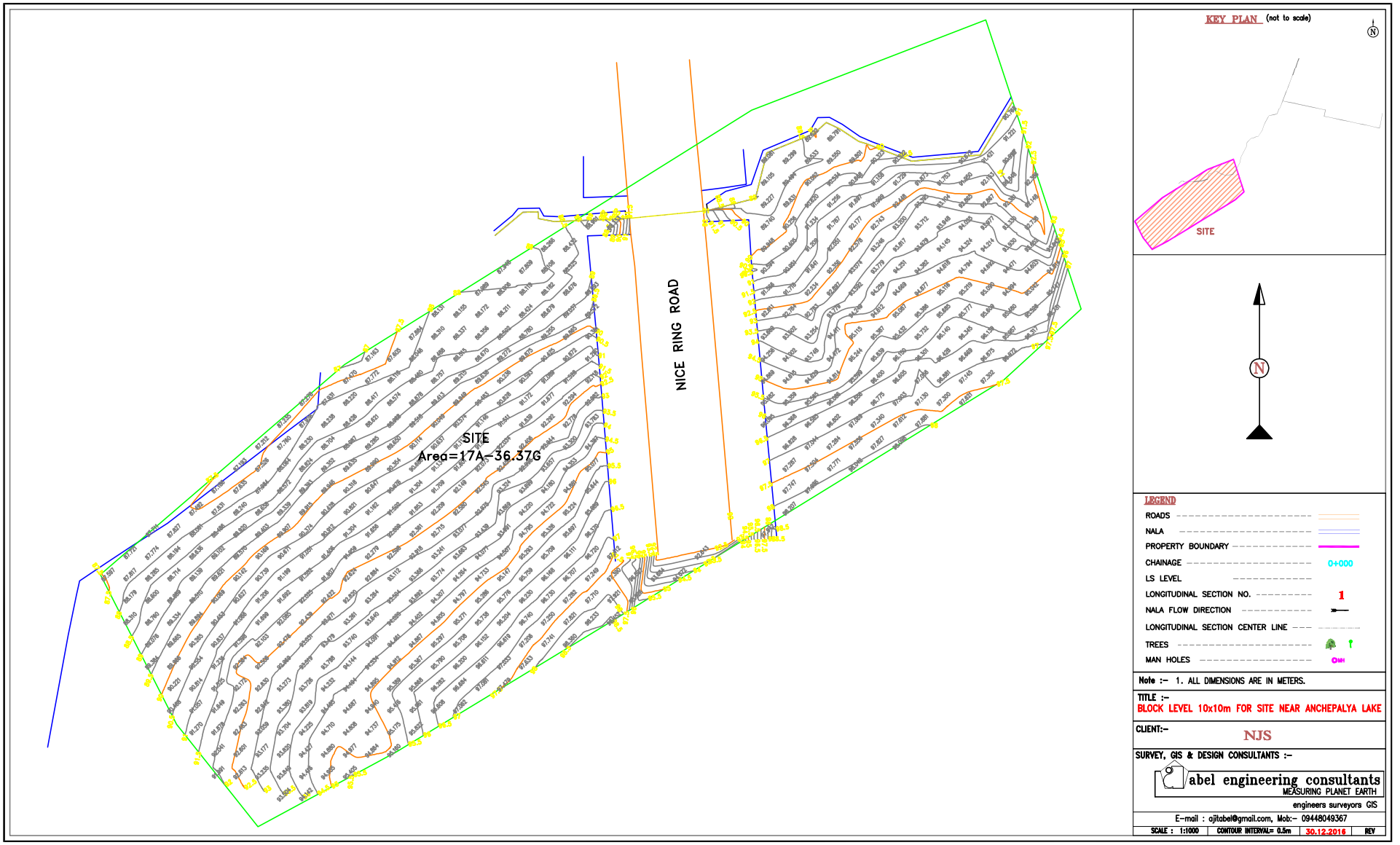

AGRICULTURAL LAND
AGRICULTURAL LAND
AGRICULTURAL LAND
AGRICULTURAL LAND
Open Land,
RESIDENTIAL BUILDING
Open Land,
Open Land,
Open Land,
Open Land,
SITE
Area=3A-1.91G
School
TANK
KEY PLAN (not to scale)
N
SITE
SITE
N
LEGEND
ROADS
NALA
PROPERTY BOUNDARY
CHAINAGE 0+000
LS LEVEL 160.372
LONGITUDINAL SECTION NO. 1
NALA FLOW DIRECTION
LONGITUDINAL SECTION CENTER LINE
TREES
MAN HOLES MH
Note :- 1. ALL DIMENSIONS ARE IN METERS.
TITLE :-
BLOCK LEVEL 5x5m FOR SITE NEAR ANCHEPALYA LAKE
CLIENT:-
NJS
SURVEY, GIS & DESIGN CONSULTANTS :-
abel engineering consultants
MEASURING PLANET EARTH
engineers surveyors GIS
E-mail : ajitabel@gmail.com, Mob:- 09448049367
SCALE : 1:1000
CONTOUR INTERVAL= 0.5m
30.12.2016
REV

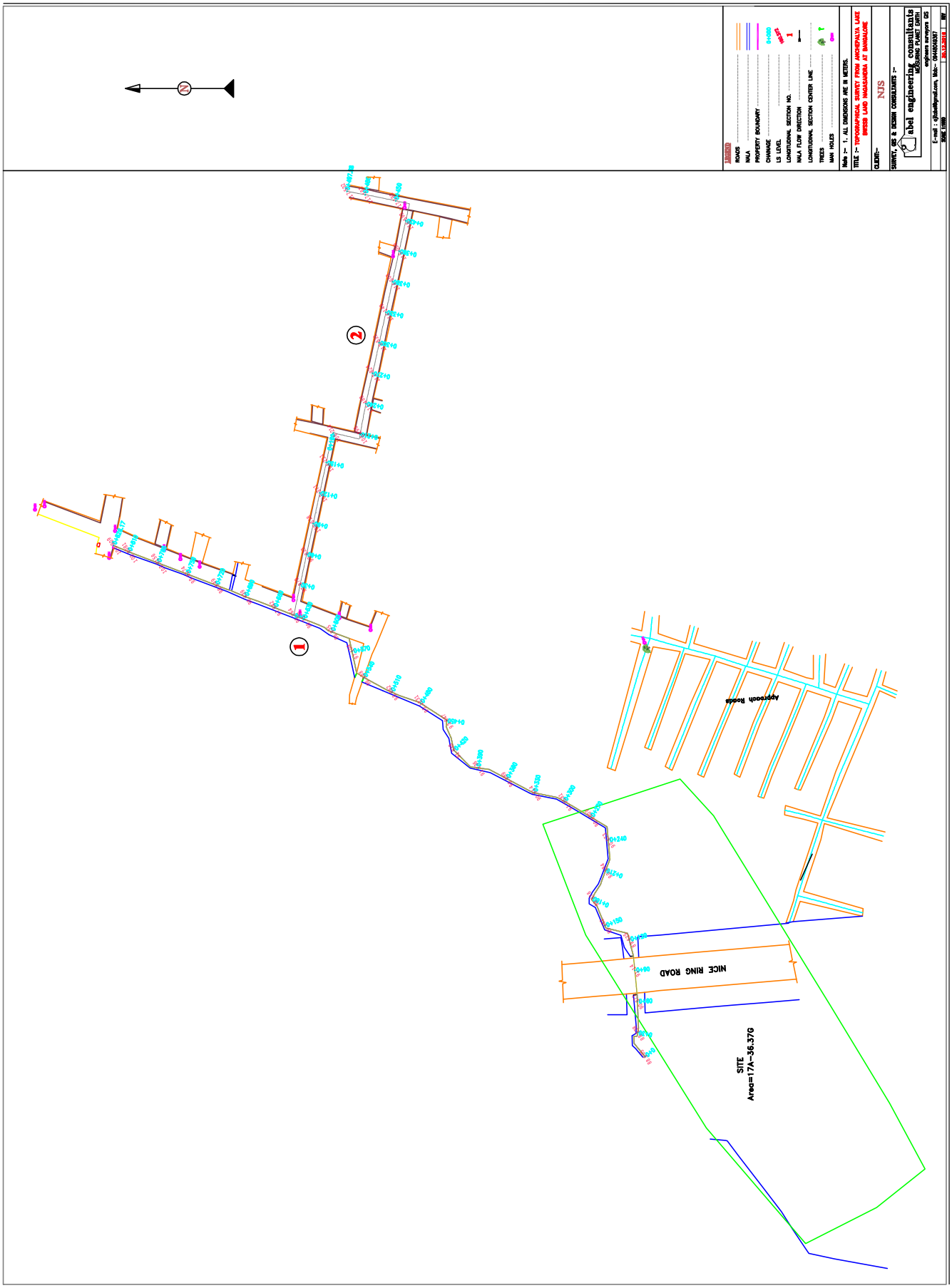

PLAN OF LONGITUDINAL SECTION-1
SCALE 1:1000
HEDIHALLI S T P
Area
64-38.00
VALMIKIYA
JAIN TEMPLE
PLAY GROUND
STATE BANK OF INDIA
BHARATH MAHAL MAIN ROAD BIDARAHALLI
APPARTMENT
BIDARAHALLI MAIN ROAD
LONGITUDINAL SECTION OF TIPPENA HALLI MAIN ROAD (CHAINAGE 00.00 TO 1253.981)
HORIZONTAL SCALE : 1:1000 OR 1.00cm=10.00M
VERTICAL SCALE : 1:100 OR 1.00cm=1.00M

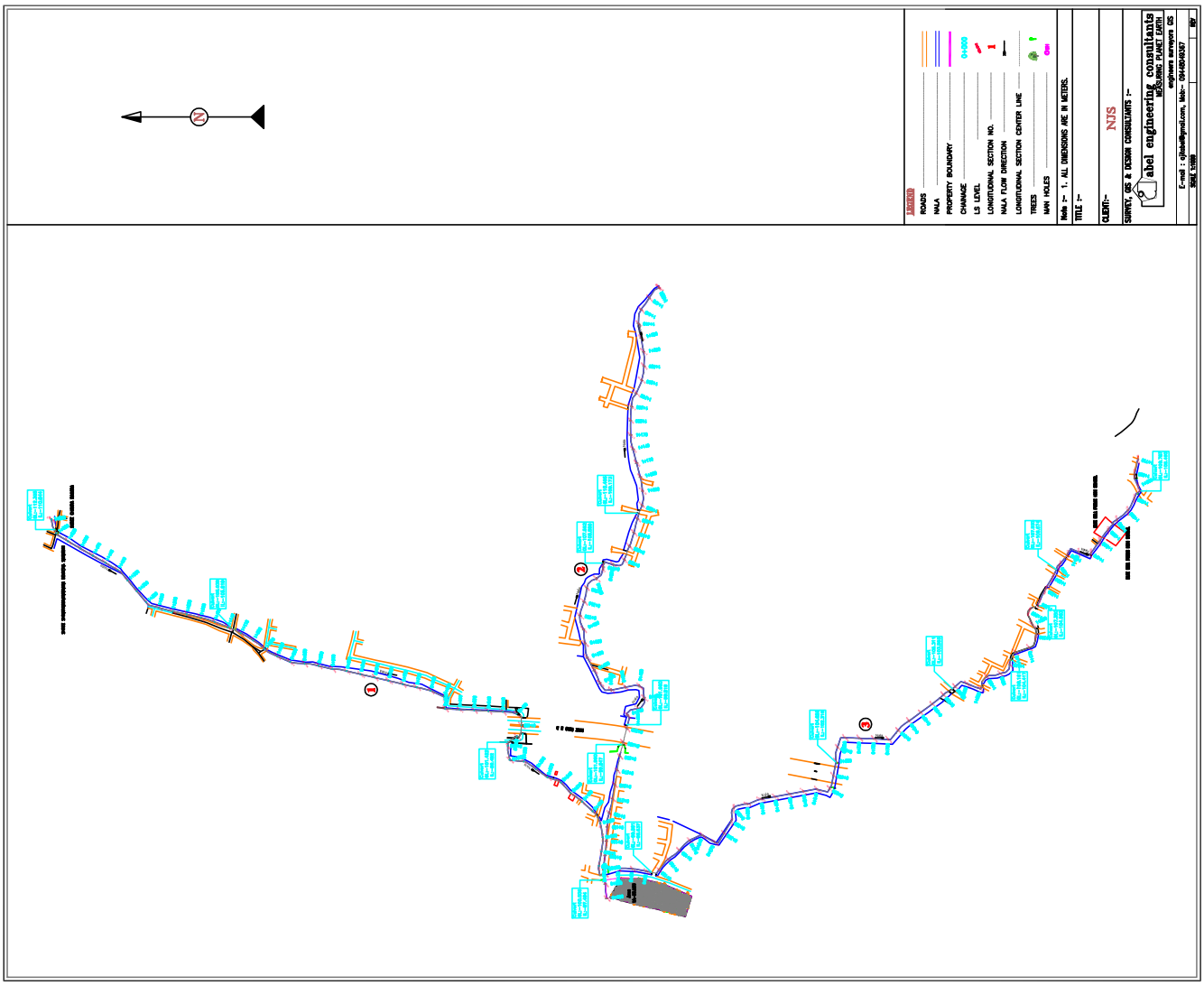

KEY PLAN (not to scale)
LEGEND
ROADS
NALA
PROPERTY BOUNDARY
CHAINAGE
LS LEVEL
LONGITUDINAL SECTION NO.
NALA FLOW DIRECTION
LONGITUDINAL SECTION CENTER LINE
TREES
MAN HOLES
Note :- 1. ALL DIMENSIONS ARE IN METERS.
TITLE :- BLOCK LEVEL 5x5m FOR HEROHALLI ISPS
CLIENT:- NJS
SURVEY, GIS & DESIGN CONSULTANTS :-
abel engineering consultants
MEASURING PLANET EARTH
engineers surveyors GIS
E-mail : ajitabel@gmail.com, Mob:- 09448049367
SCALE : 1:1000
CONTOUR INTERVAL= 0.5m
REV
HEROHALLI S T P
Area
5A-28.69G
MH
MH
MH
MH

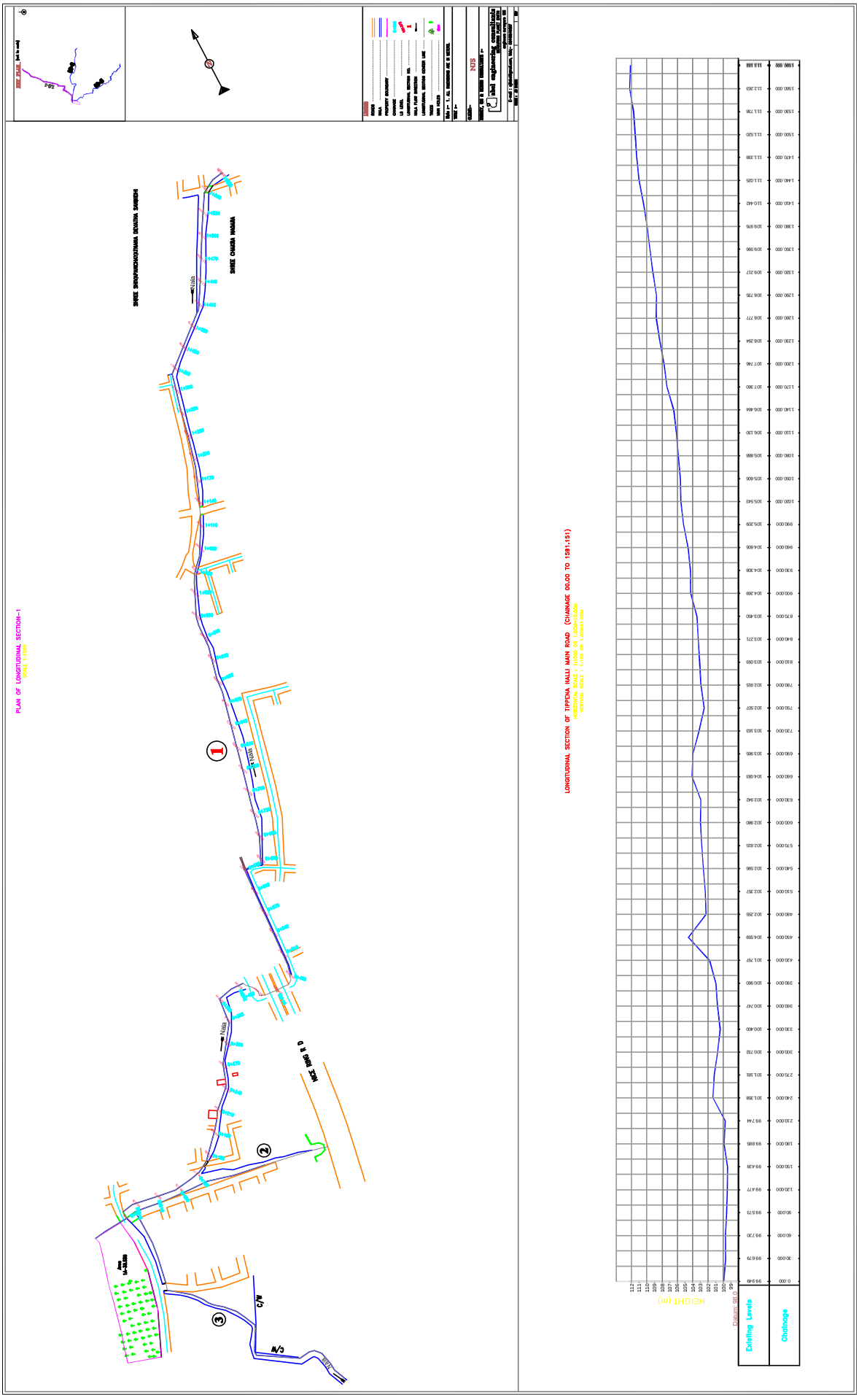

PLAN OF LONGITUDINAL SECTION-1
SCALE 1:5000
1
2
NICE RING ROAD
SITE
Area=17A-36.37G
LONGITUDINAL SECTION OF NALA (CHAINAGE 00.00 TO 826.169)
HORIZONTAL SCALE 1:1000 OR 1CM=10.00M
VERTICAL SCALE 1:1000 OR 1.0CM=1.00M
HEIGHT (m)
Datum: 87.0
Existing Levels
Chainage

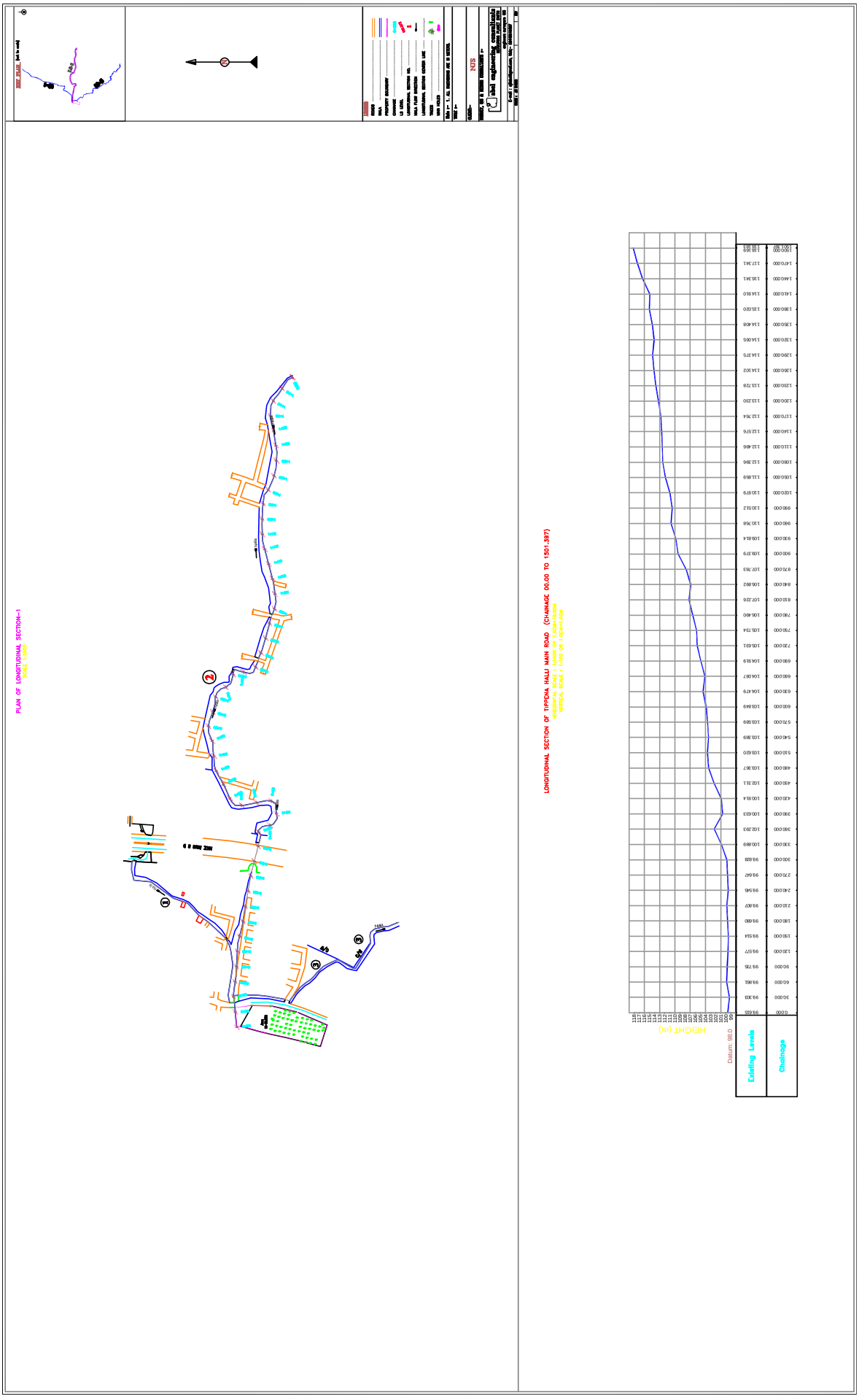

INDEX PLAN
PLAN OF LONGITUDINAL SECTION-2
SCALE 1:1000
LONGITUDINAL SECTION OF NALA (CHAINAGE 00.00 TO 497.880)
HORIZONTAL SCALE : 1:1000 OR 1 inch=1000ft
VERTICAL SCALE : 1:100 OR 1 inch=100ft
HEIGHT (m)
Datum: 95.0
Existing Levels
Chainage
NJS
1
2

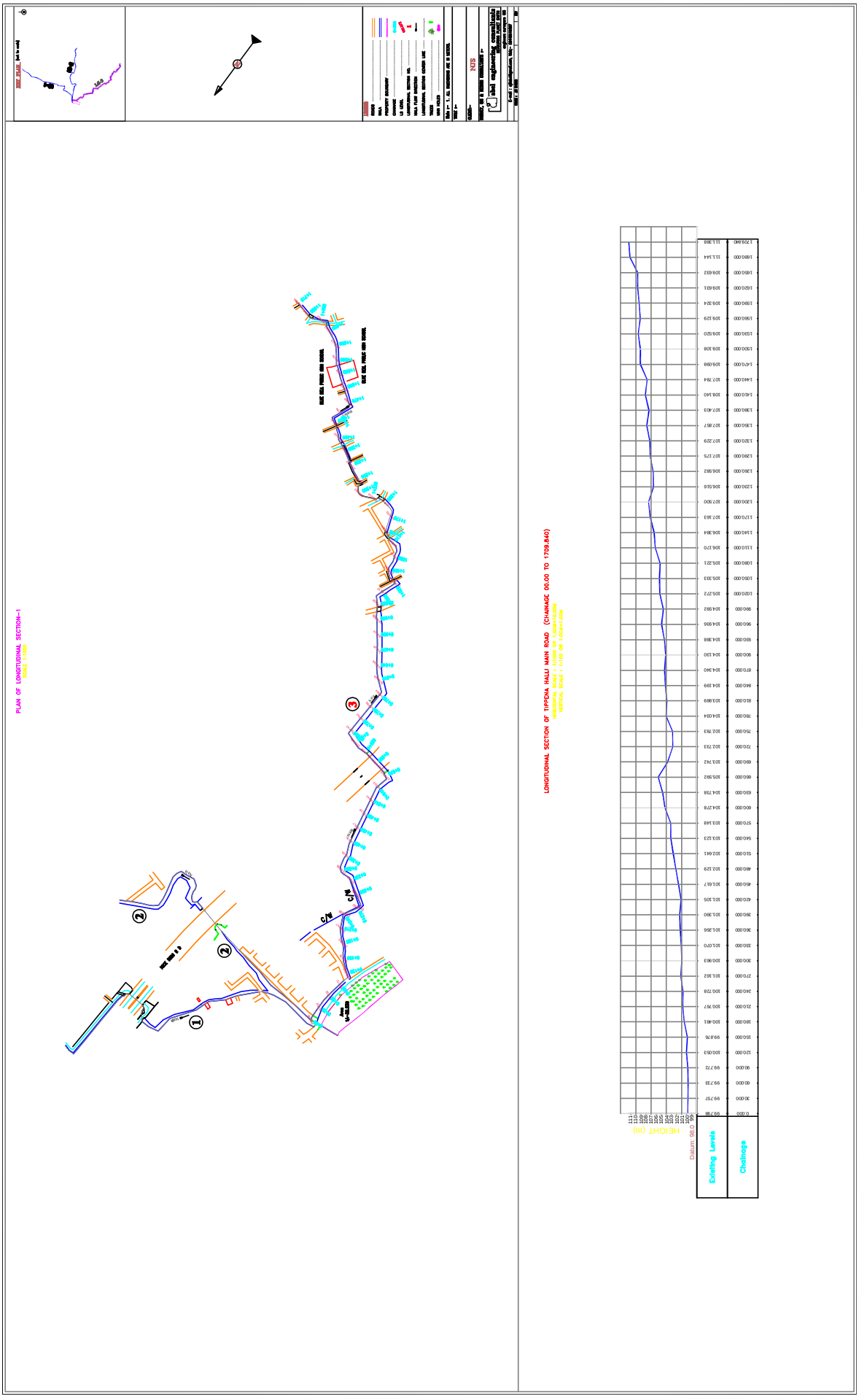

NICE RING ROAD
SITE
Area=17A-36.37G
KEY PLAN (not to scale)
SITE
N
LEGEND
ROADS
NALA
PROPERTY BOUNDARY
CHAINAGE
0+000
LS LEVEL
LONGITUDINAL SECTION NO.
1
NALA FLOW DIRECTION
LONGITUDINAL SECTION CENTER LINE
TREES
MAN HOLES
Note :- 1. ALL DIMENSIONS ARE IN METERS.
TITLE :-
BLOCK LEVEL 10x10m FOR SITE NEAR ANCHEPALYA LAKE
CLIENT:-
NJS
SURVEY, GIS & DESIGN CONSULTANTS :-
abel engineering consultants
MEASURING PLANET EARTH
engineers surveyors GIS
E-mail : ajitabel@gmail.com, Mob:- 09448049367
SCALE : 1:1000
CONTOUR INTERVAL= 0.5m
30.12.2016
REV

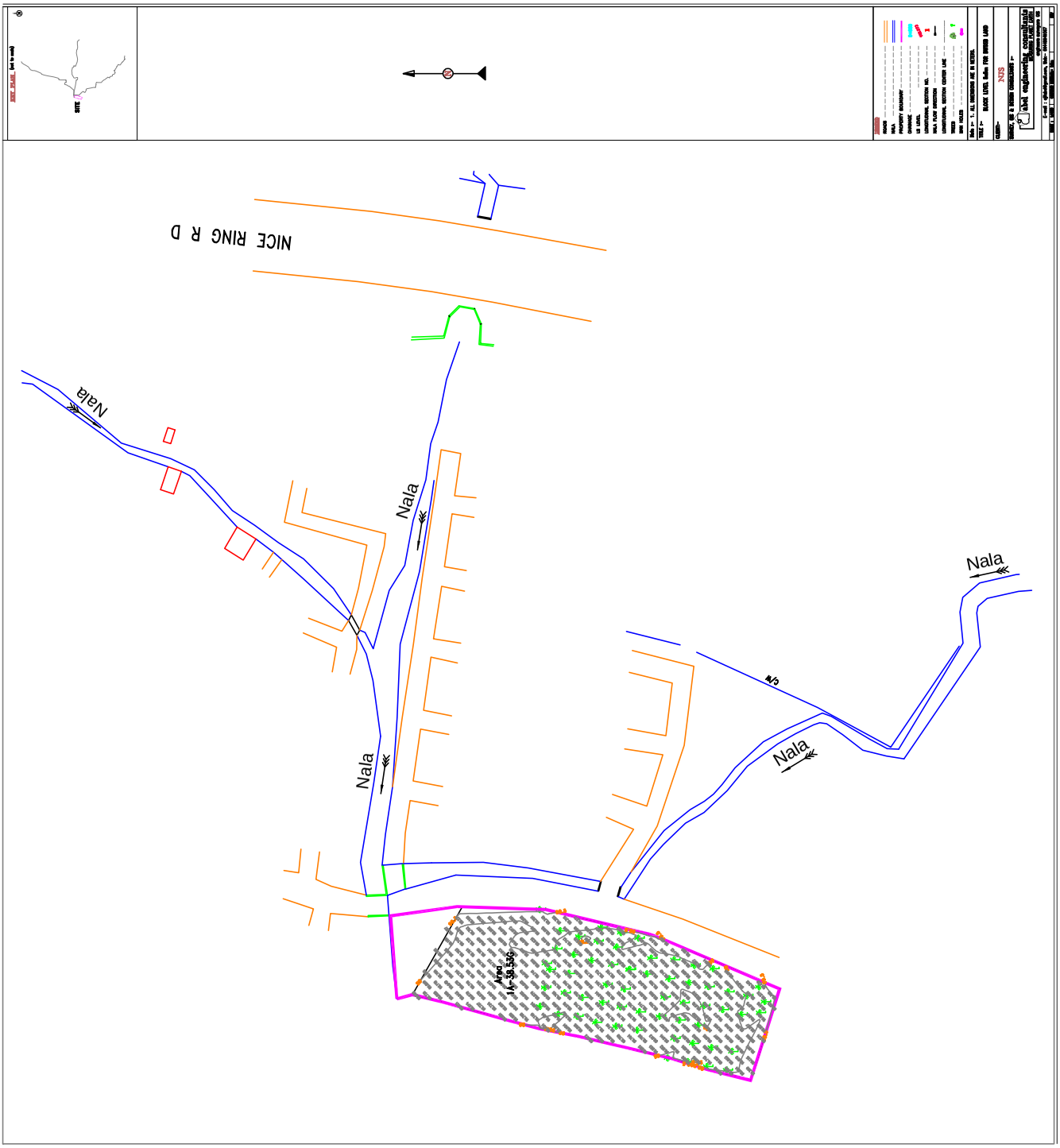

LEGEND
ROADS
NALA
PROPERTY BOUNDARY
CHAINAGE
LS LEVEL
LONGITUDINAL SECTION NO.
NALA FLOW DIRECTION
LONGITUDINAL SECTION CENTER LINE
TREES
MAN HOLES
Note :- 1. ALL DIMENSIONS ARE IN METERS.
TITLE :- TOPOGRAPHICAL SURVEY FROM ANCHEPALYA LAKE BYPASS LAND NAGASANDRA AT BANGALORE
CLIENT:-
NJS
SURVEY, GIS & DESIGN CONSULTANTS :-
abel engineering consultants
MEASURING PLANET EARTH
engineers surveyors GIS
E-mail : @abelengg@gmail.com, Mob:- 09448049097
Approach Roads
NICE RING ROAD
SITE
Area=17A-36.37G

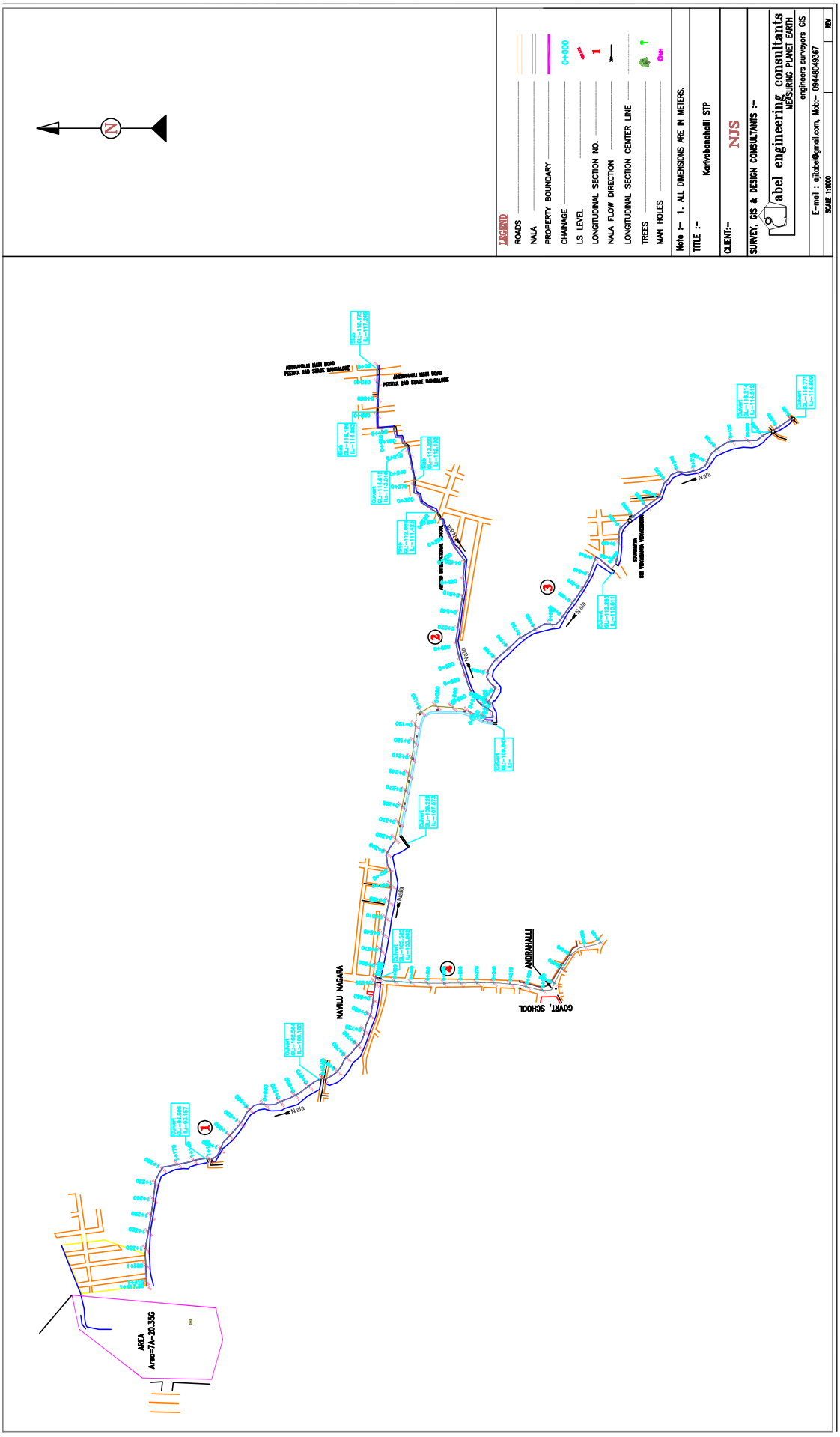

LEGEND
ROADS
NALA
PROPERTY BOUNDARY
CHAINAGE
0+000
LS LEVEL
LONGITUDINAL SECTION NO.
NALA FLOW DIRECTION
LONGITUDINAL SECTION CENTER LINE
TREES
MAN HOLES
Note :- 1. ALL DIMENSIONS ARE IN METERS.
TITLE :-
CLIENT:-
NJS
SURVEY, GIS & DESIGN CONSULTANTS :-
abel engineering consultants
MEASURING PLANET EARTH
engineers surveyors GIS
E-mail : @abelsk@gmail.com, Mob:- 09448045057
SHEET NO.
REF

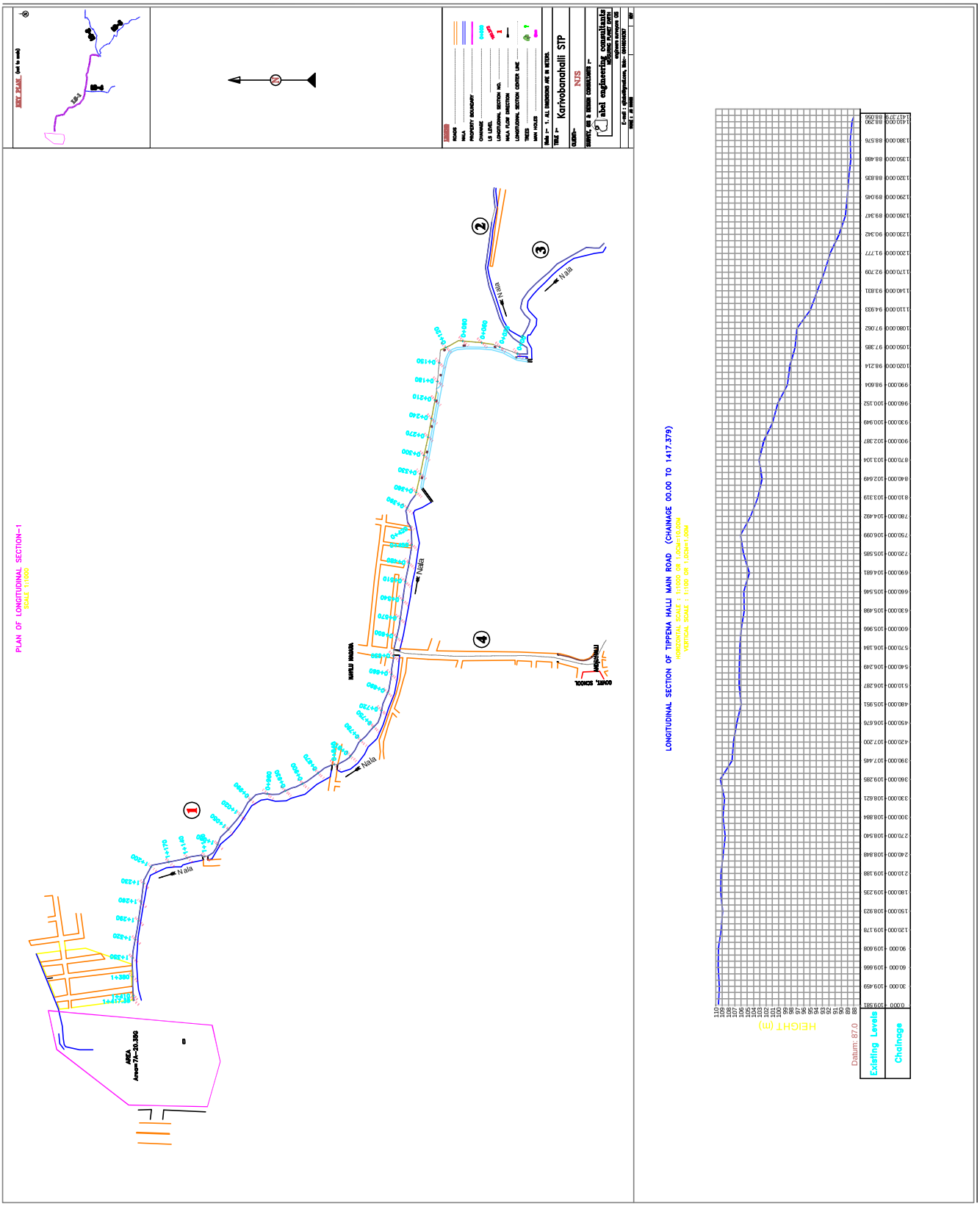

PLAN OF LONGITUDINAL SECTION-1
SCALE 1:2000
LONGITUDINAL SECTION OF TIPPENA HALLI MAIN ROAD (CHAINAGE 00.00 TO 1591.151)
HORIZONTAL SCALE : 1:2000
VERTICAL SCALE : 1:100 OR 1:200
HEIGHT (m)
Datum - 98.0
Existing Levels
Chainage
SHREE SHIVAPANCHAYATHANA DEVATHA SAMEDHA
SHREE CHAMDA MAHANA
NEW ROAD 9.0
C/D
C/D

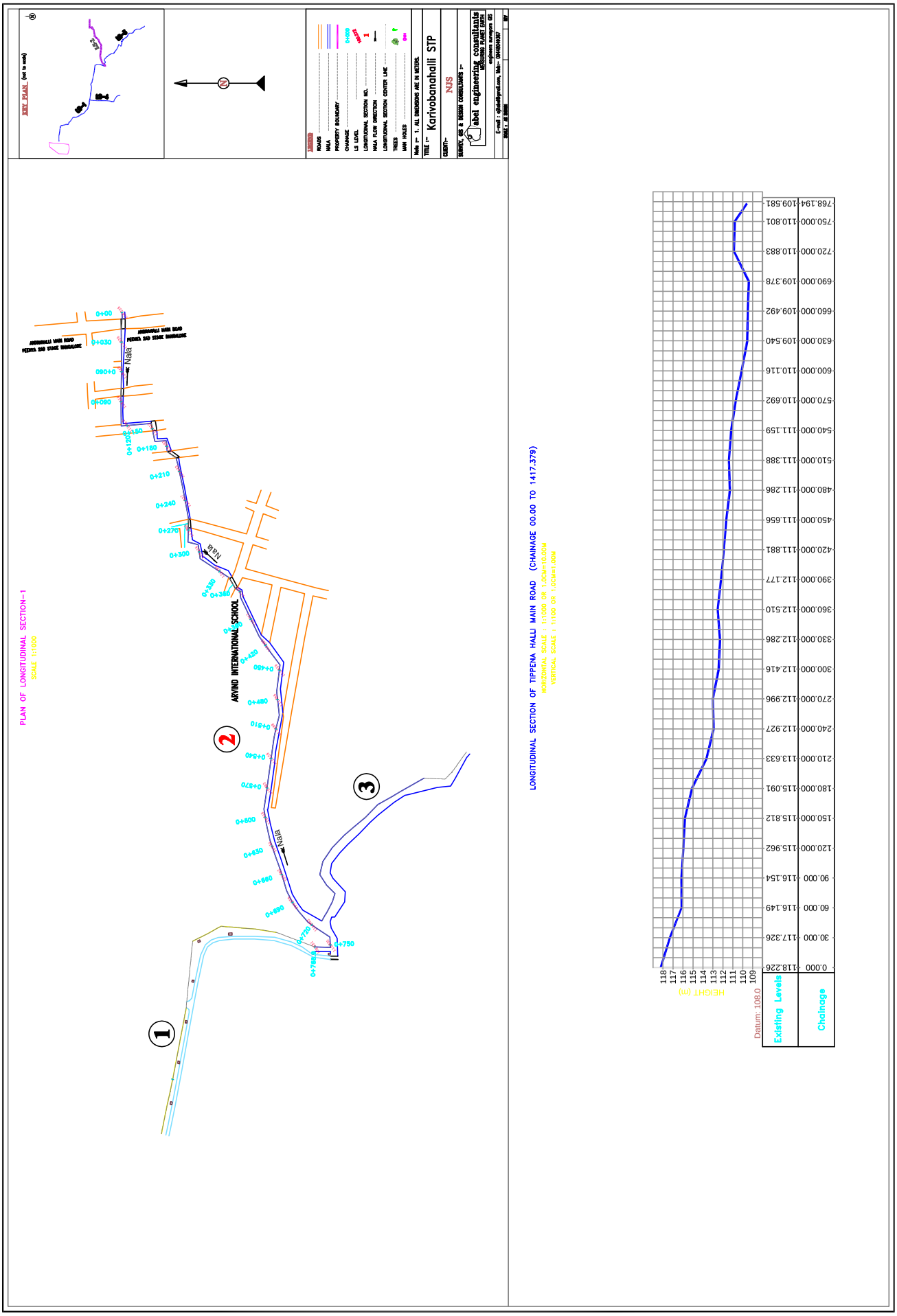

PLAN OF LONGITUDINAL SECTION-1
SCALE 1:5000
LONGITUDINAL SECTION OF TIPPENA MALU MAIN ROAD (CHAINAGE 00.00 TO 1501.397)
HORIZONTAL SCALE : 1 INCH = 100 FEET OR 1:1200
VERTICAL SCALE : 1 INCH OR 1:100
HEIGHT (m)
Datum: 88.0
Existing Levels
Chainage
NJS
engineering consultants

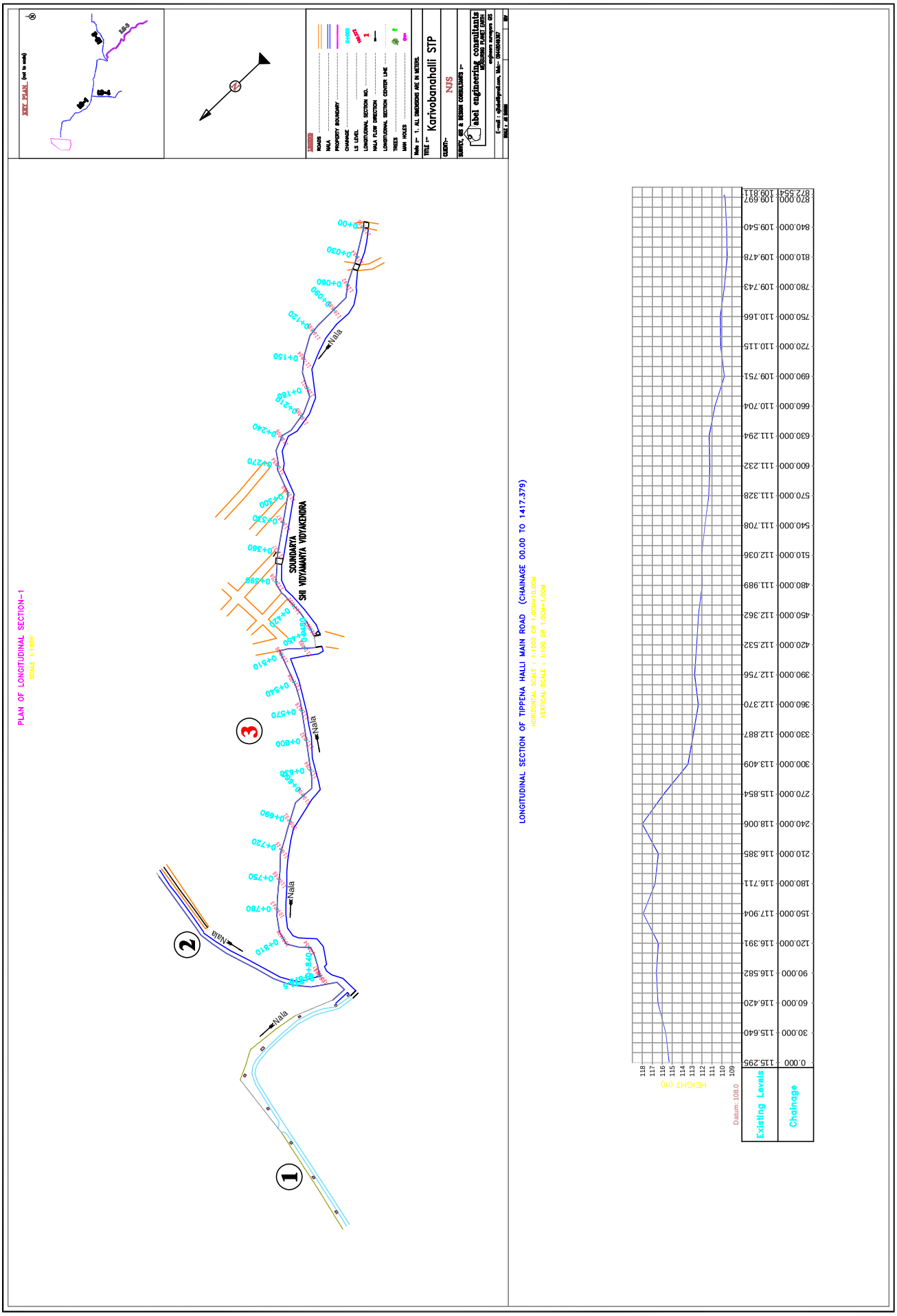

PLAN OF LONGITUDINAL SECTION-1
LONGITUDINAL SECTION OF TIPPENA MALU MAIN ROAD (CHAINAGE 00.00 TO 1709.840)
HORIZONTAL SCALE : 1000 m / 100mm
VERTICAL SCALE : 1m / 100 m / 100mm
HEIGHT (m)
Datum: 80.0
Existing Levels
Chainage


NICE RING R D
Nala
Nala
Nala
Nala
Nala
Area
1A-38.536

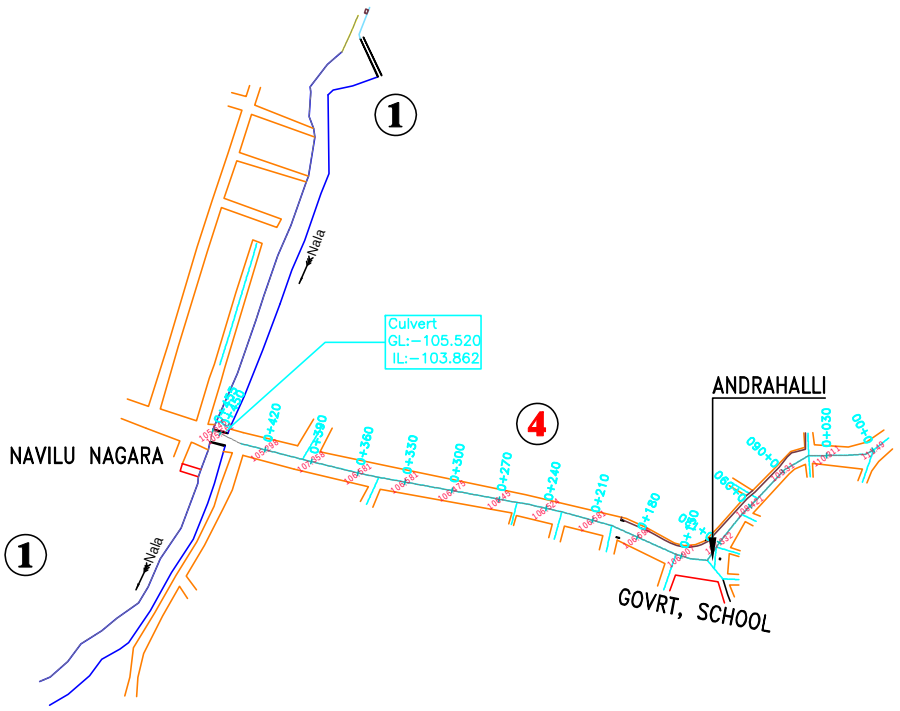

LEGEND
ROADS
NALA
PROPERTY BOUNDARY
CHAINAGE
LS LEVEL
LONGITUDINAL SECTION NO.
NALA FLOW DIRECTION
LONGITUDINAL SECTION CENTER LINE
TREES
MAN HOLES
Note :- 1. ALL DIMENSIONS ARE IN METERS.
TITLE :- Kariwobanahalli STP
CLIENT:- NJS
SURVEY, GIS & DESIGN CONSULTANTS :-
abel engineering consultants
MEASURING PLANET EARTH
engineers surveyors GIS
E-mail : ajitabel@gmail.com, Mob:- 09448049367
SCALE 1:1000
REV
ANDRAHALLI MAIN ROAD
PEENYA 2ND STAGE BANGALORE
NAVILU NAGARA
GOVRT. SCHOOL
ANDRAHALLI
AREA
Area=7A-20.356

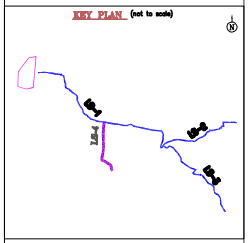

PLAN OF LONGITUDINAL SECTION-1
SCALE 1:1000
LONGITUDINAL SECTION OF TIPPENA HALLI MAIN ROAD (CHAINAGE 00.00 TO 1417.379)
HORIZONTAL SCALE : 1:1000 OR 1 INCH=1000
VERTICAL SCALE : 1:100 OR 1 INCH=1.0000
HEIGHT (m)
Datum: 87.0
Existing Levels
Chainage
Karivobanahalli STP
NJS
Nala
AREA
Area=79.-20.300

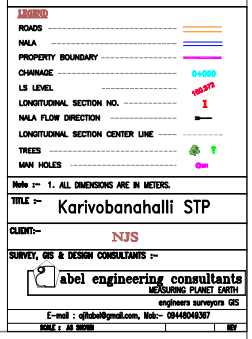

KEY PLAN (not to scale)
SYMBOLS
ROADS
NALA
PROPERTY BOUNDARY
CHAINAGE
LS LEVEL
LONGITUDINAL SECTION NO.
NALA FLOW DIRECTION
LONGITUDINAL SECTION CENTER LINE
TREES
MAN HOLES
Note :- 1. ALL DIMENSIONS ARE IN METERS.
TITLE :- Karivobanahalli STP
CLIENT:-
NJS
SURVEY, GIS & DESIGN CONSULTANTS :-
abel engineering consultants
E-mail : abelbe@gmail.com, Mob:- 9448044367
PLAN OF LONGITUDINAL SECTION-1
SCALE 1:1000
ARVIND INTERNATIONAL SCHOOL
Nala
LONGITUDINAL SECTION OF TIPPENA HALLI MAIN ROAD (CHAINAGE 00.00 TO 1417.379)
HORIZONTAL SCALE : 1:1000 OR 1.00cm=10.00M
VERTICAL SCALE : 1:100 OR 1.00cm=1.00M
HEIGHT (m)
Datum: 108.0
Existing Levels
Chainage


KEY PLAN (not to scale)
LEGEND
ROADS
NALA
PROPERTY BOUNDARY
CHAINAGE
LS LEVEL
LONGITUDINAL SECTION NO.
NALA FLOW DIRECTION
LONGITUDINAL SECTION CENTER LINE
TREES
MAN HOLES
Note :- 1. ALL DIMENSIONS ARE IN METERS.
TITLE :- Karivobanahalli STP
CLIENT:-
NJS
abel engineering consultants
PLAN OF LONGITUDINAL SECTION-1
SCALE 1:1000
SOUNDARYA
SHI VIDYAMANYA VIDYAKENDRA
Nala
LONGITUDINAL SECTION OF TIPPENA HALLI MAIN ROAD (CHAINAGE 00.00 TO 1417.379)
HORIZONTAL SCALE : 1:1000 OR 1.0CM=10.00M
VERTICAL SCALE : 1:100 OR 1.0CM=1.00M
HEIGHT (m)
Datum: 108.0
Existing Levels
Chainage

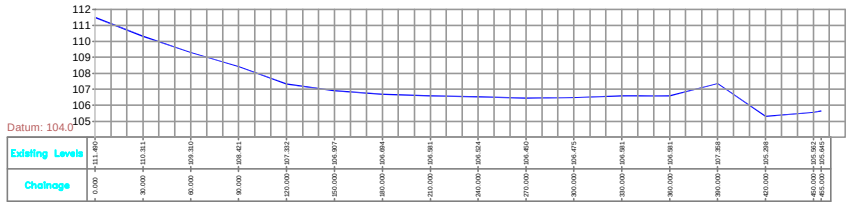
PLAN OF LONGITUDINAL SECTION-1
SCALE 1:1000

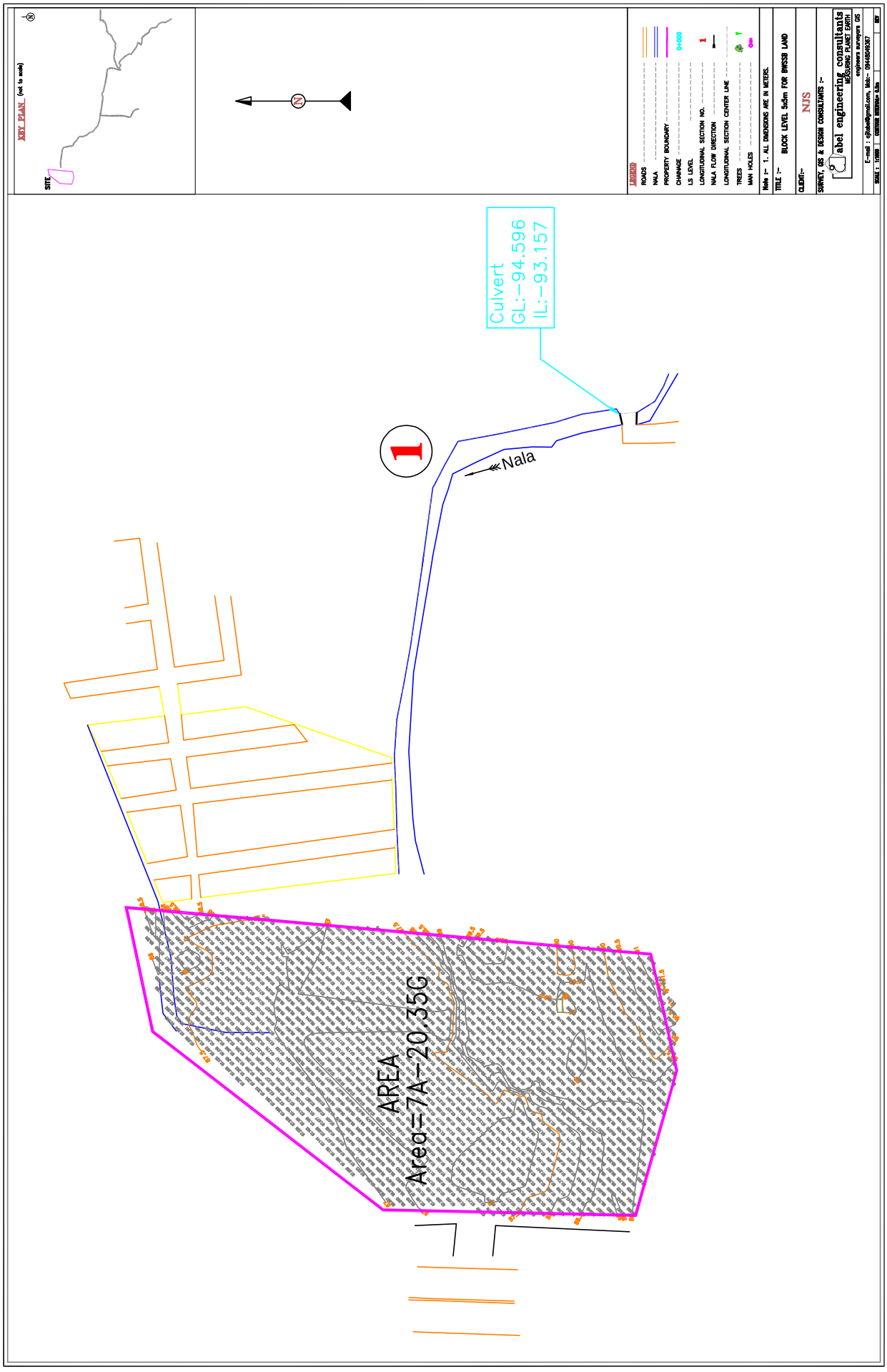


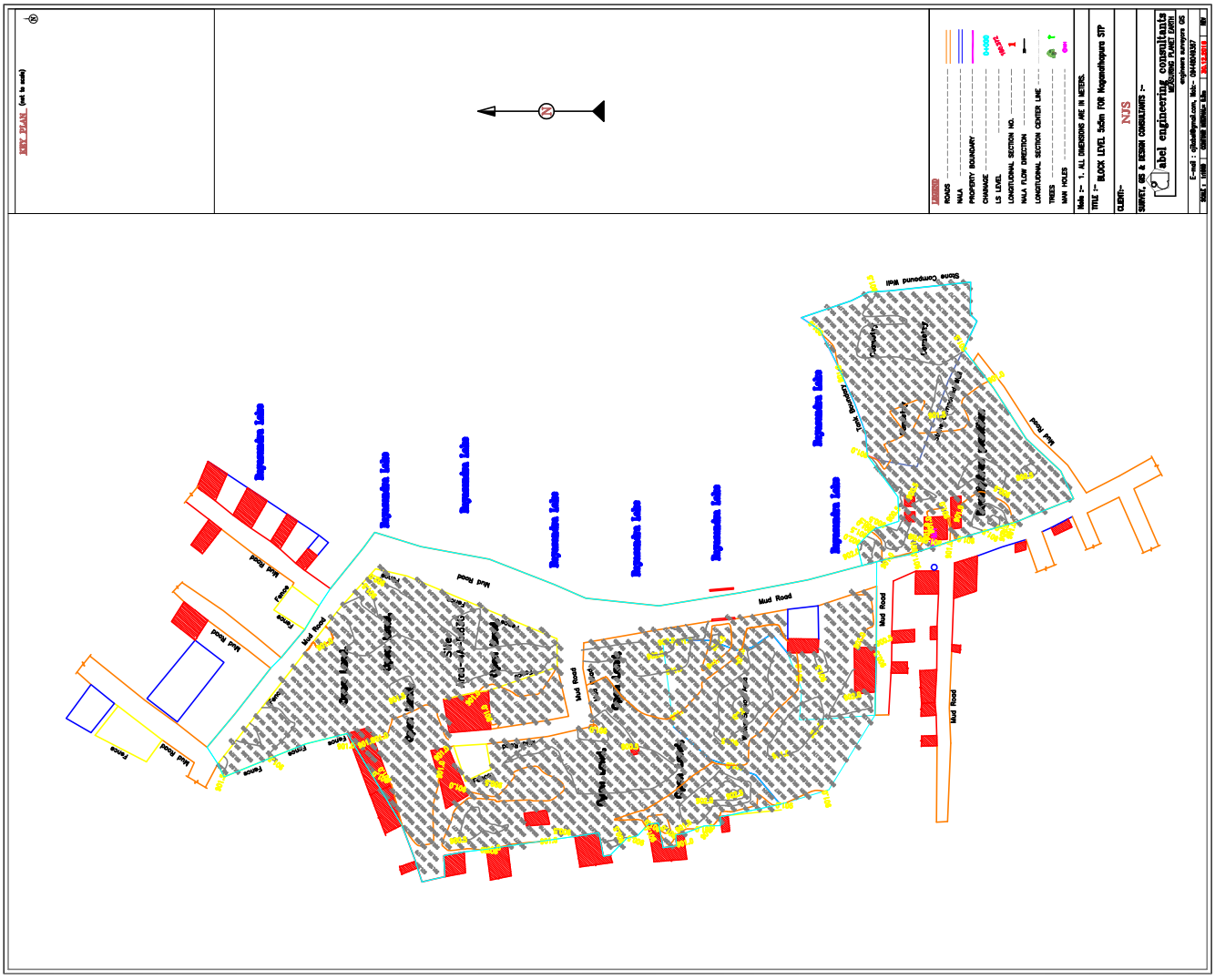


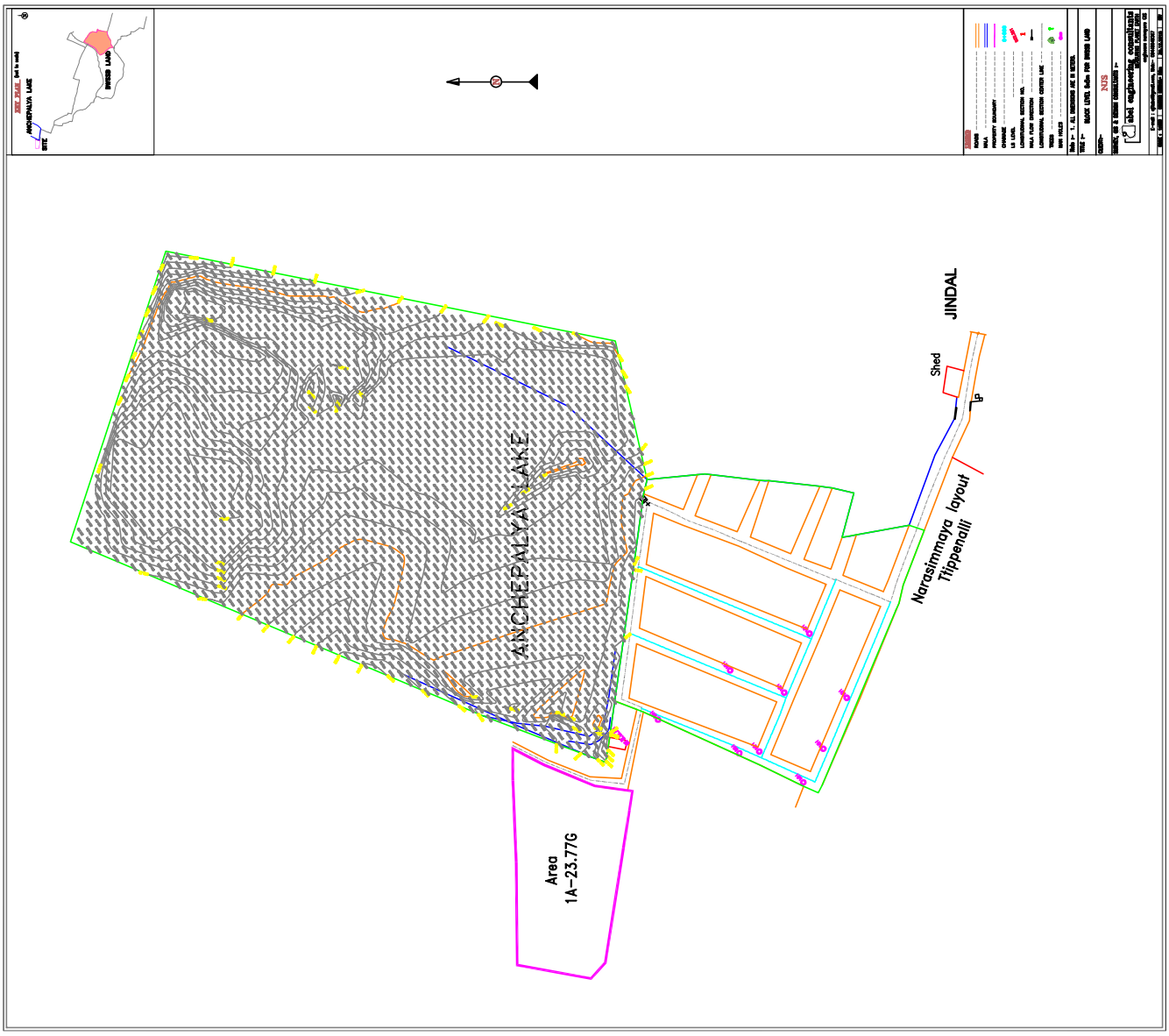


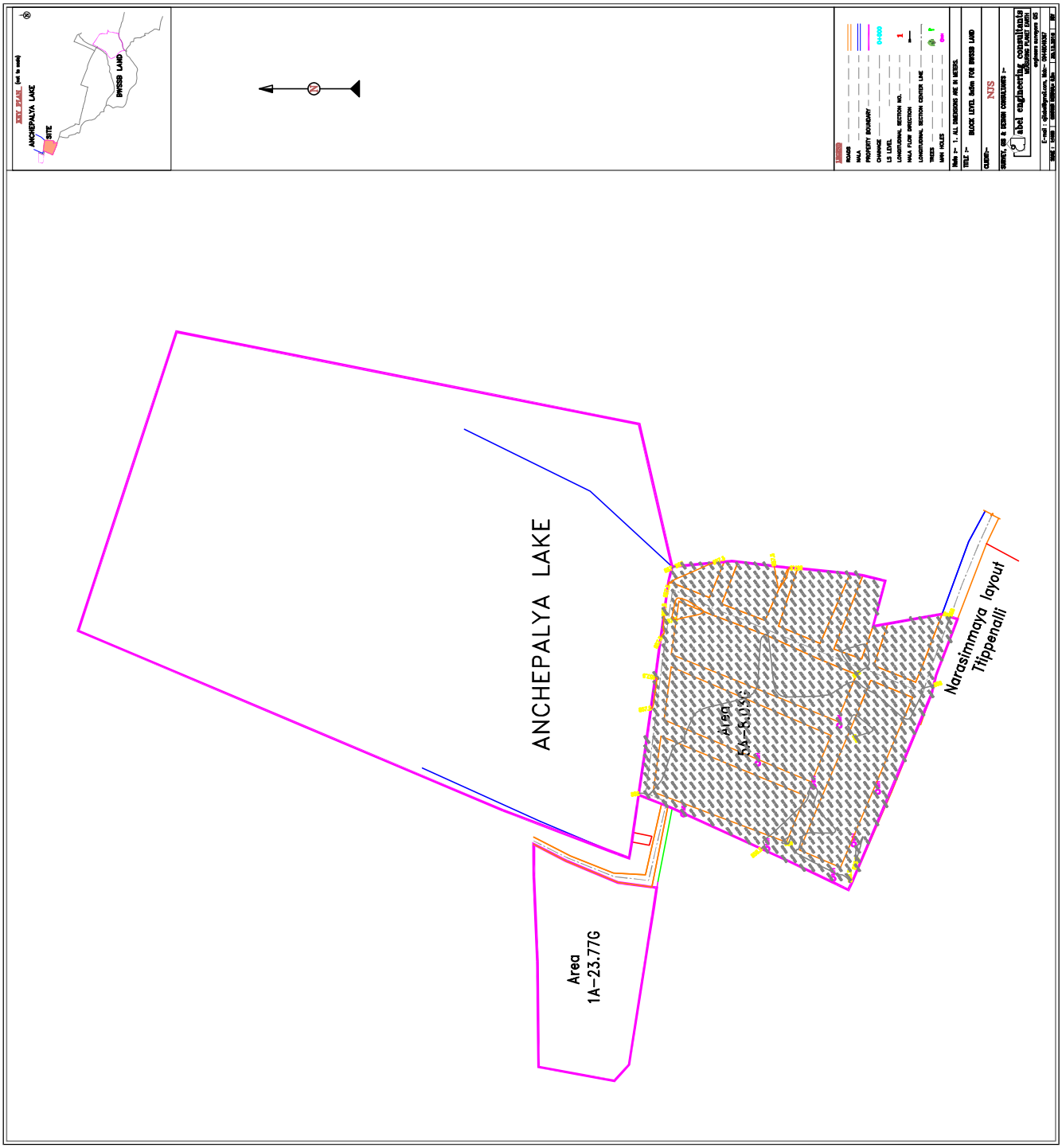
LONGITUDINAL SECTION OF TIPPENA HALLI MAIN ROAD (CHAINAGE 00.00 TO 1417.379)
HORIZONTAL SCALE : 1:1000 OR 1.0CM=10.00M
VERTICAL SCALE : 1:100 OR 1.0CM=1.00M

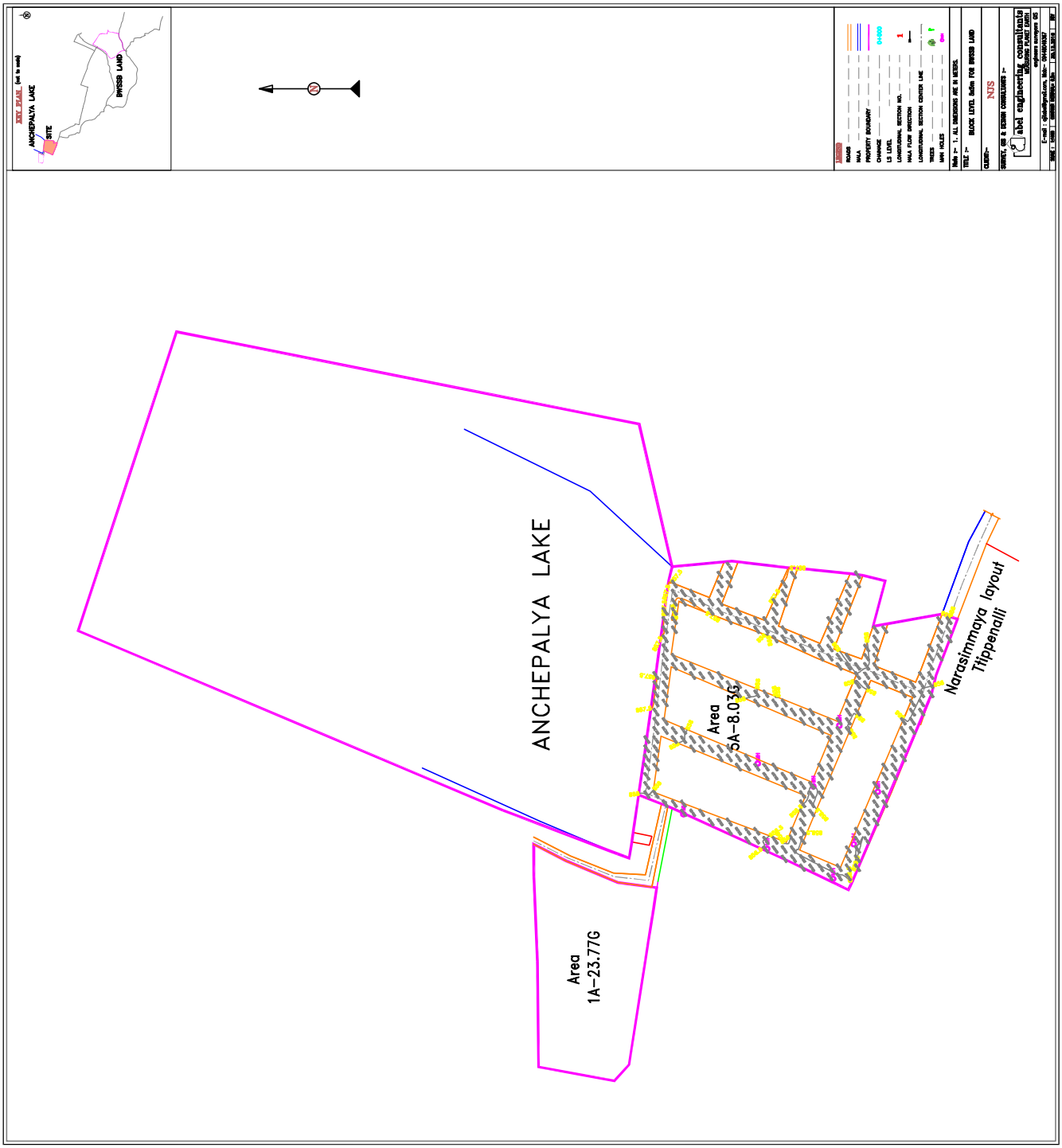


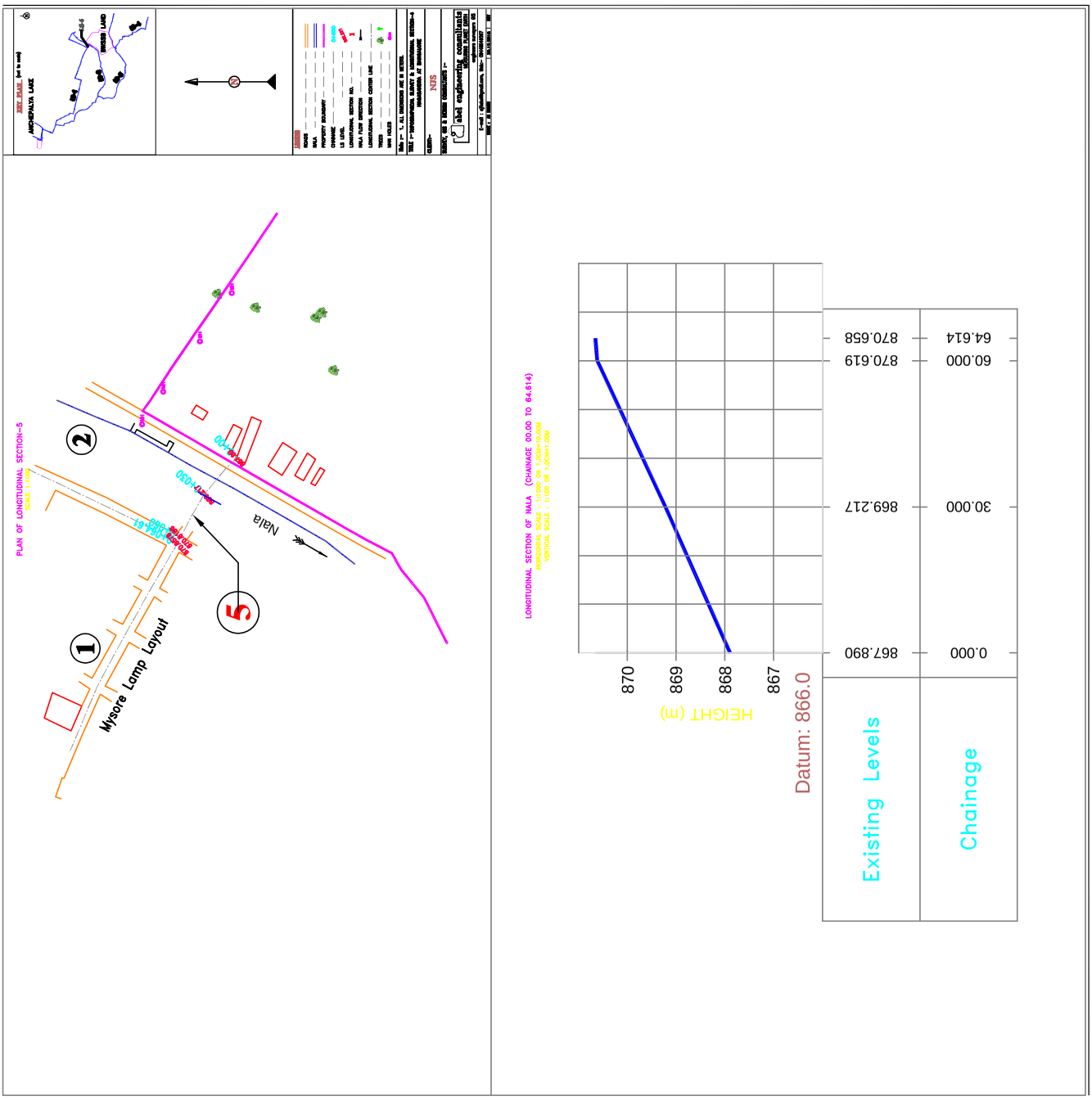

KEY PLAN (not to scale)
SITE
N
LEGEND
ROADS
NALA
PROPERTY BOUNDARY
CHAINAGE
LS LEVEL
LONGITUDINAL SECTION NO.
NALA FLOW DIRECTION
LONGITUDINAL SECTION CENTER LINE
TREES
MAN HOLES
Note :- 1. ALL DIMENSIONS ARE IN METERS.
TITLE :- BLOCK LEVEL 5x5m FOR BWSSB LAND
CLIENT:- NJS
SURVEY, GIS & DESIGN CONSULTANTS :-
abel engineering consultants
engineers surveyors GIS
Culvert
GL:-94.596
IL:-93.157
Nala
AREA
Area=7A-20.35G

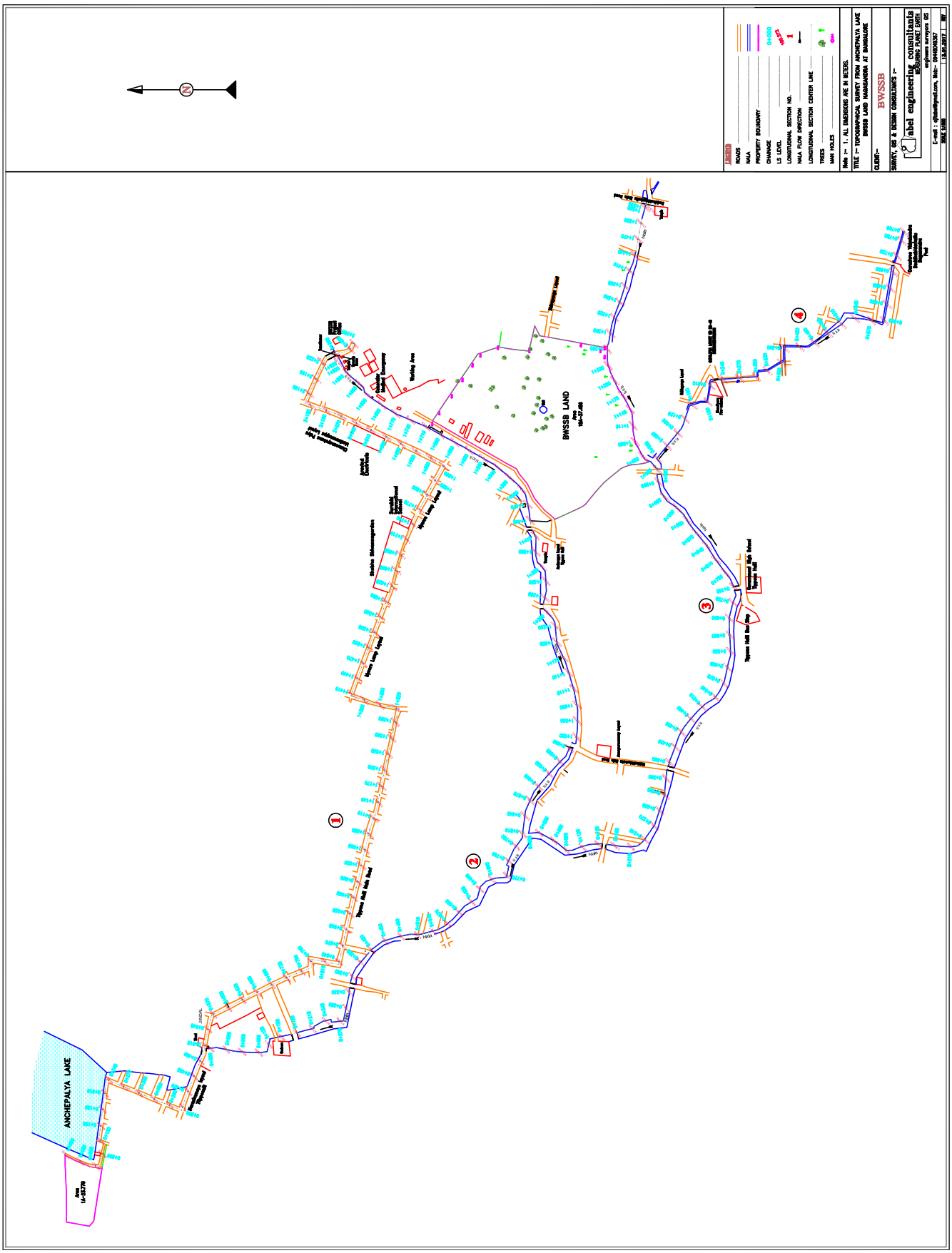

KEY PLAN (not to scale)
LEGEND
ROADS
NALA
PROPERTY BOUNDARY
CHAINAGE
LS LEVEL
LONGITUDINAL SECTION NO.
NALA FLOW DIRECTION
LONGITUDINAL SECTION CENTER LINE
TREES
MAN HOLES
Note :- 1. ALL DIMENSIONS ARE IN METERS.
TITLE :- BLOCK LEVEL Scale FOR Nagarathnapura STP
CLIENT:-
NJS
SURVEY, GIS & DESIGN CONSULTANTS :-
abel engineering consultants
engineers surveyors GIS
Shona Compound Wall
Agrahara Lake
Mud Road
Open Lands
Site
Fence
Pond Lands
Cultivated Lands

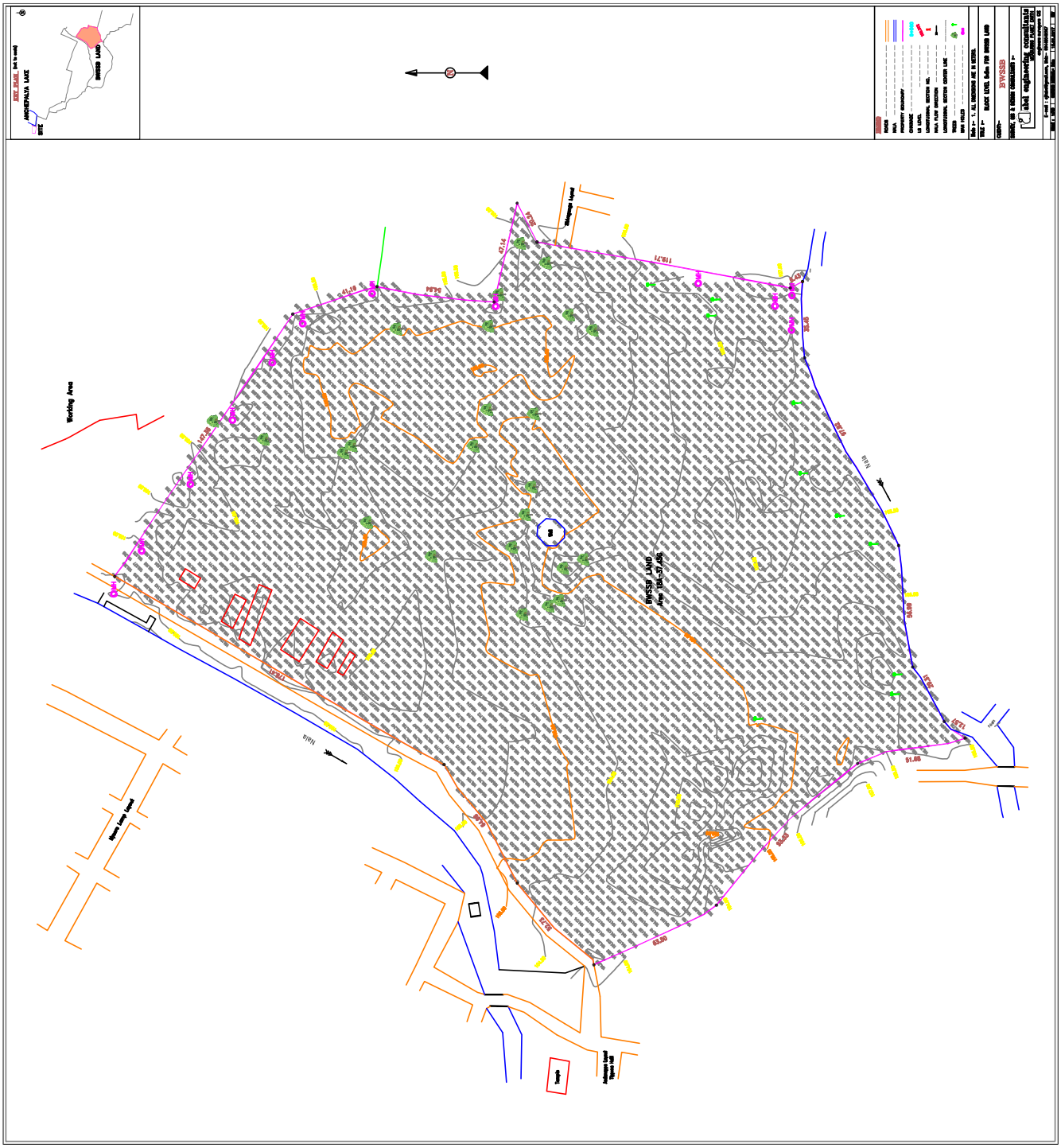

ANCHEPALYA LAKE
JINDAL
Shed
Narasimmaya layout
Tiippenalli
Area
1A-23.77G

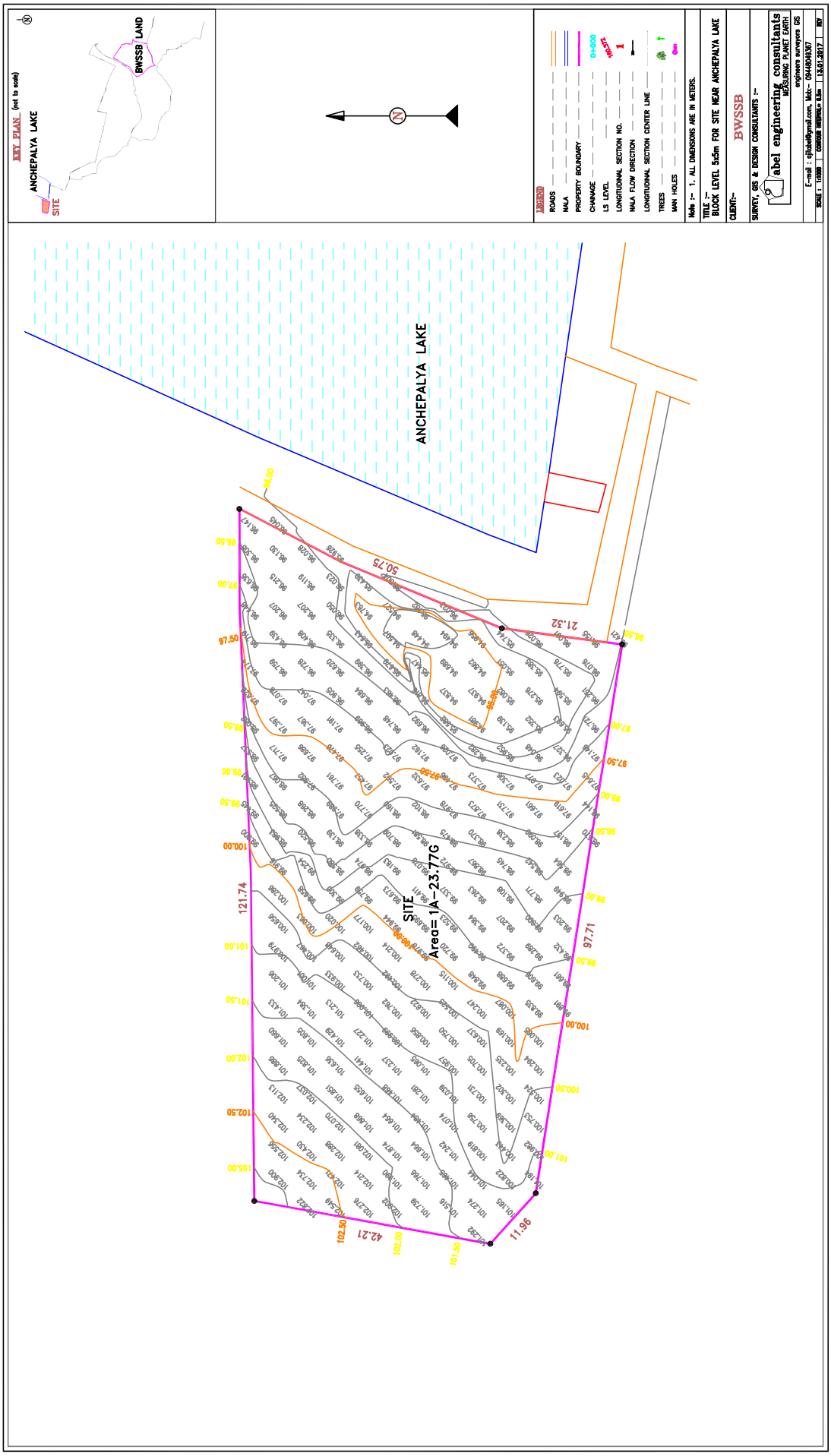

KEY PLAN
ANCHEPALYA LAKE
SITE
BWSSB LAND
ANCHEPALYA LAKE
Area
1A-23.77G
Area
5A-8.05G
Narasimmaya layout
Thippenalli

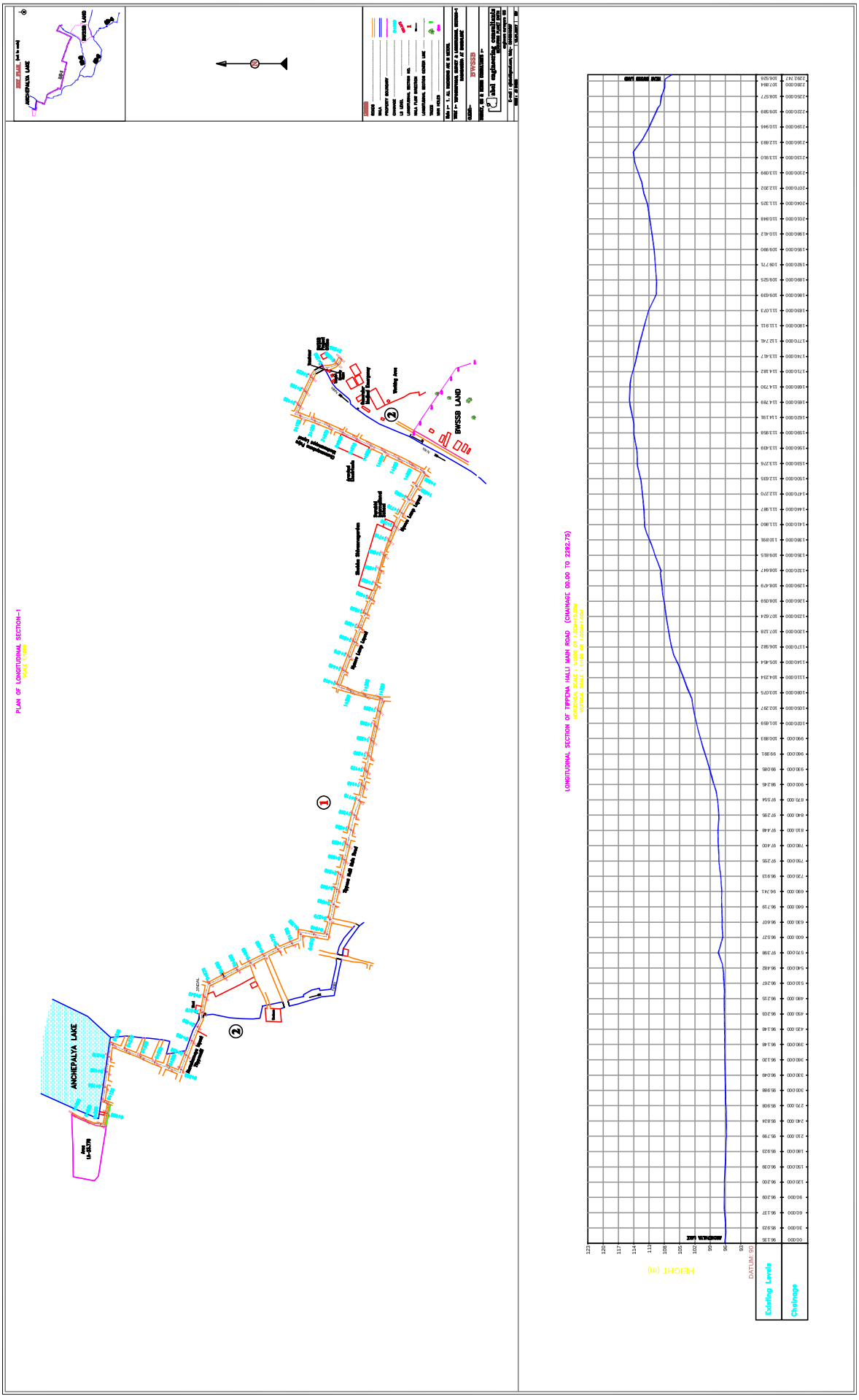

KEY PLAN (not to scale)
ANCHEPALYA LAKE
SITE
BWSSB LAND
ANCHEPALYA LAKE
Area
5A-8.03G
Area
1A-23.77G
Narasimmaya layout
Thippenalli

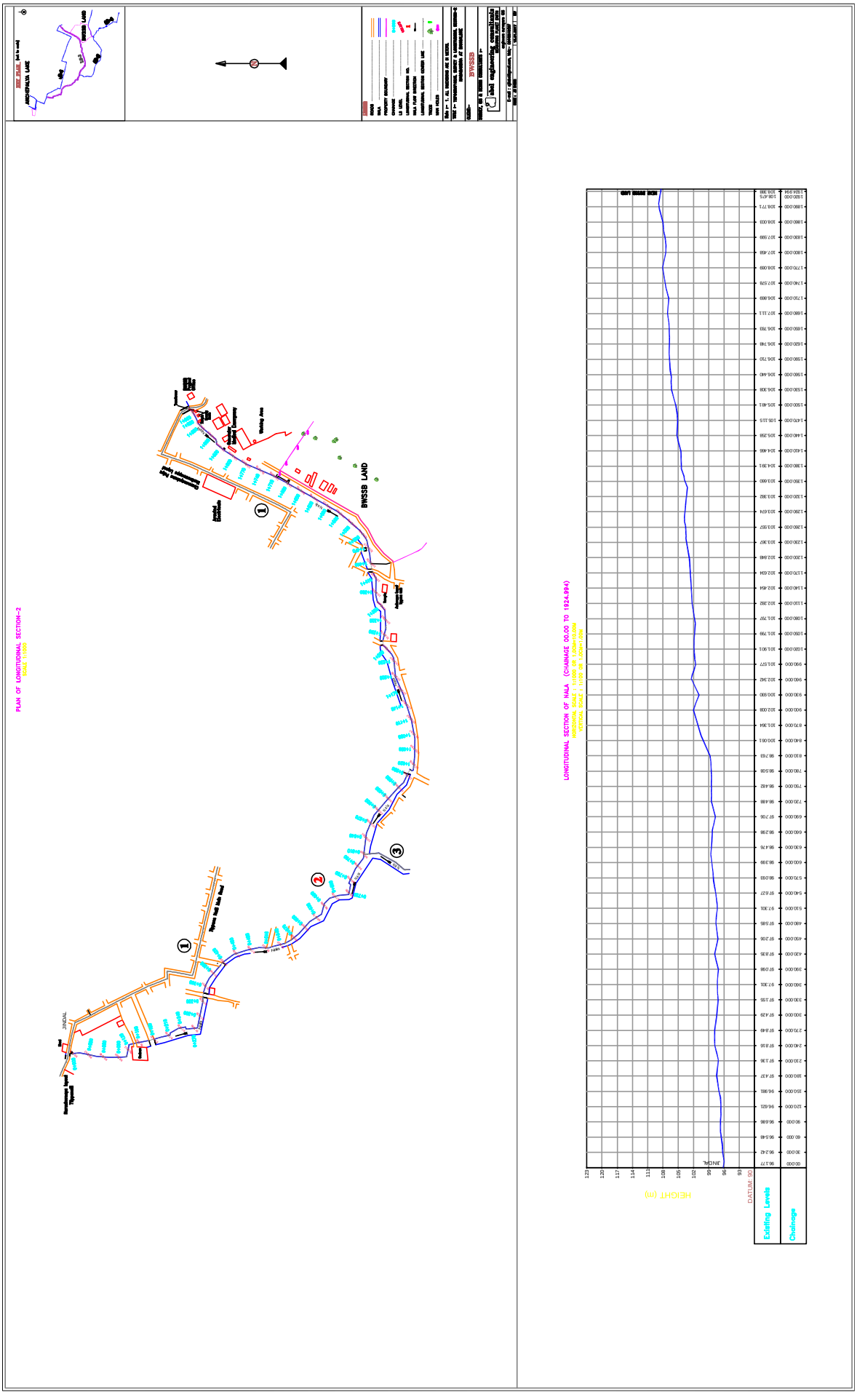

PLAN OF LONGITUDINAL SECTION-5
SCALE 1:1000
Mysore Lamp Layout
Nala
1
2
5
LONGITUDINAL SECTION OF NALA (CHAINAGE 00.00 TO 64.614)
HEIGHT (m)
870
869
868
867
Datum: 866.0
Existing Levels
Chainage
867.890
869.217
870.619
870.658
0.000
30.000
60.000
64.614

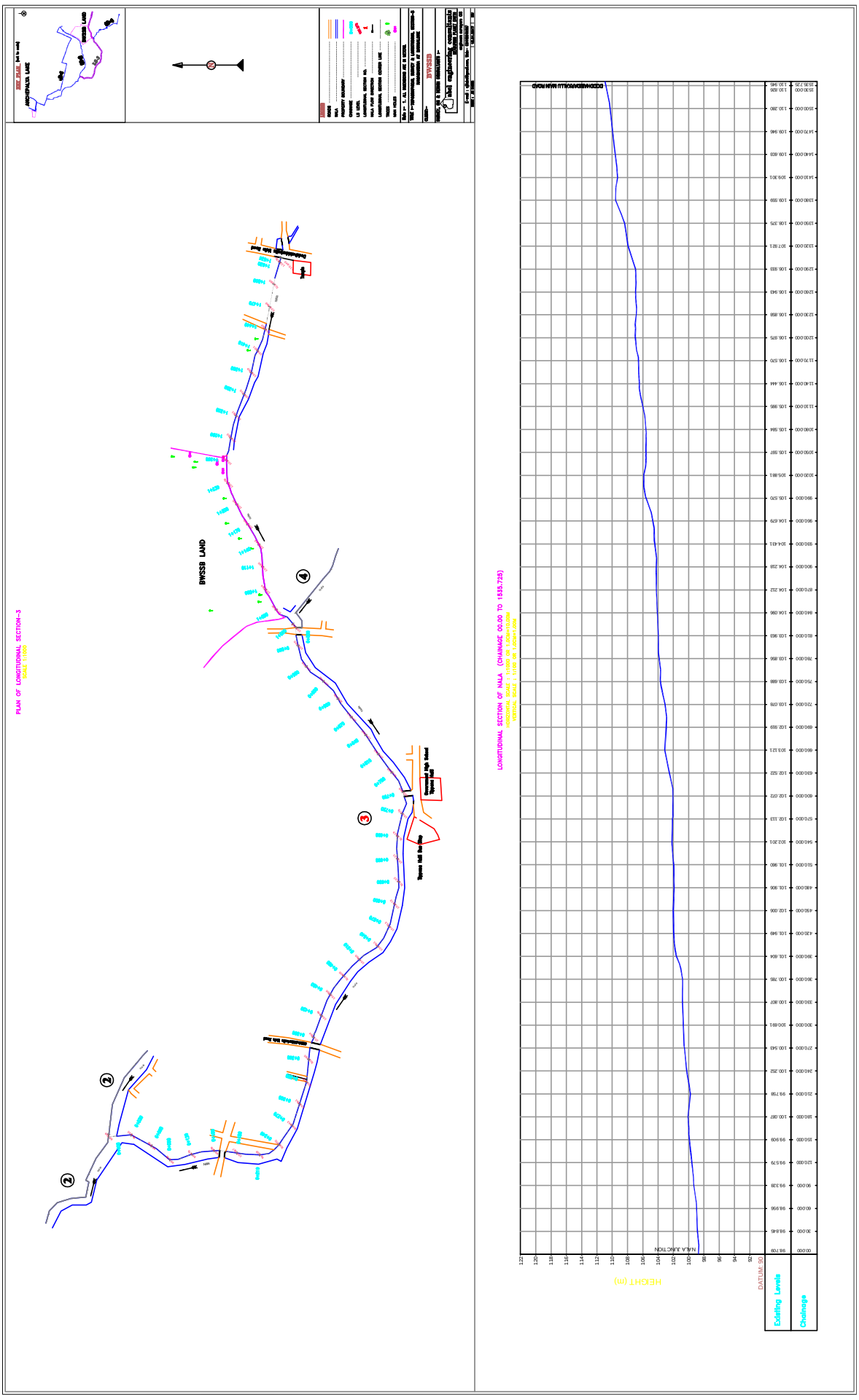

ROADS
NALA
PROPERTY BOUNDARY
CHAINAGE
LS LEVEL
LONGITUDINAL SECTION NO.
NALA FLOW DIRECTION
LONGITUDINAL SECTION CENTER LINE
TREES
MAN HOLES
Note :- 1. ALL DIMENSIONS ARE IN METERS.
TITLE :- TOPOGRAPHICAL SURVEY FROM ANCHEPALYA LAKE BWSSB LAND NAGAVARDHA AT BANGALORE
CLIENT:- BWSSB
SURVEY, GIS & DESIGN CONSULTANTS :-
abel engineering consultants
ANCHEPALYA LAKE
BWSSB LAND
Nagavara Main Road
Nagavara Lower Road
Nagavara Valley Road
Nagavara Hall Road
Nagavara Hall Jn
Nagavara Hall School
Nagavara Hall Bus Stop
Nagavara Hall Temple
Nagavara Hall Jn
Nagavara Hall Bus Stop
Nagavara Hall Temple

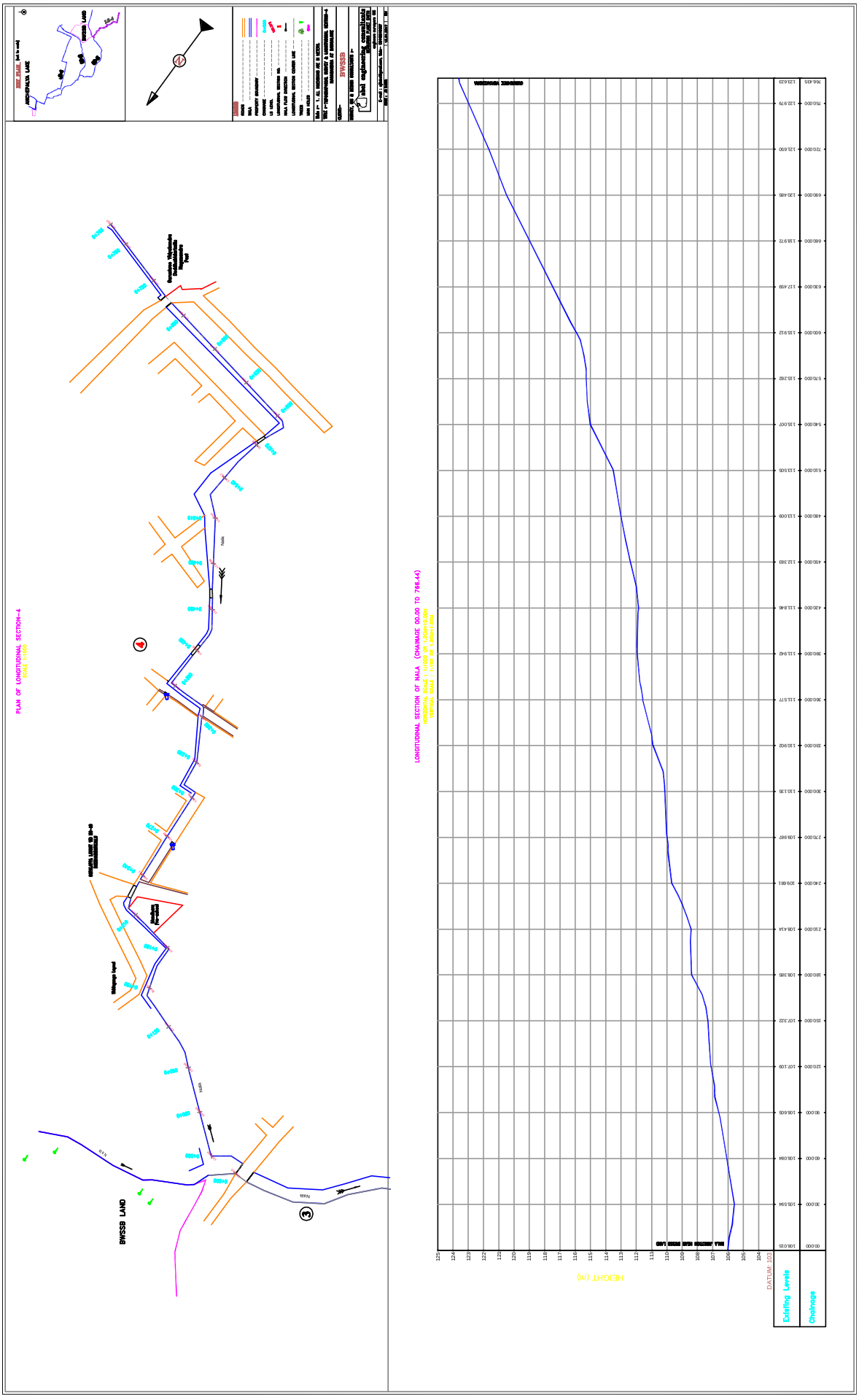

Working Area
BWSSB LAND
Area 154-37-456
Nala
Nala
Water Loss Land
Water Loss
Tank

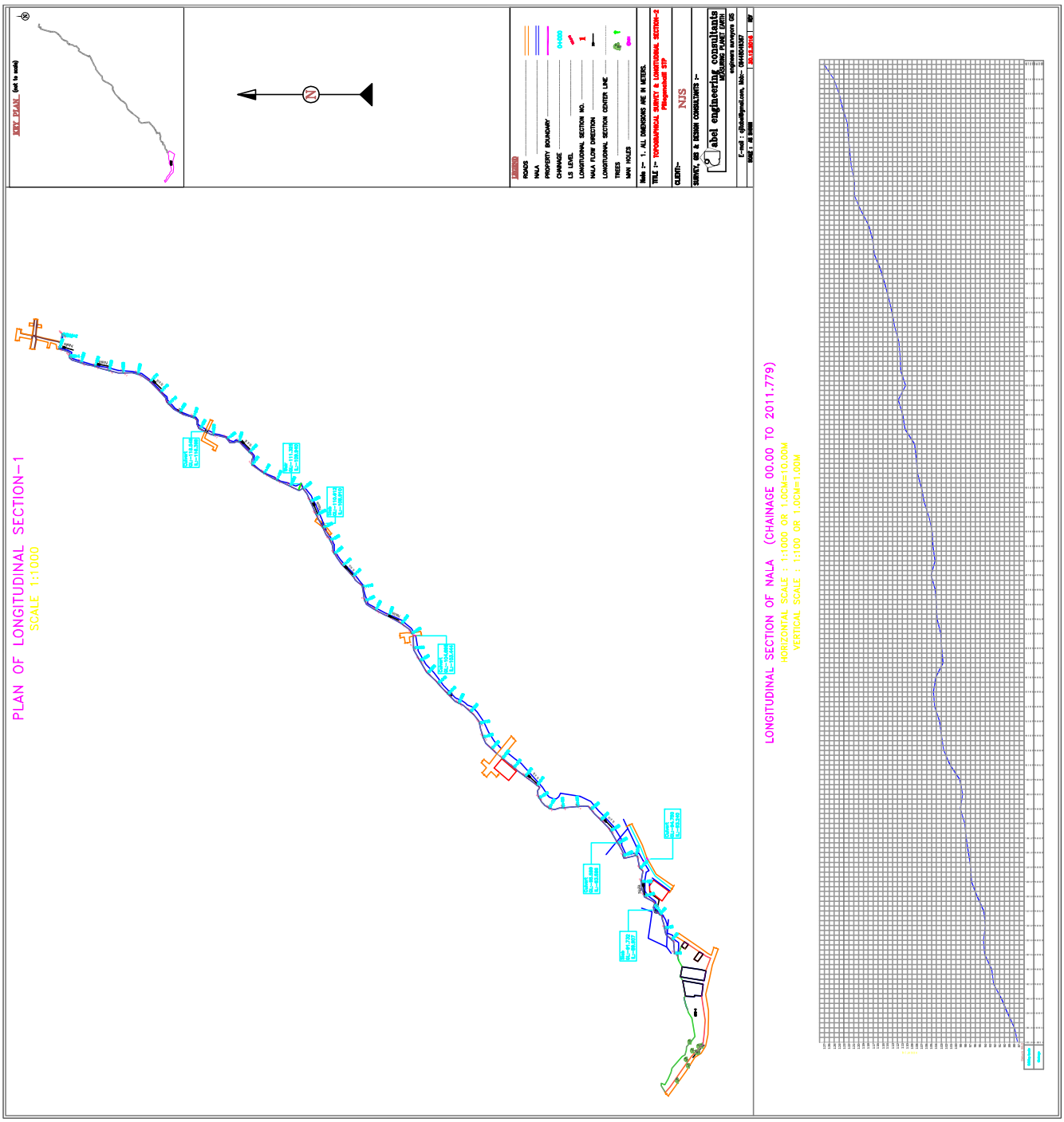

KEY PLAN (not to scale)
ANCHEPALYA LAKE
BWSSB LAND
SITE
LEGEND
ROADS
NALA
PROPERTY BOUNDARY
CHAINAGE
LS LEVEL
LONGITUDINAL SECTION NO.
NALA FLOW DIRECTION
LONGITUDINAL SECTION CENTER LINE
TREES
MAN HOLES
Note :- 1. ALL DIMENSIONS ARE IN METERS.
TITLE :-
BLOCK LEVEL 5x5m FOR SITE NEAR ANCHEPALYA LAKE
CLIENT:-
BWSSB
SURVEY, GIS & DESIGN CONSULTANTS :-
abel engineering consultants
MEASURING PLANET EARTH
engineers surveyors GIS
E-mail : njlabel@gmail.com, Mob:- 09448048367
SCALE : 1:500
CONTOUR INTERVAL= 0.5m
15.01.2017
REV
ANCHEPALYA LAKE
SITE
Area= 1A-23.77G
50.75
21.32
121.74
97.71
42.21
11.96

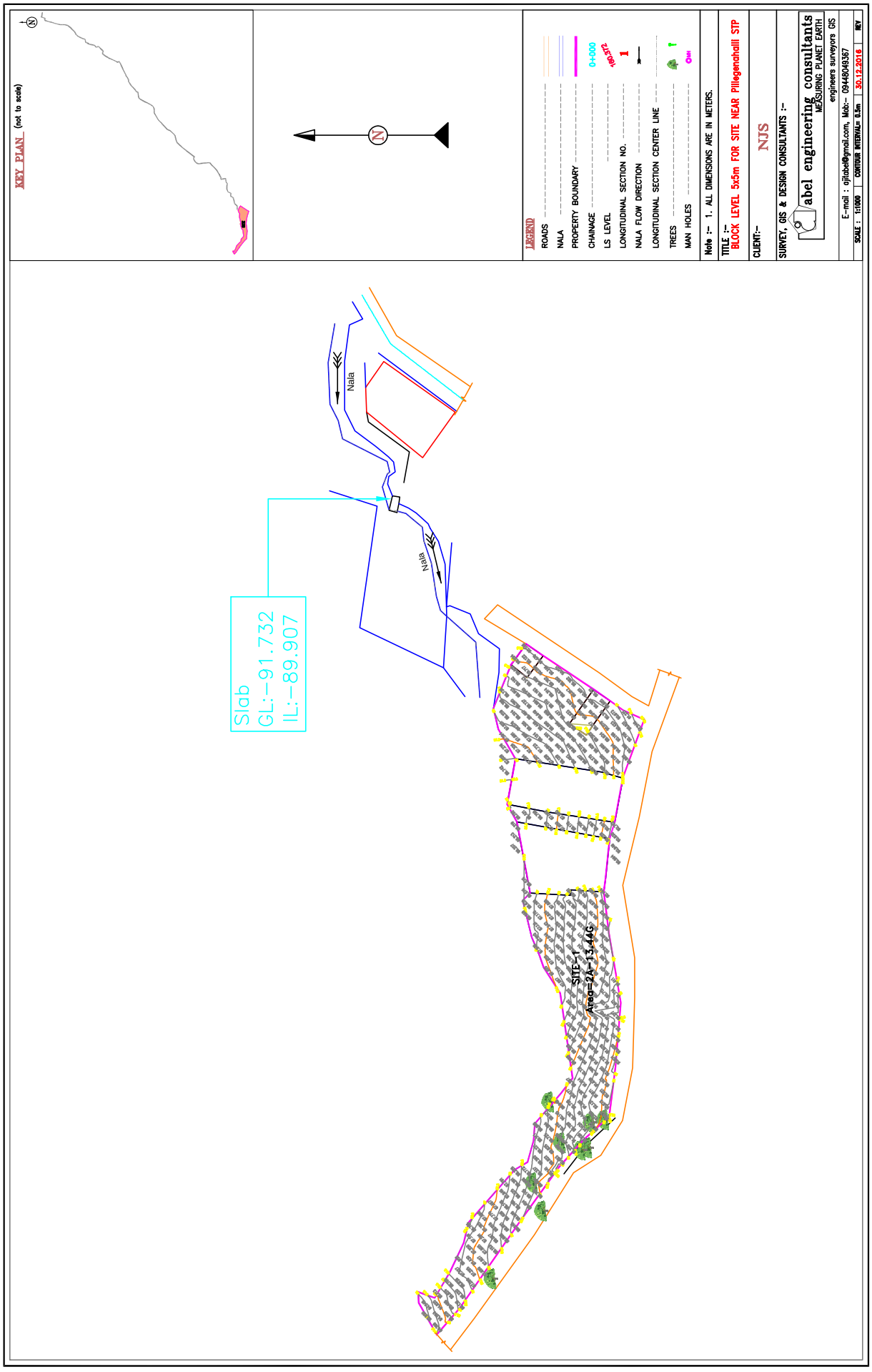

PLAN OF LONGITUDINAL SECTION-1
LONGITUDINAL SECTION OF TRIPPENA HALLI MAIN ROAD (CHAINAGE 00.00 TO 2282.75)
ANCHEPALYA LAKE
BWSSB LAND
Existing Levels
Chainage
HEIGHT (m)
DATUM (M)

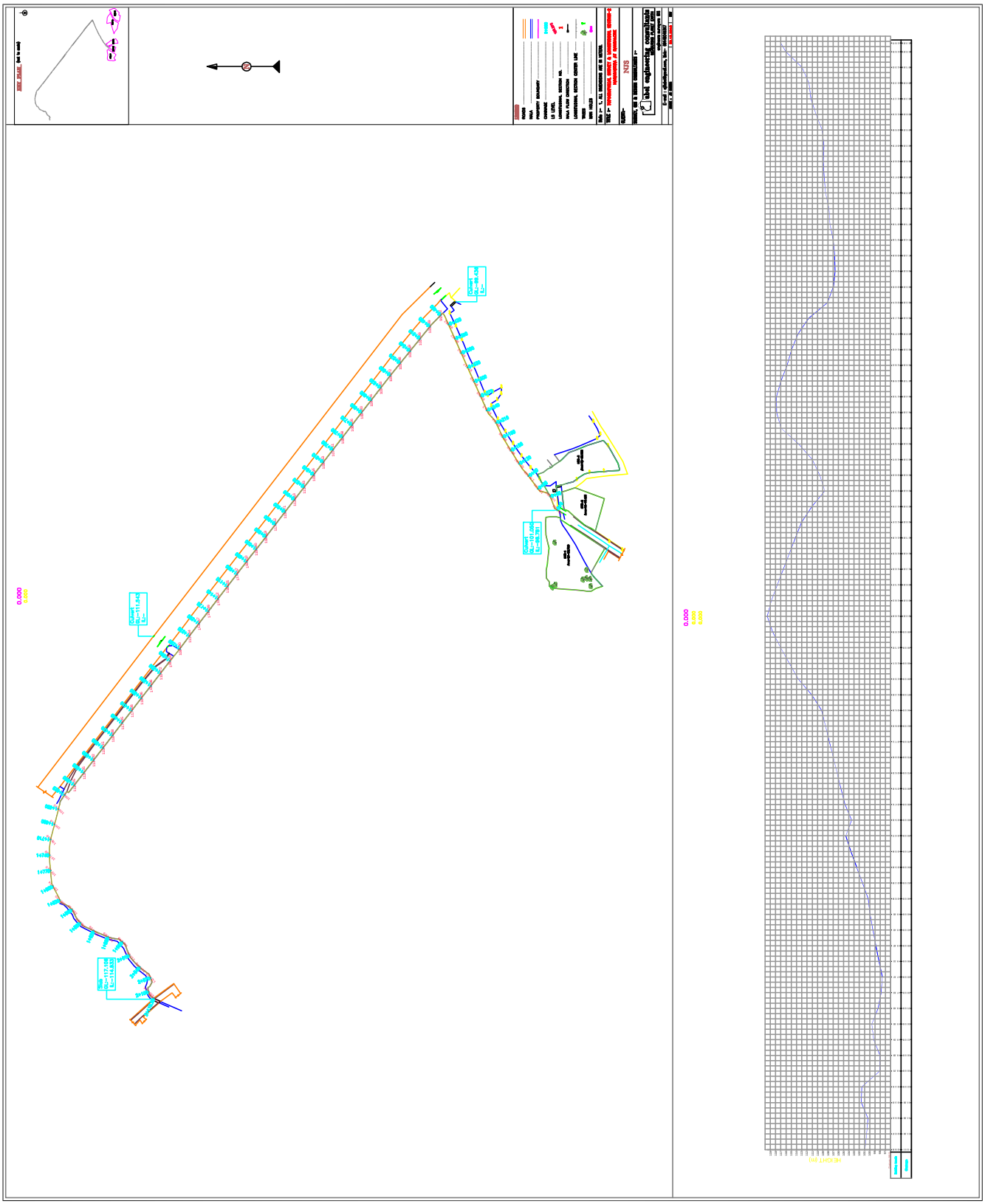

PLAN OF LONGITUDINAL SECTION-2
BWSSB LAND
JINDAL
LONGITUDINAL SECTION OF NALA (CHAINAGE 00.00 TO 1924.994)
HEIGHT (m)
DATUM (m)
Existing Levels
Chainage

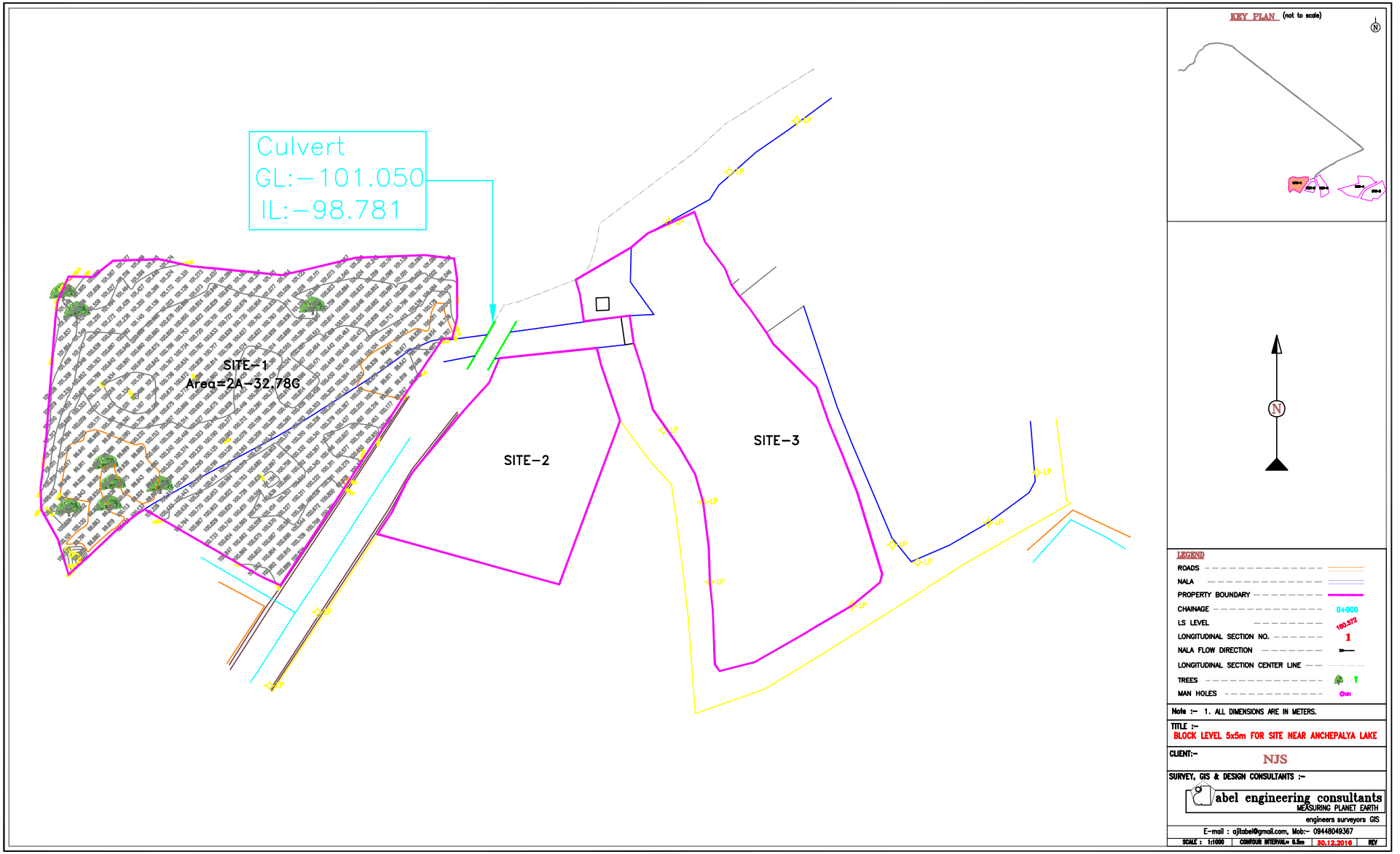

PLAN OF LONGITUDINAL SECTION-3
BWSSB LAND
LONGITUDINAL SECTION OF NALA (CHAINAGE 00.00 TO 1535.725)
HEIGHT (m)
NALA JUNCTION
DATUM : 90
Existing Levels
Chainage

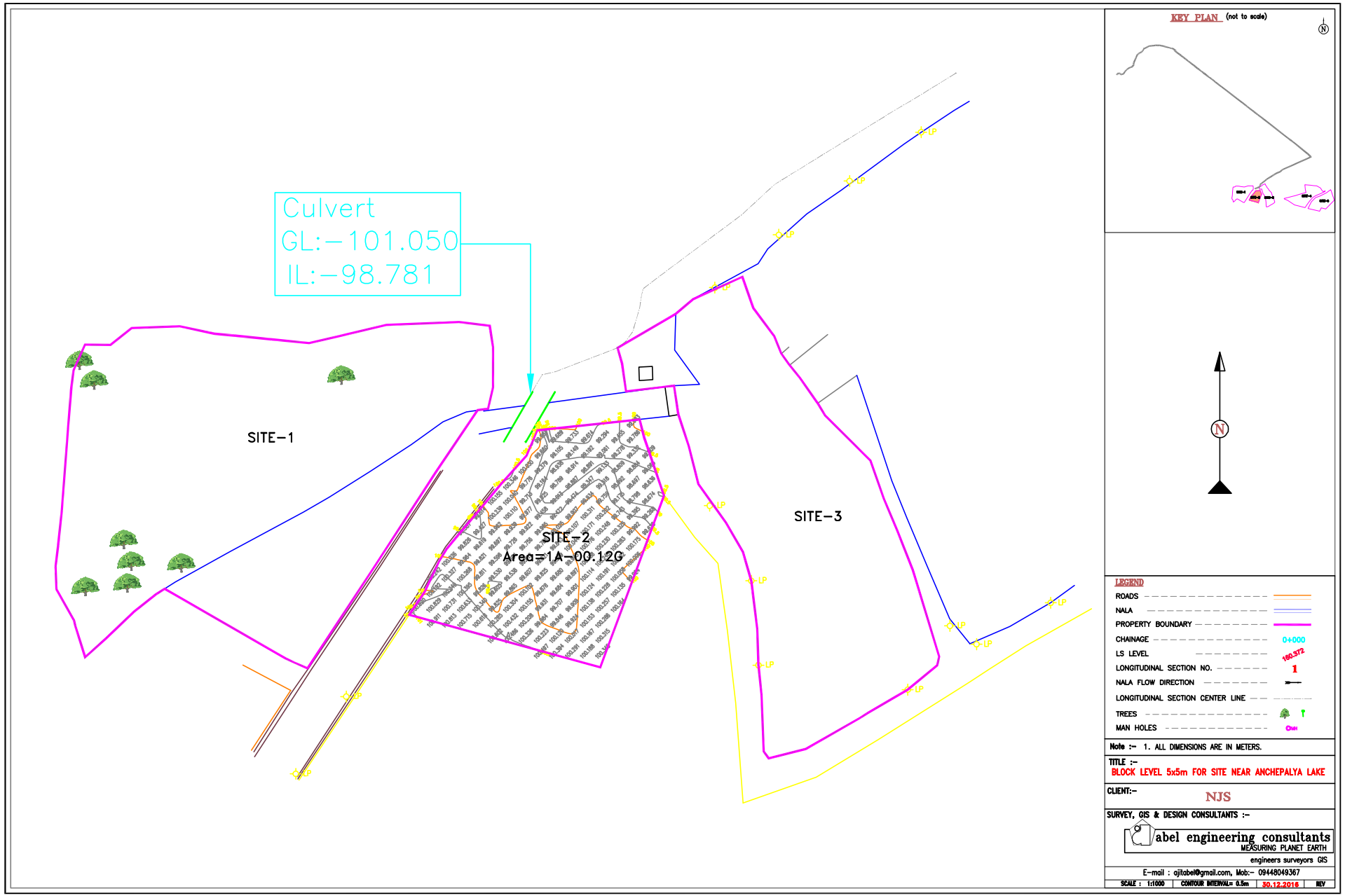

PLAN OF LONGITUDINAL SECTION-4
LONGITUDINAL SECTION OF NALA (CHAINAGE 00.00 TO 788.44)
BWSSB LAND
Nala
HEIGHT (m)
DATUM-103
Existing Levels
Chainage

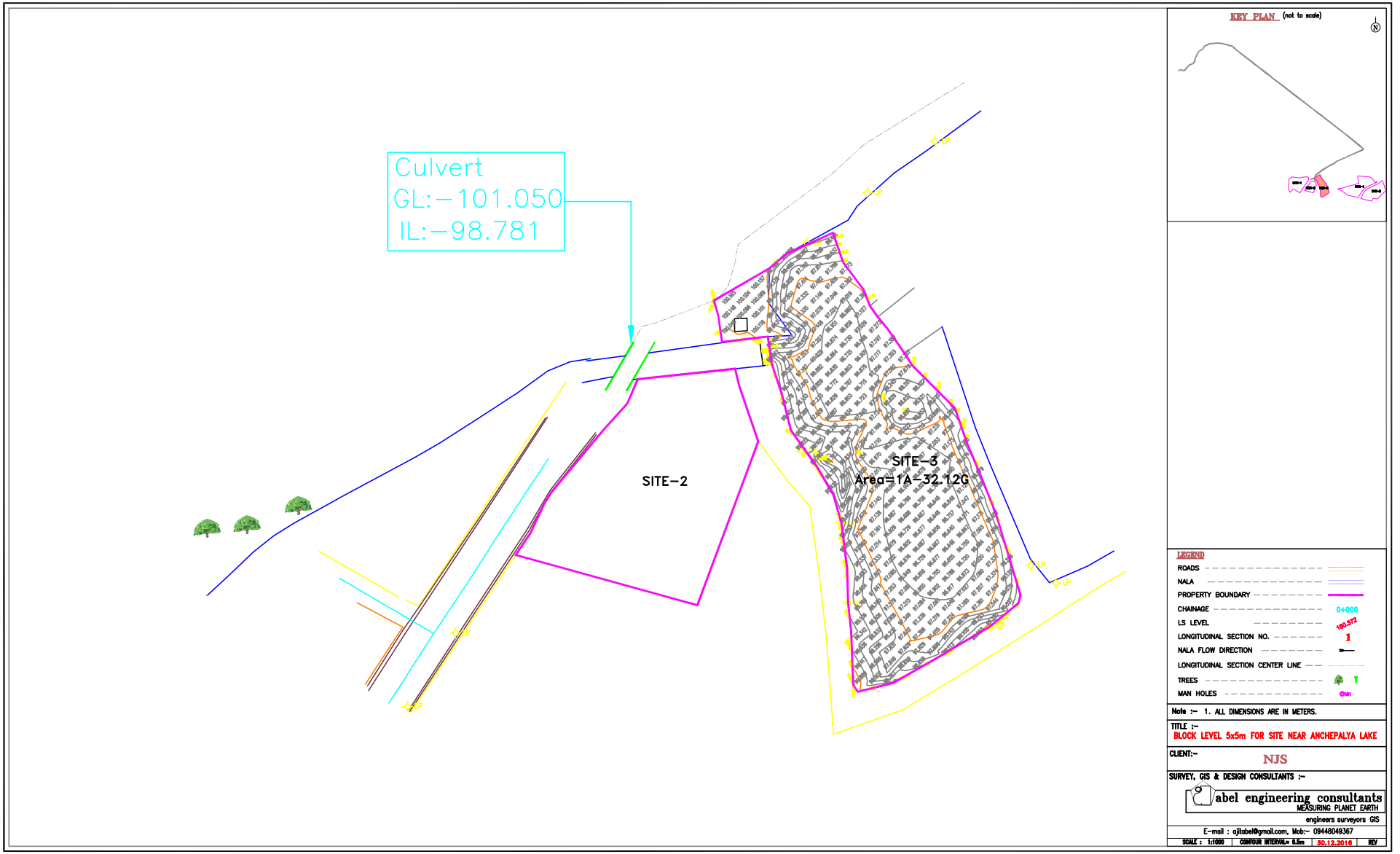

KEY PLAN (not to scale)
LEGEND
ROADS
NALA
PROPERTY BOUNDARY
CHAINAGE
LS LEVEL
LONGITUDINAL SECTION NO.
NALA FLOW DIRECTION
LONGITUDINAL SECTION CENTER LINE
TREES
MAN HOLES
Note :- 1. ALL DIMENSIONS ARE IN METERS.
TITLE :- TOPOGRAPHICAL SURVEY & LONGITUDINAL SECTION-2
Pillegumbodi STP
CLIENT:-
NJS
SURVEY, GIS & DESIGN CONSULTANTS :-
abel engineering consultants
engineers surveyors GIS
PLAN OF LONGITUDINAL SECTION-1
SCALE 1:1000
LONGITUDINAL SECTION OF NALA (CHAINAGE 00.00 TO 2011.779)
HORIZONTAL SCALE : 1:1000 OR 1.0CM=10.00M
VERTICAL SCALE : 1:100 OR 1.0CM=1.00M

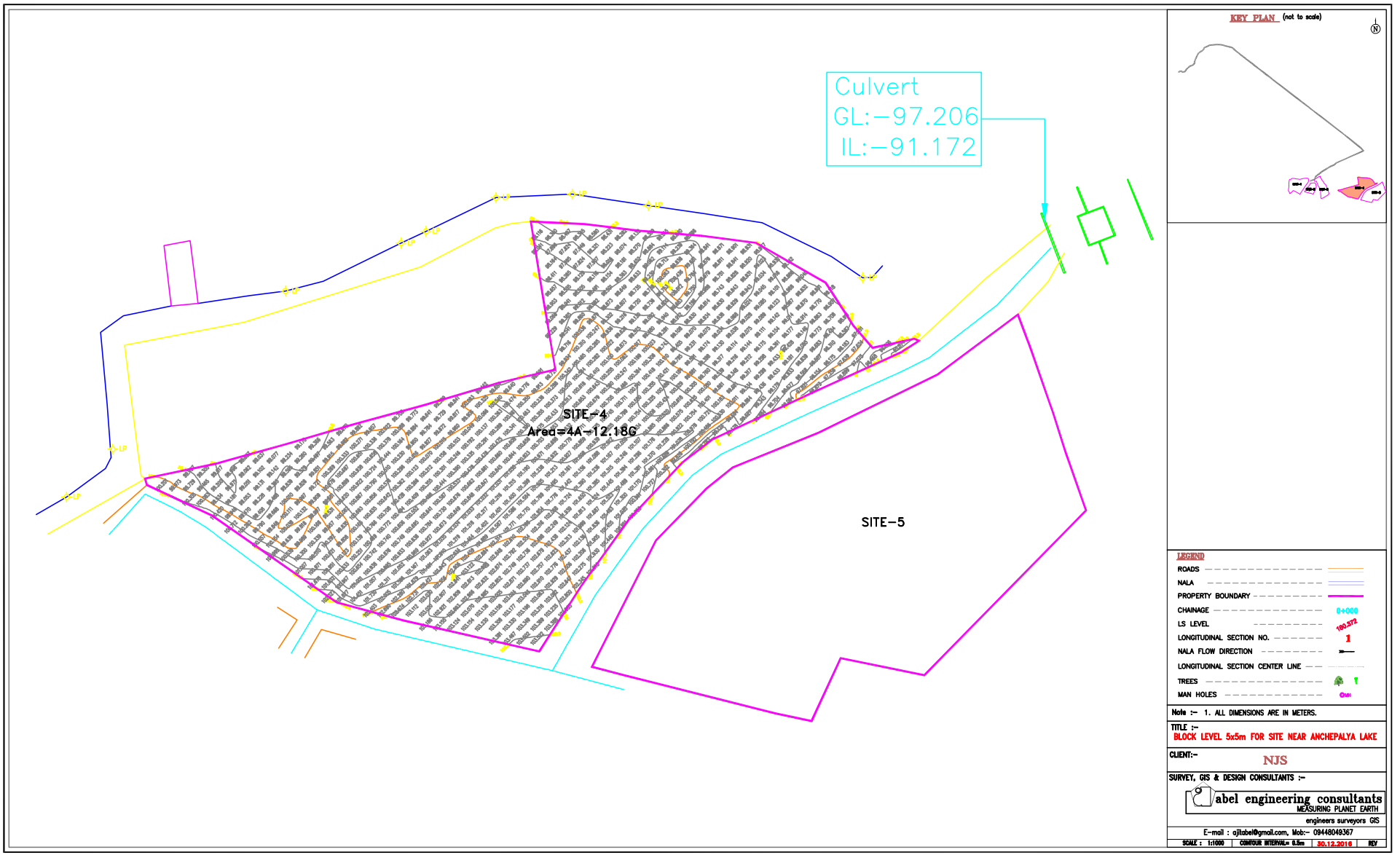

KEY PLAN (not to scale)
LEGEND
ROADS
NALA
PROPERTY BOUNDARY
CHAINAGE 0+000
LS LEVEL 100.572
LONGITUDINAL SECTION NO. 1
NALA FLOW DIRECTION
LONGITUDINAL SECTION CENTER LINE
TREES
MAN HOLES
Note :- 1. ALL DIMENSIONS ARE IN METERS.
TITLE :-
BLOCK LEVEL 5x5m FOR SITE NEAR Pillegenahalli STP
CLIENT:-
NJS
SURVEY, GIS & DESIGN CONSULTANTS :-
abel engineering consultants
MEASURING PLANET EARTH
engineers surveyors GIS
E-mail : ajitabel@gmail.com, Mob:- 09448049367
SCALE : 1:1000
CONTOUR INTERVAL= 0.5m
30.12.2016
REV
Nala
Nala
Slab
GL:-91.732
IL:-89.907
SITE-1
AREA=24-13.446

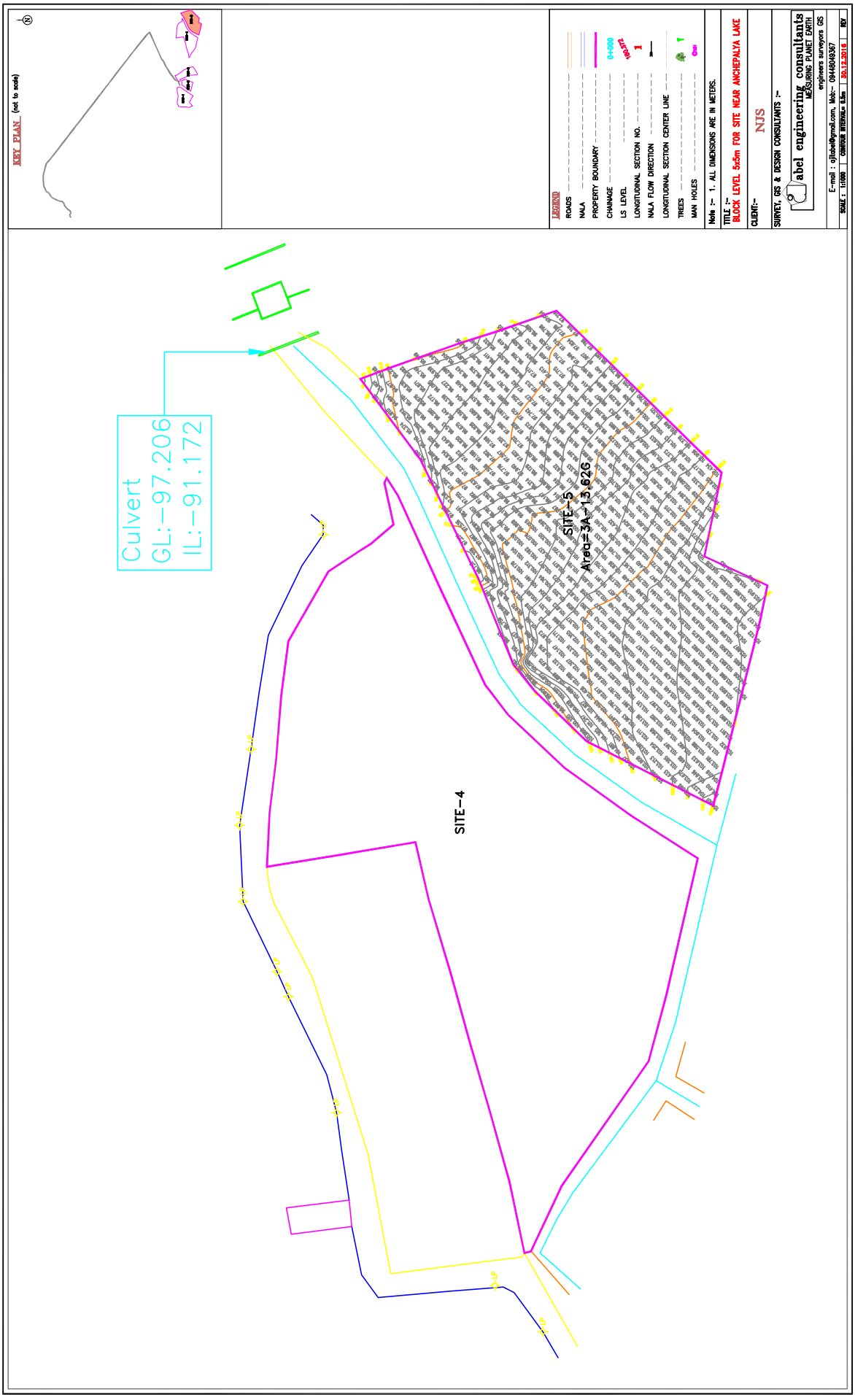


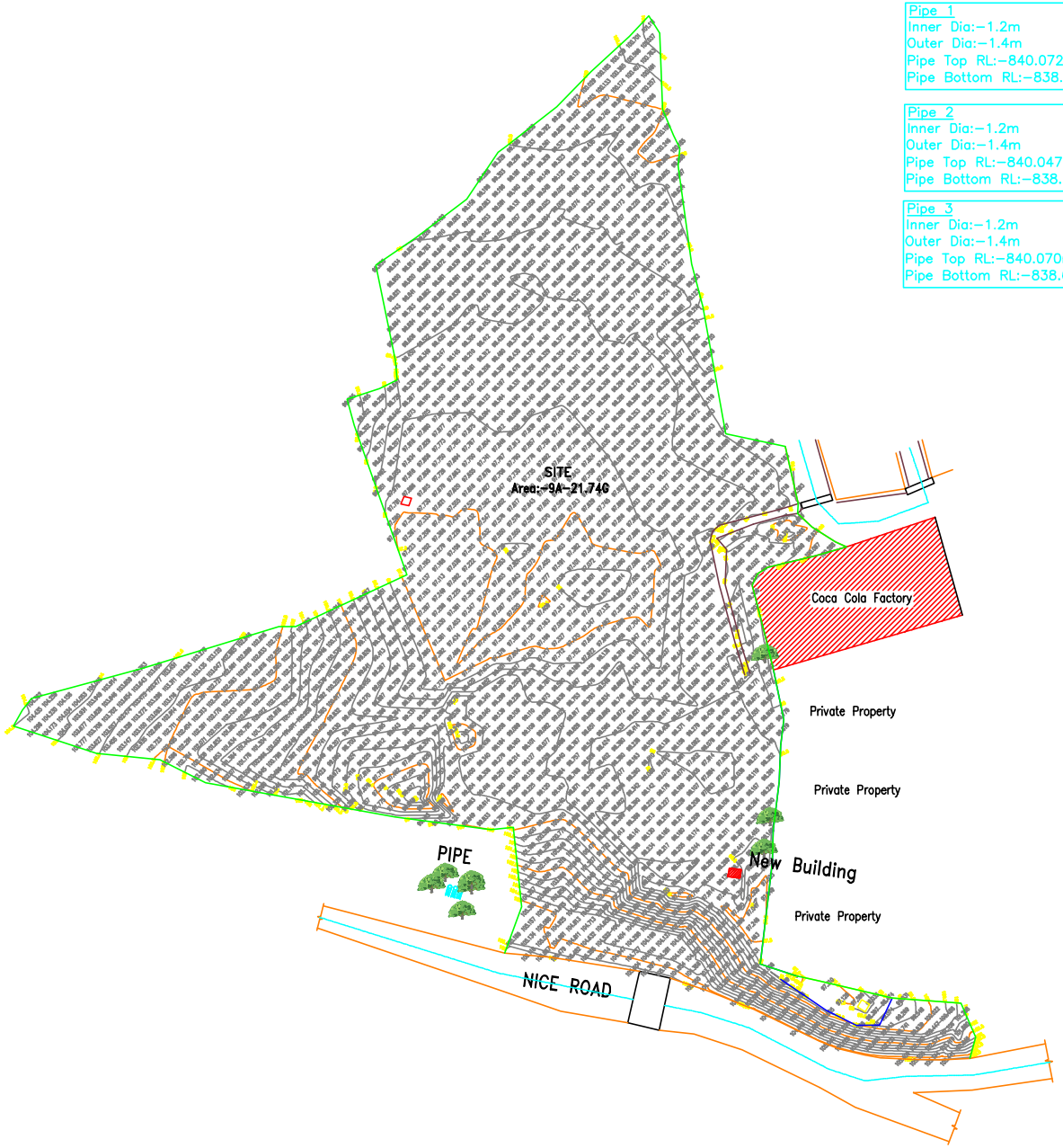

Culvert
GL:-101.050
IL:-98.781
SITE-1
Area=2A-32.78G
SITE-2
SITE-3
KEY PLAN (not to scale)
LEGEND
ROADS
NALA
PROPERTY BOUNDARY
CHAINAGE
LS LEVEL
LONGITUDINAL SECTION NO.
NALA FLOW DIRECTION
LONGITUDINAL SECTION CENTER LINE
TREES
MAN HOLES
Note :- 1. ALL DIMENSIONS ARE IN METERS.
TITLE :-
BLOCK LEVEL 5x5m FOR SITE NEAR ANCHEPALYA LAKE
CLIENT:-
NJS
SURVEY, GIS & DESIGN CONSULTANTS :-
abel engineering consultants
MEASURING PLANET EARTH
engineers surveyors GIS
E-mail : ajitabel@gmail.com, Mob:- 09448049367
SCALE : 1:1000
CONTOUR INTERVAL= 0.5m
30.12.2016
REV


Culvert
GL:-101.050
IL:-98.781
SITE-1
SITE-2
Area=1A-00.12G
SITE-3
KEY PLAN (not to scale)
LEGEND
ROADS
NALA
PROPERTY BOUNDARY
CHAINAGE
0+000
LS LEVEL
160.372
LONGITUDINAL SECTION NO.
1
NALA FLOW DIRECTION
LONGITUDINAL SECTION CENTER LINE
TREES
MAN HOLES
Note :- 1. ALL DIMENSIONS ARE IN METERS.
TITLE :-
BLOCK LEVEL 5x5m FOR SITE NEAR ANCHEPALYA LAKE
CLIENT:-
NJS
SURVEY, GIS & DESIGN CONSULTANTS :-
abel engineering consultants
MEASURING PLANET EARTH
engineers surveyors GIS
E-mail : ajitabel@gmail.com, Mob:- 09448049367
SCALE : 1:1000
CONTOUR INTERVAL= 0.5m
30.12.2016
REV


Culvert
GL:-101.050
IL:-98.781
SITE-2
SITE-3
Area=1A-32.12G
KEY PLAN (not to scale)
LEGEND
ROADS
NALA
PROPERTY BOUNDARY
CHAINAGE
0+000
LS LEVEL
100.572
LONGITUDINAL SECTION NO.
1
NALA FLOW DIRECTION
LONGITUDINAL SECTION CENTER LINE
TREES
MAN HOLES
Note :- 1. ALL DIMENSIONS ARE IN METERS.
TITLE :-
BLOCK LEVEL 5x5m FOR SITE NEAR ANCHEPALYA LAKE
CLIENT:-
NJS
SURVEY, GIS & DESIGN CONSULTANTS :-
abel engineering consultants
MEASURING PLANET EARTH
engineers surveyors GIS
E-mail : ajitabel@gmail.com, Mob:- 09448049367
SCALE : 1:1000
CONTOUR INTERVAL= 0.5m
30.12.2016
REV


Culvert
GL:-97.206
IL:-91.172
SITE-4
Area=4A-12.18G
SITE-5
KEY PLAN (not to scale)
LEGEND
ROADS
NALA
PROPERTY BOUNDARY
CHAINAGE
0+000
LS LEVEL
160.572
LONGITUDINAL SECTION NO.
1
NALA FLOW DIRECTION
LONGITUDINAL SECTION CENTER LINE
TREES
MAN HOLES
Note :- 1. ALL DIMENSIONS ARE IN METERS.
TITLE :-
BLOCK LEVEL 5x5m FOR SITE NEAR ANCHEPALYA LAKE
CLIENT:-
NJS
SURVEY, GIS & DESIGN CONSULTANTS :-
abel engineering consultants
MEASURING PLANET EARTH
engineers surveyors GIS
E-mail : ajitabel@gmail.com, Mob:- 09448049367
SCALE : 1:1000
CONTOUR INTERVAL= 0.5m
30.12.2016
REV


KEY PLAN (not to scale)
LEGEND
ROADS
NALA
PROPERTY BOUNDARY
CHAINAGE
0+000
LS LEVEL
HEIGHT
LONGITUDINAL SECTION NO.
1
NALA FLOW DIRECTION
LONGITUDINAL SECTION CENTER LINE
TREES
MAN HOLES
Dia
Note :- 1. ALL DIMENSIONS ARE IN METERS.
TITLE :-
BLOCK LEVEL 5x5m FOR SITE NEAR ANCHEPALYA LAKE
CLIENT:-
NJS
SURVEY, GIS & DESIGN CONSULTANTS :-
abel engineering consultants
MEASURING PLANET EARTH
engineers surveyors GIS
E-mail : ajitabel@gmail.com, Mob:- 09448048367
SCALE : 1:1000
CONTOUR INTERVAL 0.5m
30.12.2016
REV
Culvert
GL:-97.206
IL:-91.172
SITE-5
Area=3A-13.626
SITE-4



Pipe 1
Inner Dia:-1.2m
Outer Dia:-1.4m
Pipe Top RL:-840.072m
Pipe Bottom RL:-838.705m

Pipe 2
Inner Dia:-1.2m
Outer Dia:-1.4m
Pipe Top RL:-840.047m
Pipe Bottom RL:-838.669m

Pipe 3
Inner Dia:-1.2m
Outer Dia:-1.4m
Pipe Top RL:-840.070m
Pipe Bottom RL:-838.682m

KEY PLAN (not to scale)

N

N

LEGEND

ROADS	
NALA	
PROPERTY BOUNDARY	
CHAINAGE	0+000
LS LEVEL	160.372
LONGITUDINAL SECTION NO.	1
NALA FLOW DIRECTION	
LONGITUDINAL SECTION CENTER LINE	
TREES	
MAN HOLES	MH

Note :- 1. ALL DIMENSIONS ARE IN METERS.

TITLE :-
BLOCK LEVEL 5x5m FOR Thalaghattapura STP

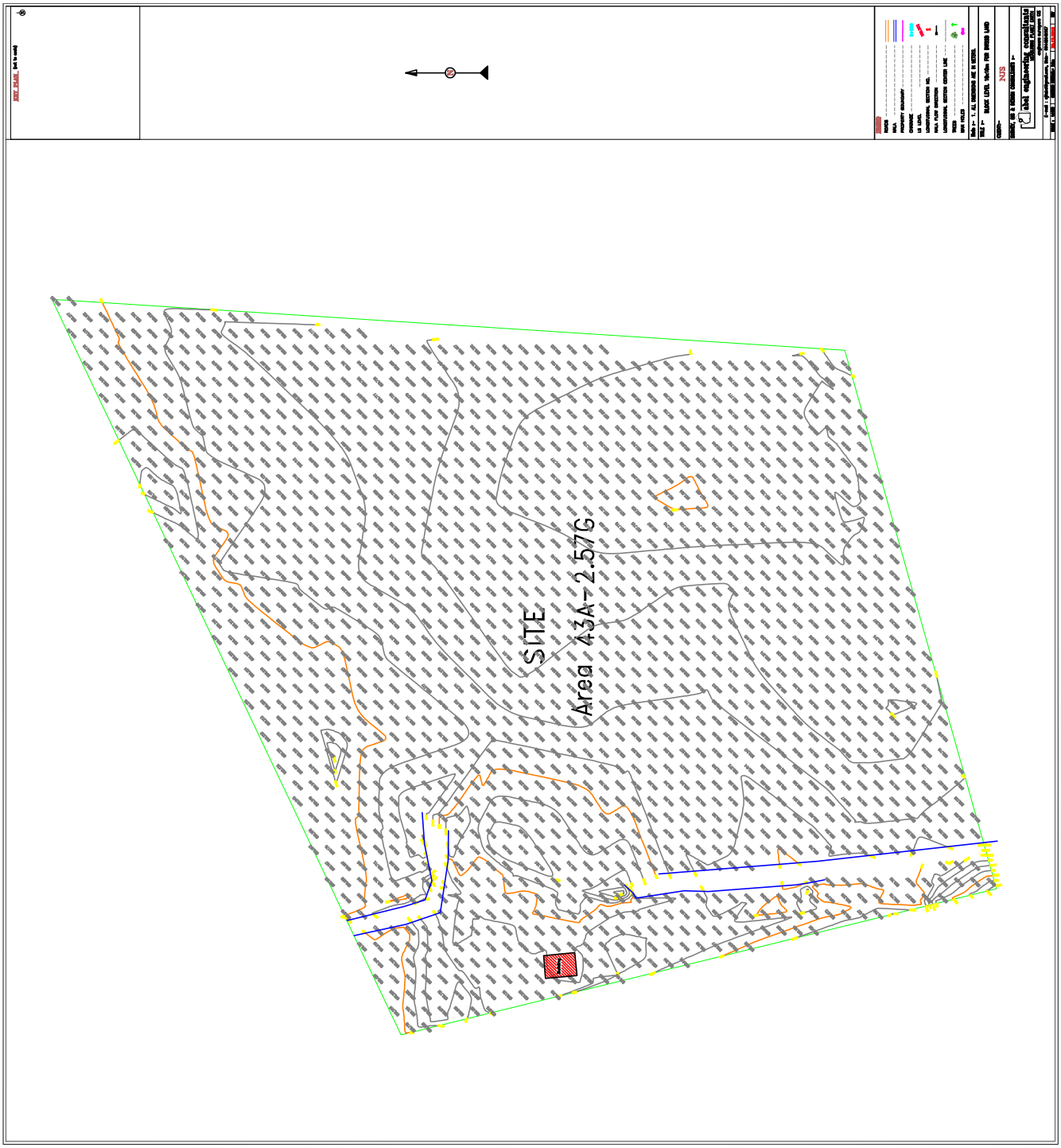
CLIENT:-
NJS

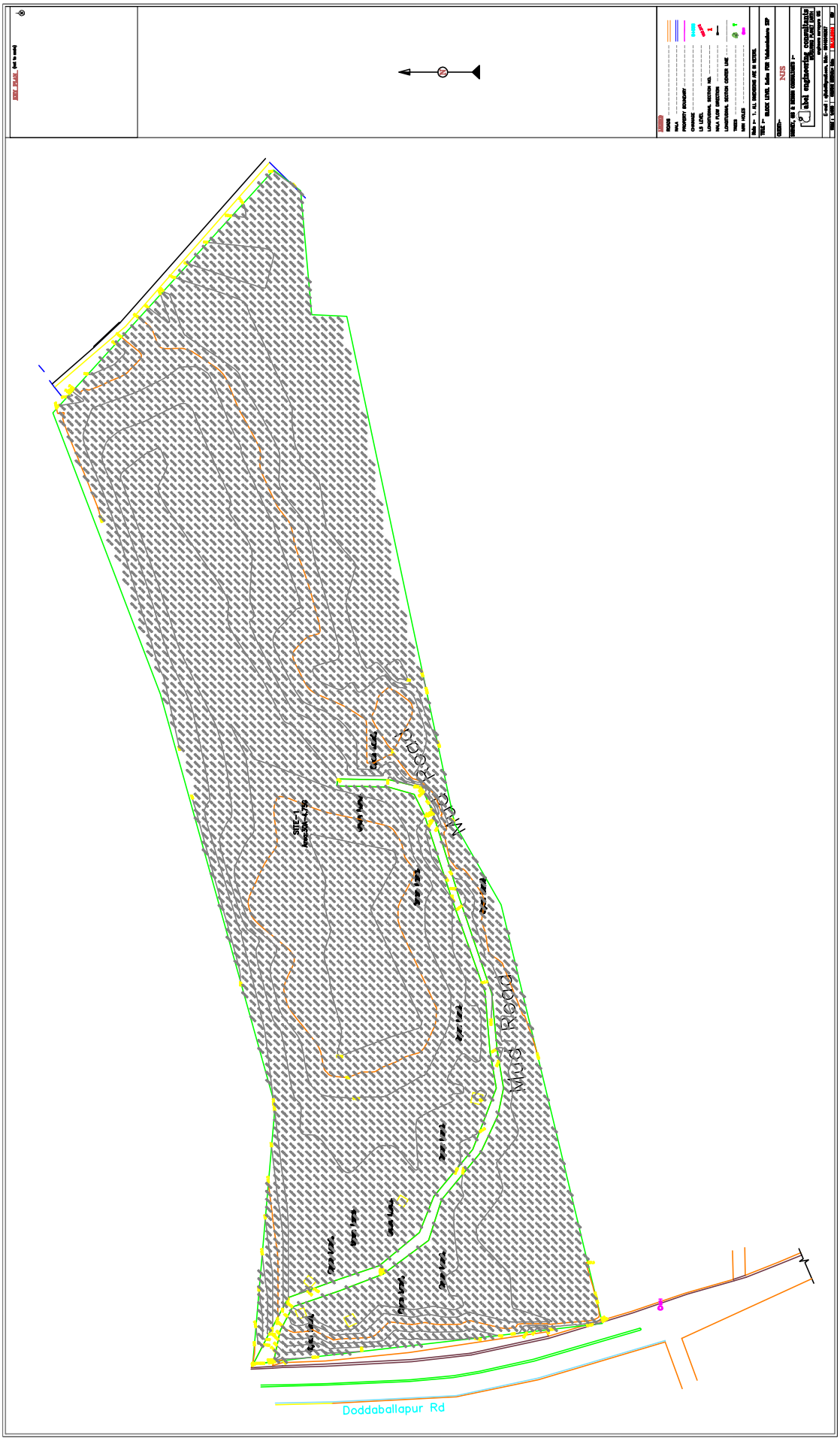
SURVEY, GIS & DESIGN CONSULTANTS :-

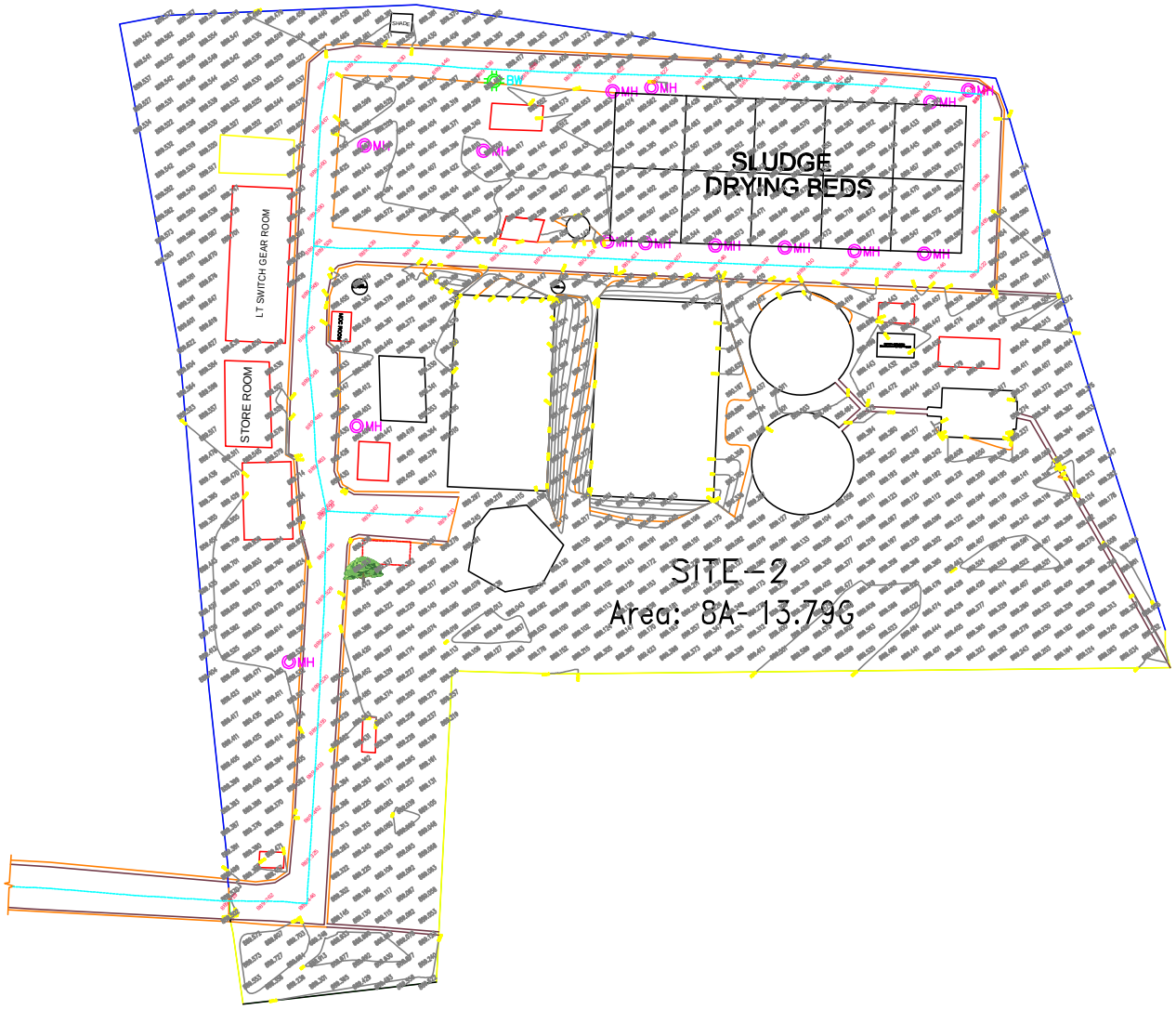
abel engineering consultants
MEASURING PLANET EARTH
engineers surveyors GIS

E-mail : ajitabel@gmail.com, Mob:- 09448049367

SCALE : 1:1000	CONTOUR INTERVAL= 0.5m	30.12.2016	REV


SITE
Area 43A-2.57G


SITE-1
MUDA Road
Doddaballapur Rd



KEY PLAN (not to scale)



LEGEND

ROADS	
NALA	
MEDIAN	
PROPERTY BOUNDARY	
COMPOUND WALL	
BUILDING	
FENCE BOUNDARY	
CHAINAGE	0+000
BLOCK LEVELS	160.372
CONTOUR LEVEL	160.0
LS LEVEL	160.372
LONGITUDINAL SECTION NO.	1
NALA FLOW DIRECTION	
LONGITUDINAL SECTION CENTER LINE	
TREES	
MAN HOLES	MH
ELECTRIC POLE	EP

Note :- 1. ALL DIMENSIONS ARE IN METERS.

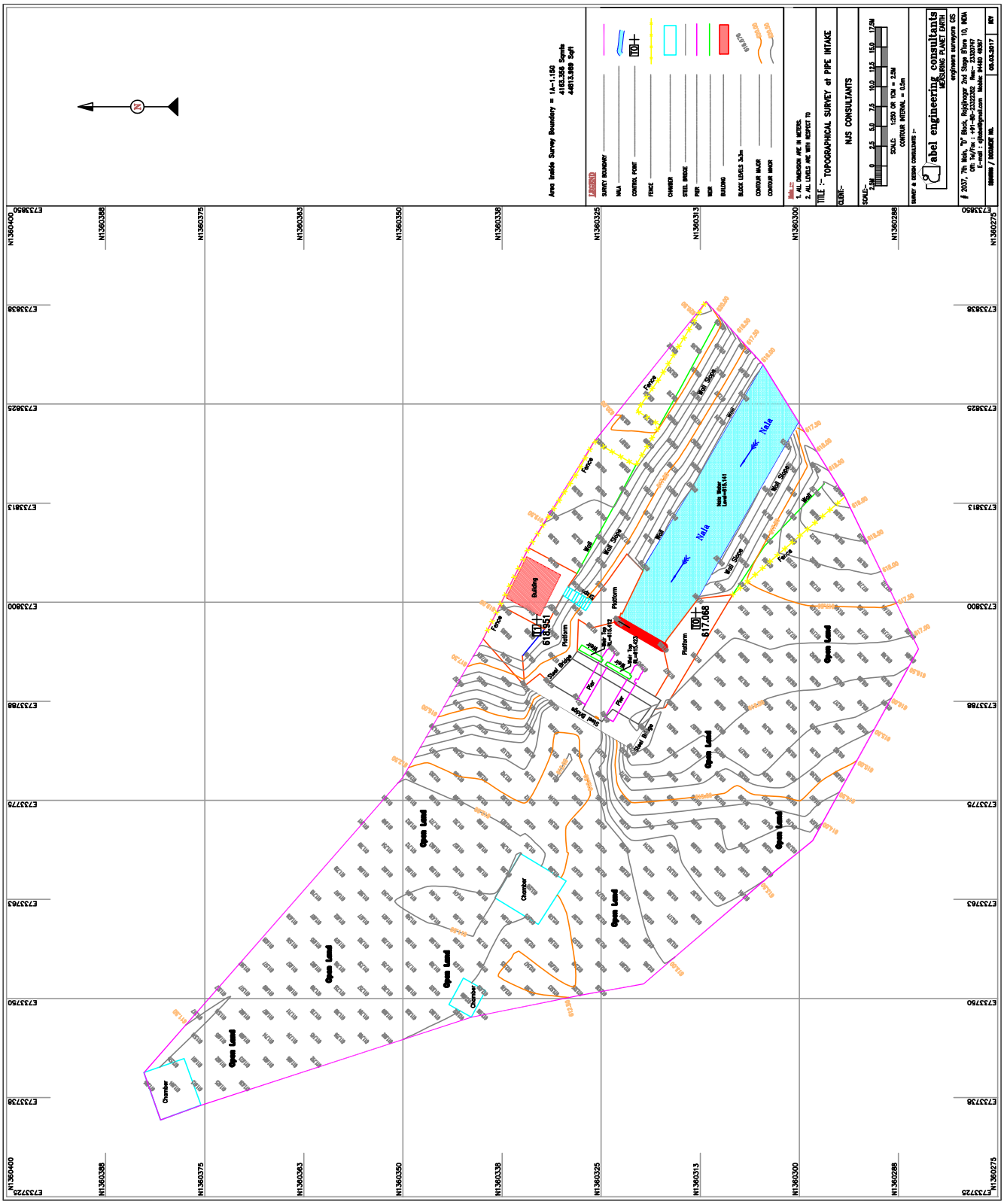
TITLE :- BLOCK LEVEL 5x5m FOR JAKKUR STP

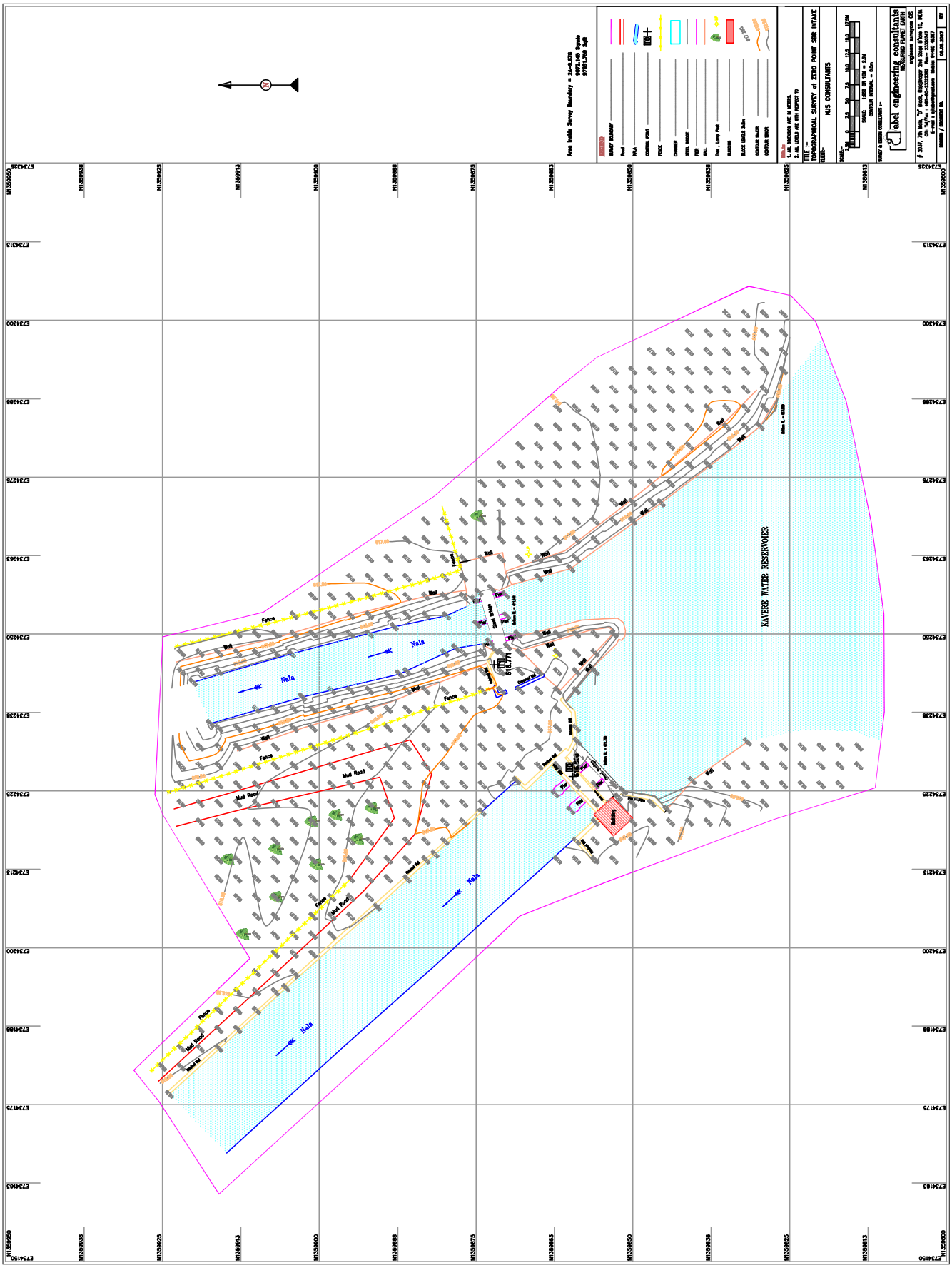
CLIENT:- NJS

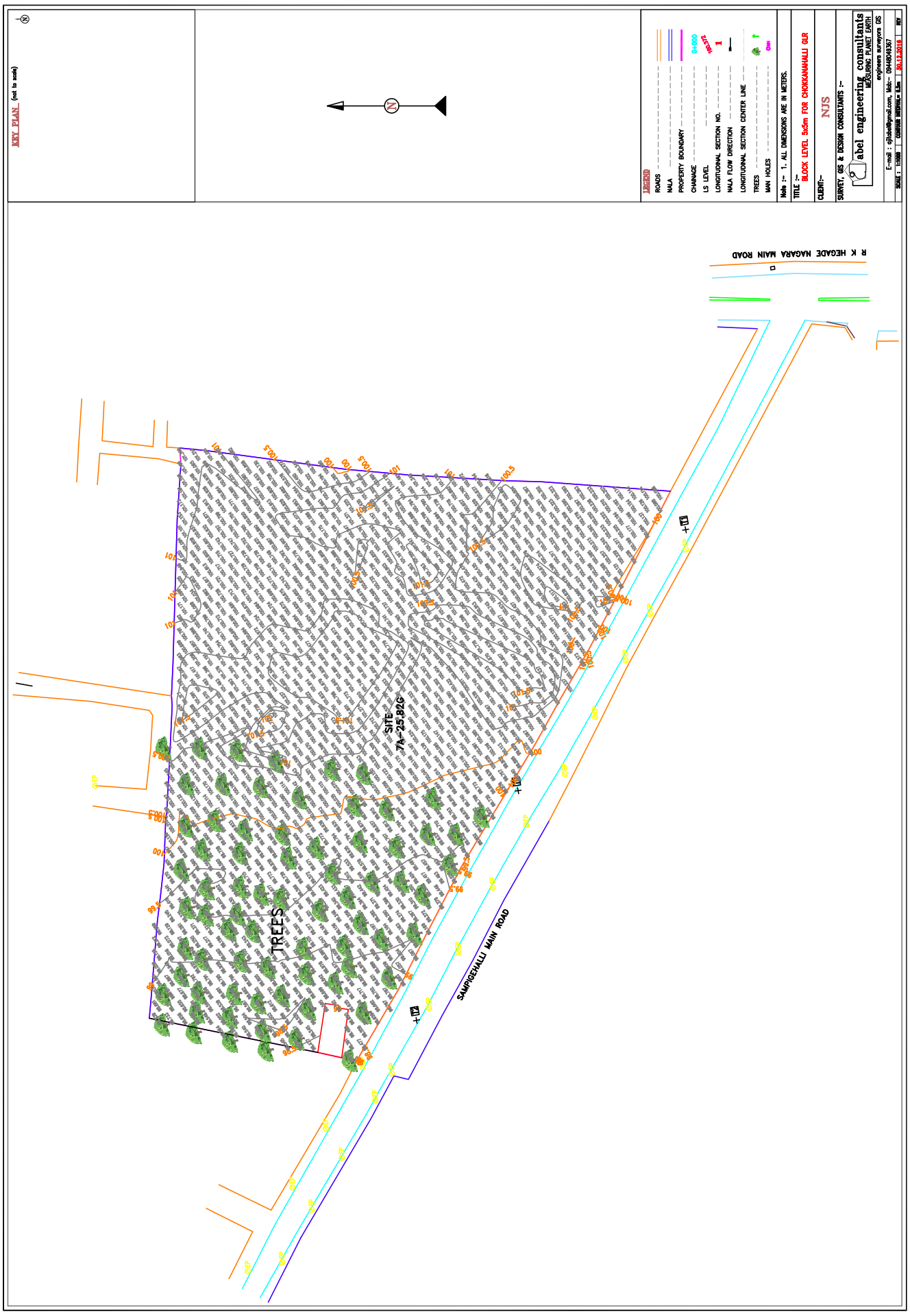
SURVEY, GIS & DESIGN CONSULTANTS :-

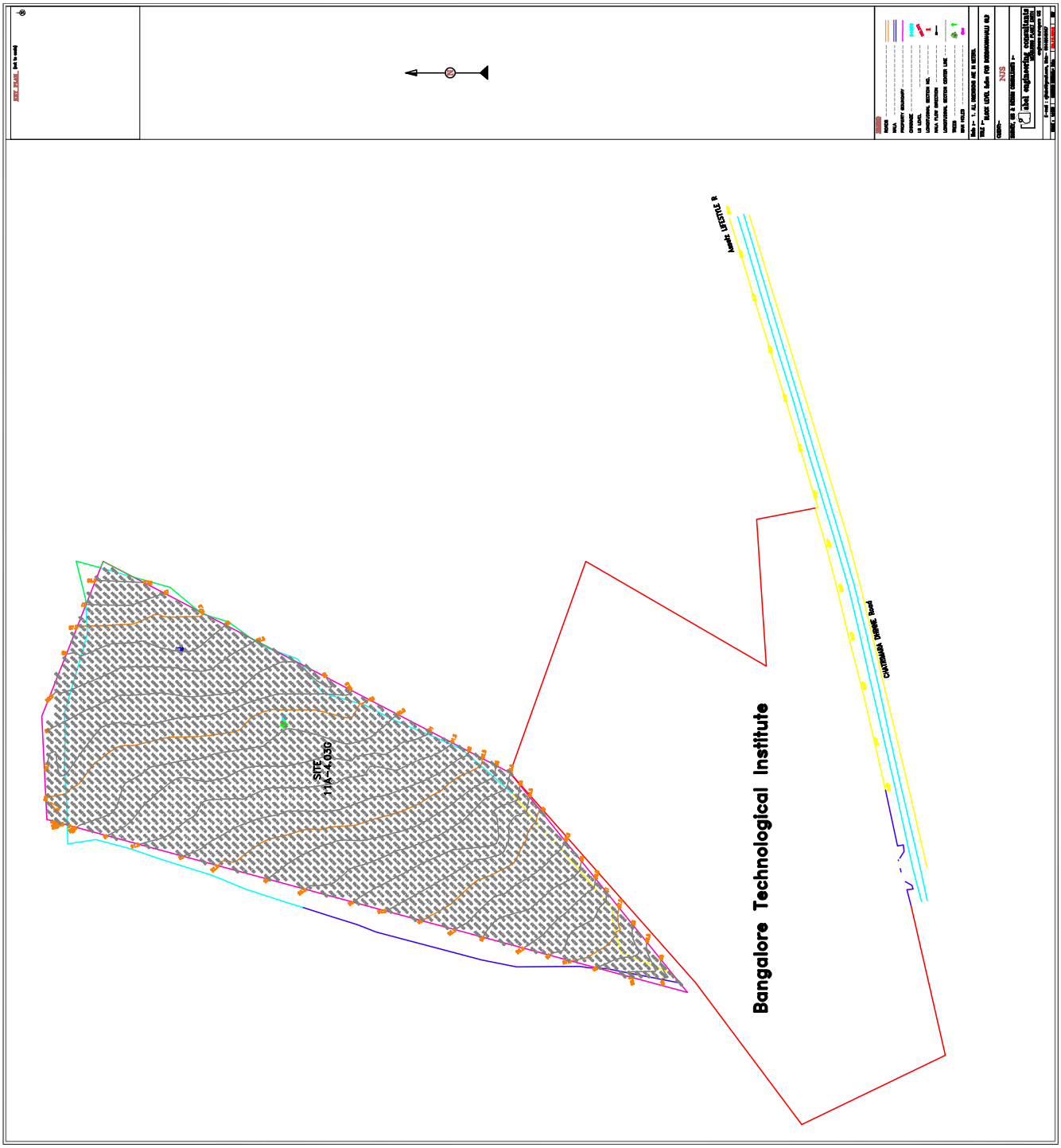
abel engineering consultants
MEASURING PLANET EARTH
engineers surveyors GIS
E-mail : ajitabel@gmail.com, Mob:- 09448049367

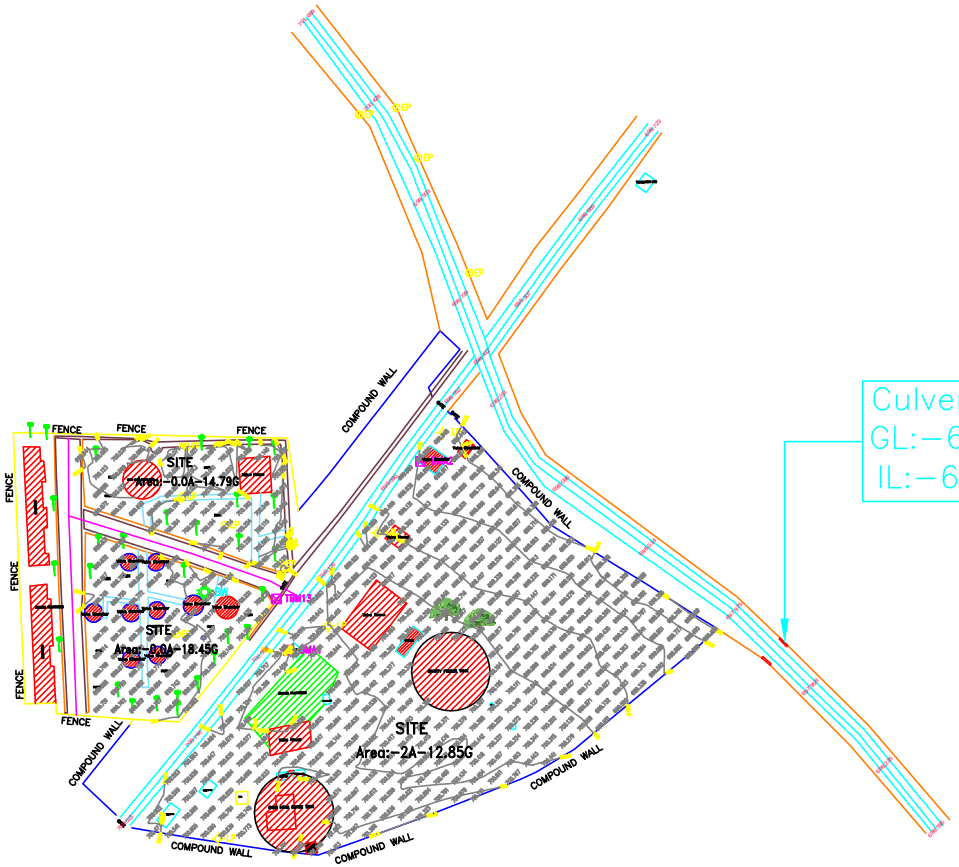
SCALE : 1:1000	CONTOUR INTERVAL= 0.5m	30.12.2016	REV


Area Inside Survey Boundary = 1A-1.150
4163.358 Sqmts
44813.569 Sqft
SURVEY BOUNDARY
NALA
CONTROL POINT
FENCE
CHAMBER
STEEL BRIDGE
PIER
WEIR
BUILDING
BLOCK LEVELS 3.5m
CONTOUR MAJOR
CONTOUR MINOR
1. ALL DIMENSION ARE IN METERS.
2. ALL LEVELS ARE WITH RESPECT TO
TITLE :- TOPOGRAPHICAL SURVEY of PIPE INTAKE
CLIENT:- NJS CONSULTANTS
SCALE:- 1:250 OR 1CM = 2.5M
CONTOUR INTERVAL = 0.5m
SURVEY & DESIGN CONSULTANTS :-
abel engineering consultants
MEASURING PLANET EARTH
engineers surveyors GIS
2037, 7th Main, "D" Block, Rajajinagar 2nd Stage Bl'lore 10, INDIA
OFF. Tel/Fax : +91-80-23322282, Mob:- 23301497
E-mail : abel@vsnl.com Website : www.abel.in
DRAWING / WORKSHEET NO.
DATE
REV
Building
Platform
Nala
Wall
Wall Slope
Fence
Steel Bridge
Pier
Open Land
Chamber
Main Water Level 618.441
Pier Top 618.951
Pier Top 617.088
617.088


KAVERI WATER RESERVOIR
Nala
Mud Road
Fence
Wall
Pipe
Building
abel engineering consultants
TOPOGRAPHICAL SURVEY of ZERO POINT SUMP INTAKE
NJS CONSULTANTS


KEY PLAN (not to scale)
LEGEND
ROADS
NALA
PROPERTY BOUNDARY
CHAINAGE
LS LEVEL
LONGITUDINAL SECTION NO.
NALA FLOW DIRECTION
LONGITUDINAL SECTION CENTER LINE
TREES
MAN HOLES
Note :- 1. ALL DIMENSIONS ARE IN METERS.
TITLE :- BLOCK LEVEL 5x5m FOR CHOKKANAHALLI GLR
CLIENT:- NJS
SURVEY, GIS & DESIGN CONSULTANTS :- abel engineering consultants
MEASURING PLANET EARTH
engineers surveyors GIS
E-mail : ajilabel@gmail.com, Mob:- 09448043367
SCALE : 1:1000 CONTOUR INTERVAL= 0.5m DATE 30.12.2016
R K HEGADE NAGARA MAIN ROAD
SITE
7A-25.82G
TREES
SAMPIGEHALLI MAIN ROAD


Bangalore Technological Institute
SITE
11A-4.030



KEY PLAN (not to scale)

N

N

LEGEND

ROADS	
NALA	
PROPERTY BOUNDARY	
CHAINAGE	0+000
LS LEVEL	160.372
LONGITUDINAL SECTION NO.	1
NALA FLOW DIRECTION	
LONGITUDINAL SECTION CENTER LINE	
TREES	
MAN HOLES	MH

Note :- 1. ALL DIMENSIONS ARE IN METERS.

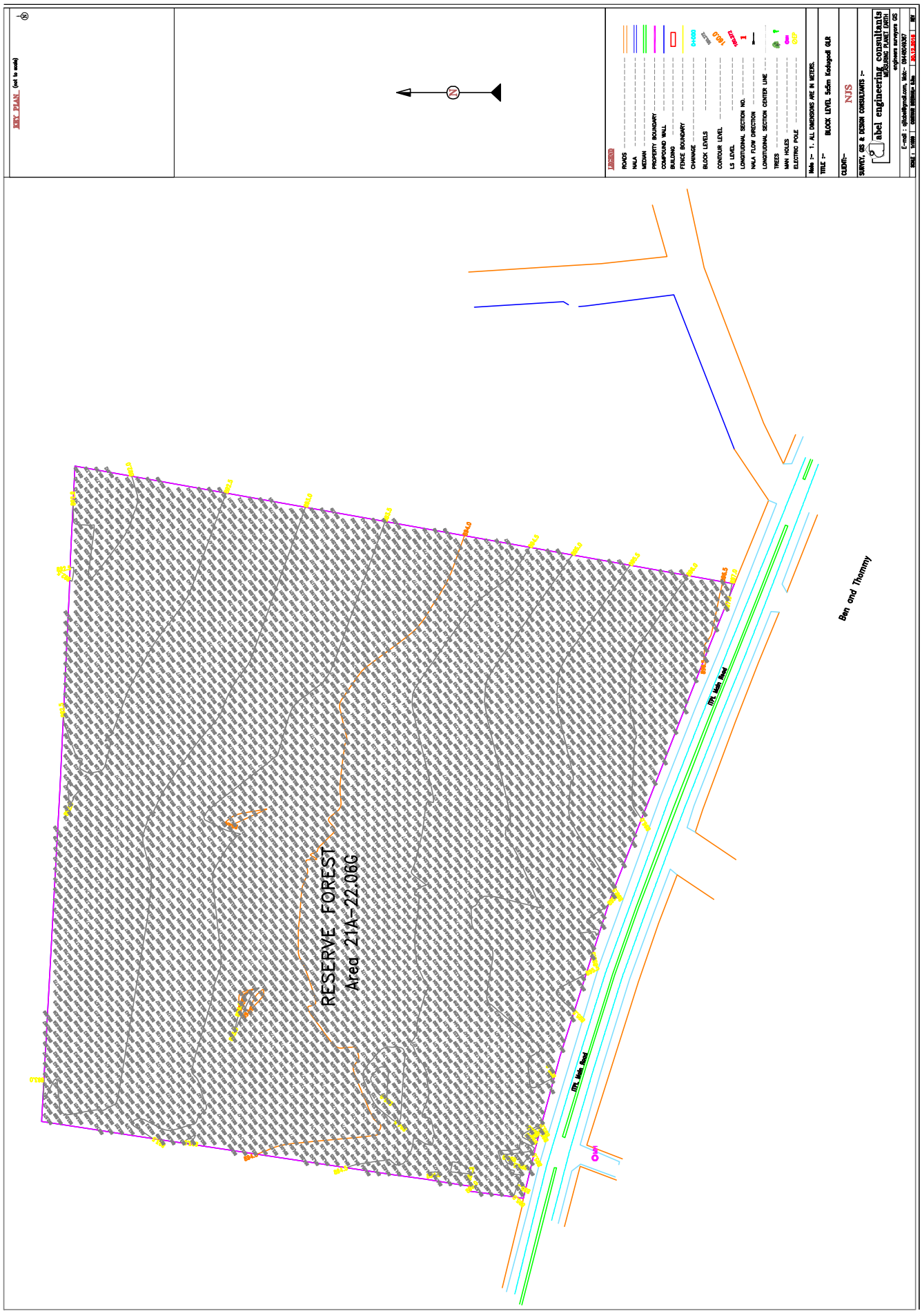
TITLE :-
BLOCK LEVEL 5x5m FOR J G Doddi Surge Tank

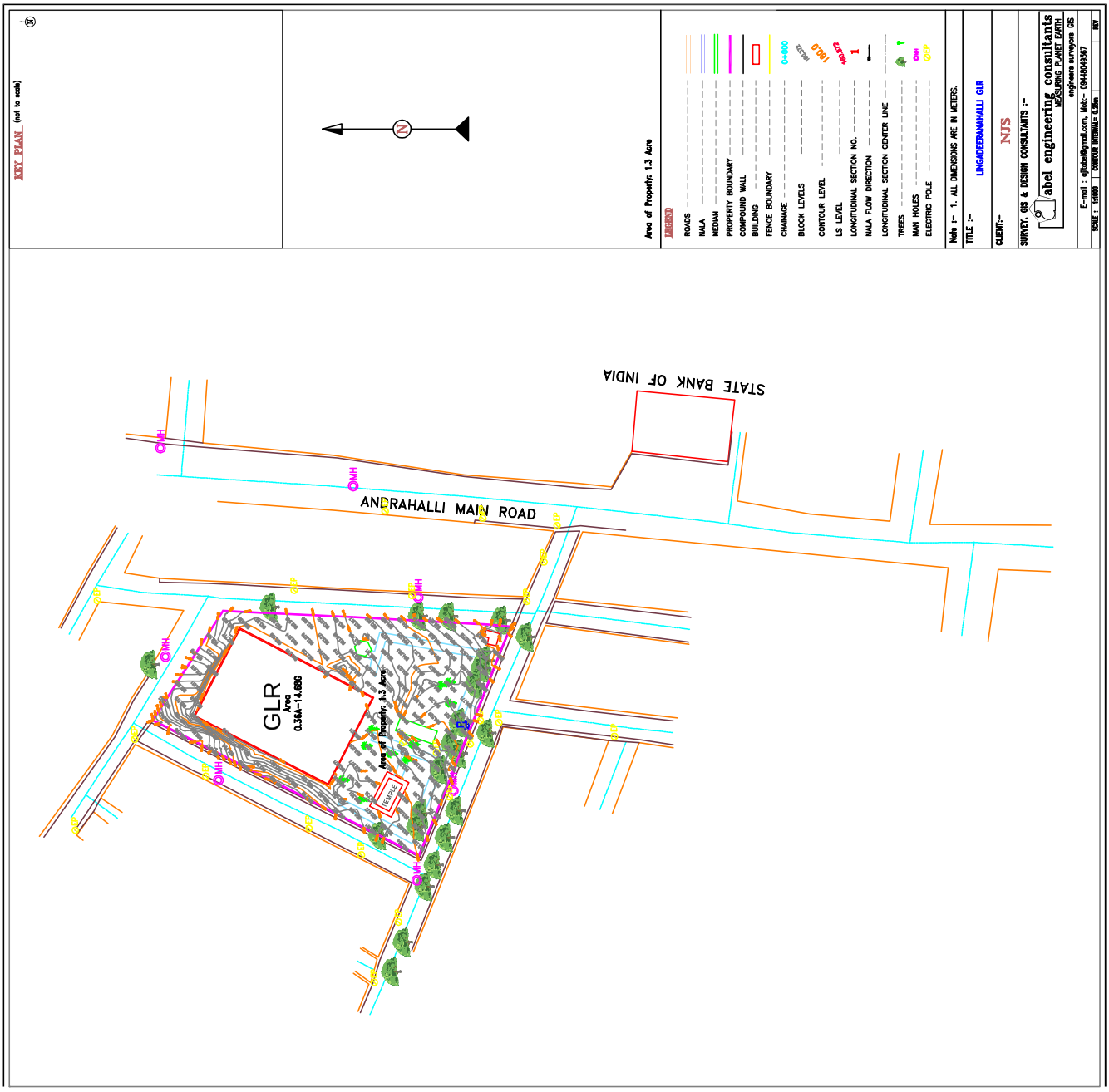
CLIENT:-
NJS

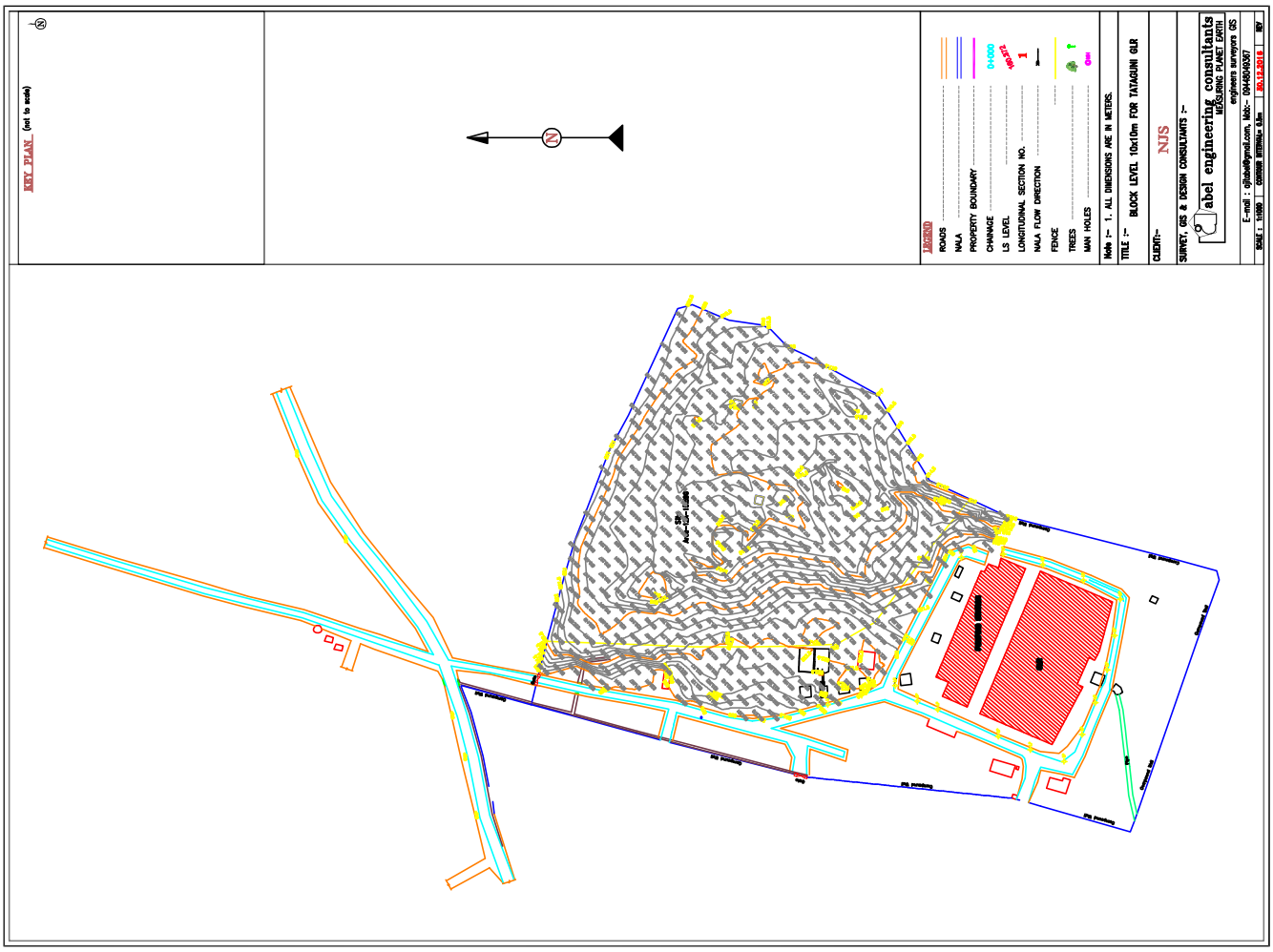
SURVEY, GIS & DESIGN CONSULTANTS :-

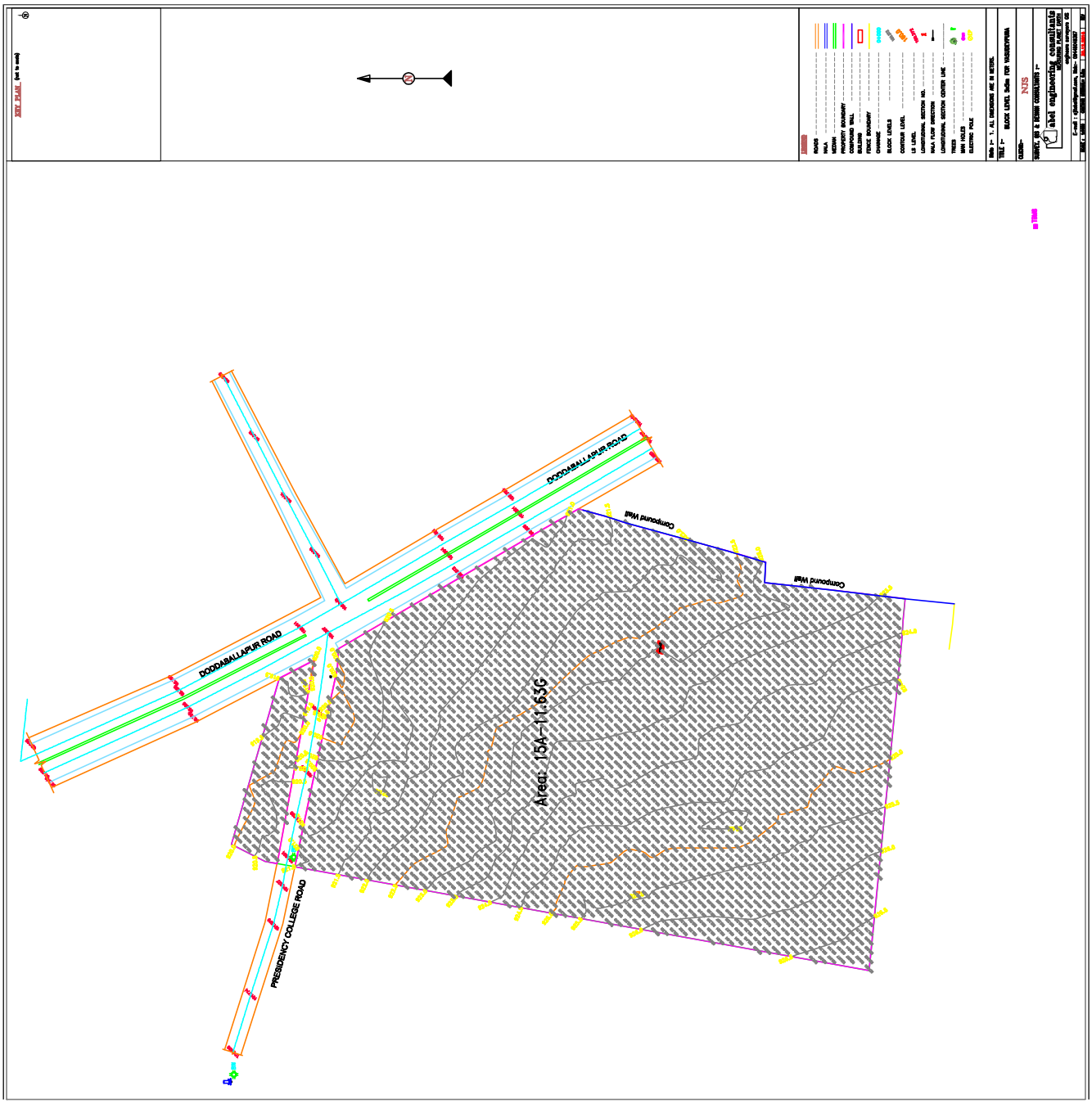
abel engineering consultants
MEASURING PLANET EARTH
engineers surveyors GIS
E-mail : ajitabel@gmail.com, Mob:- 09448049367

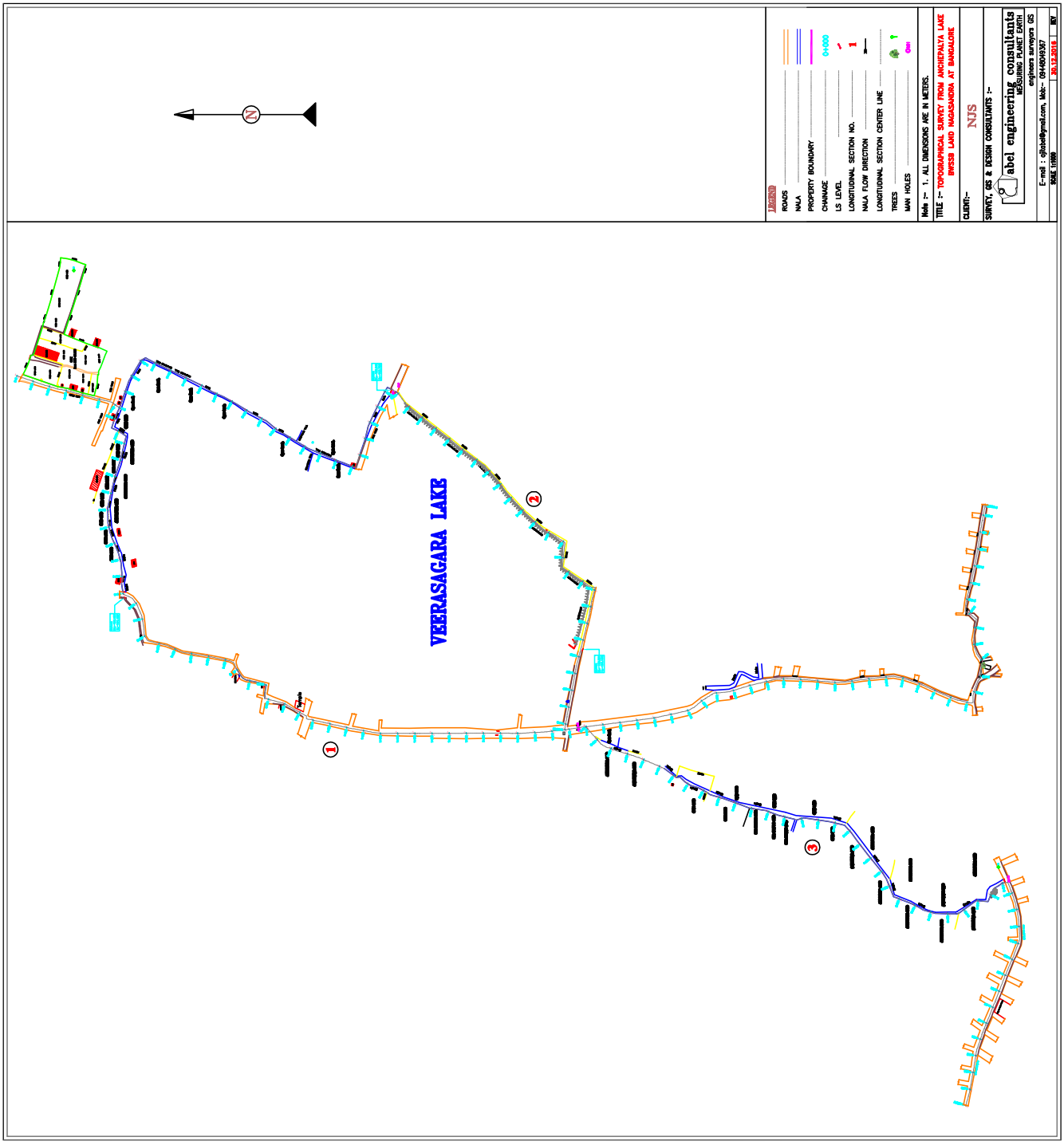
SCALE : 1:1000	CONTOUR INTERVAL= 0.5m	30.12.2016	REV


KEY PLAN (not to scale)
LEGEND
ROADS
NALA
MEDIAN
PROPERTY BOUNDARY
COMPOUND WALL
BUILDING
FENCE BOUNDARY
CHAINAGE
BLOCK LEVELS
CONTOUR LEVEL
LS LEVEL
LONGITUDINAL SECTION NO.
NALA FLOW DIRECTION
LONGITUDINAL SECTION CENTER LINE
TREES
MAN HOLES
ELECTRIC POLE
Note :- 1. ALL DIMENSIONS ARE IN METERS.
TITLE :- BLOCK LEVEL 5x5m Kodagodi GLR
CLIENT:-
NJS
SURVEY, GIS & DESIGN CONSULTANTS :-
abel engineering consultants
MEASURING PLANET EARTH
engineers surveyors GIS
RESERVE FOREST
Area 21A-22.06G
Ben and Thommy
Main Road
Main Road


KEY PLAN (not to scale)
Area of Property: 1.3 Acre
LEGEND
ROADS
NALA
MEDIAN
PROPERTY BOUNDARY
COMPOUND WALL
BUILDING
FENCE BOUNDARY
CHAINAGE
BLOCK LEVELS
CONTOUR LEVEL
LS LEVEL
LONGITUDINAL SECTION NO.
NALA FLOW DIRECTION
LONGITUDINAL SECTION CENTER LINE
TREES
MAN HOLES
ELECTRIC POLE
Note :- 1. ALL DIMENSIONS ARE IN METERS.
TITLE :- LINGADEERANAHALLI GLR
CLIENT:-
NJS
SURVEY, GIS & DESIGN CONSULTANTS :-
abel engineering consultants
MEASURING PLANET EARTH
engineers surveyors GIS
E-mail : gibelt@gmail.com Mob:- 09448049357
SCALE : 1:1000 | CONTOUR INTERVAL 0.5m
STATE BANK OF INDIA
ANDRAHALLI MAIN ROAD
GLR
Area
0.36A-14.88G
Area of Property: 1.3 Acre
TEMPLE
OWH
GSP


KEY PLAN (not to scale)
Legend
ROADS
NALA
PROPERTY BOUNDARY
CHAINAGE
LS LEVEL
LONGITUDINAL SECTION NO.
NALA FLOW DIRECTION
FENCE
TREES
MAN HOLES
Note :- 1. ALL DIMENSIONS ARE IN METERS.
TITLE :- BLOCK LEVEL 10x10m FOR TATAGUNI GLR
CLIENT:-
NJS
SURVEY, GIS & DESIGN CONSULTANTS :-
abel engineering consultants
MEASURING PLANET EARTH
engineers surveyors GIS
E-mail : ajitabel@gmail.com, Mobi:- 09448049087
SCALE : 1:500
DATE OF SURVEY : 05.12.2014
REV


DODDABALLAPUR ROAD
DODDABALLAPUR ROAD
PRESIDENCY COLLEGE ROAD
Compound Wall
Compound Wall
Area: 154-11.63G


LEGEND
ROADS
NALA
PROPERTY BOUNDARY
CHAINAGE
LS LEVEL
LONGITUDINAL SECTION NO.
NALA FLOW DIRECTION
LONGITUDINAL SECTION CENTER LINE
TREES
MAN HOLES
Note :- 1. ALL DIMENSIONS ARE IN METERS.
TITLE :- TOPOGRAPHICAL SURVEY FROM AKKENALYA LAKE
BYPASS LAND NAGASANDRA AT BANGALORE
CLIENT:-
NJS
SURVEY, GIS & DESIGN CONSULTANTS :-
abel engineering consultants
MEASURING PLANET EARTH
engineers surveyors GIS
E-mail : ajabel@gmail.com, Mob:- 09448049367
SCALE 1:1000
REV
VEERASAGARA LAKE